

An Introduction to Parallel Programming with OpenMP

Tim Mattson



Download tutorial materials onto your laptop:
`git clone https://github.com/CERN-STEAM-Academy/26-PARALLEL-COMPUTING.git`

An Introduction to me

I'm just a simple kayak instructor

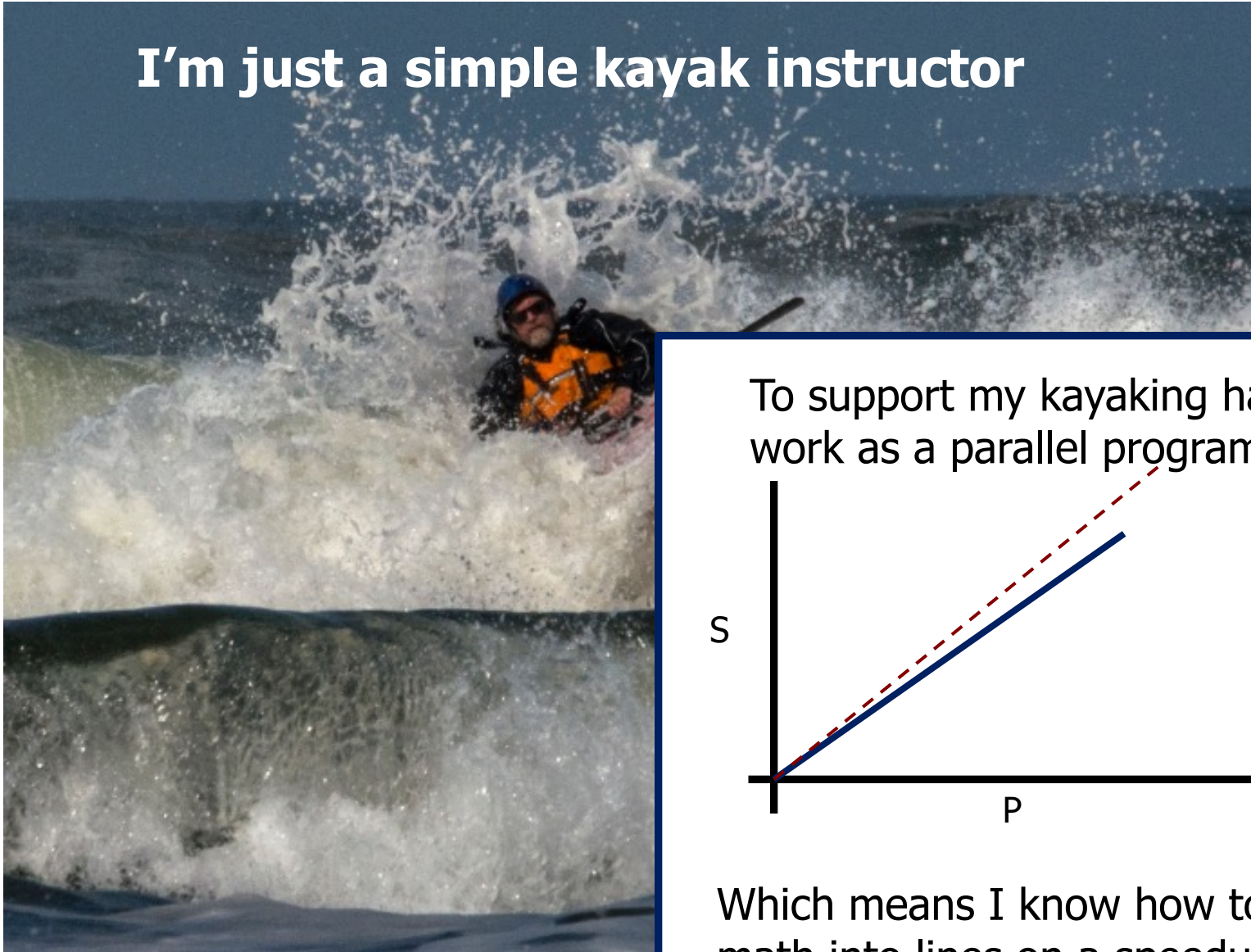
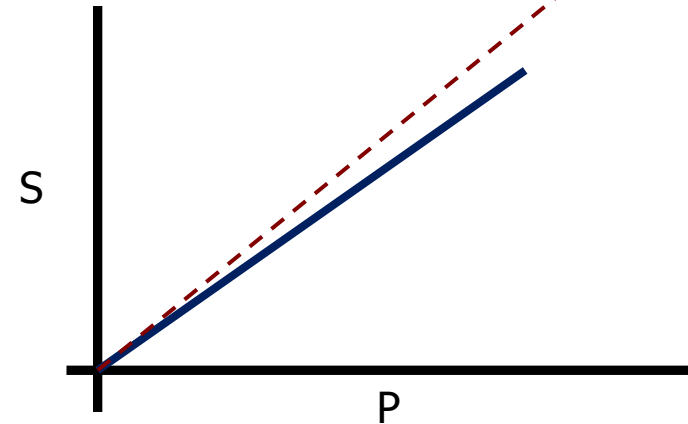


Photo © by Greg Clopton, 2014

To support my kayaking habit, I work as a parallel programmer



Which means I know how to turn math into lines on a speedup plot

Preliminaries:

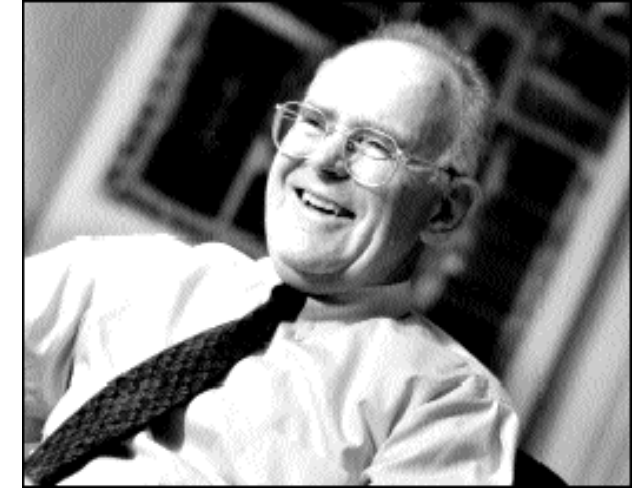
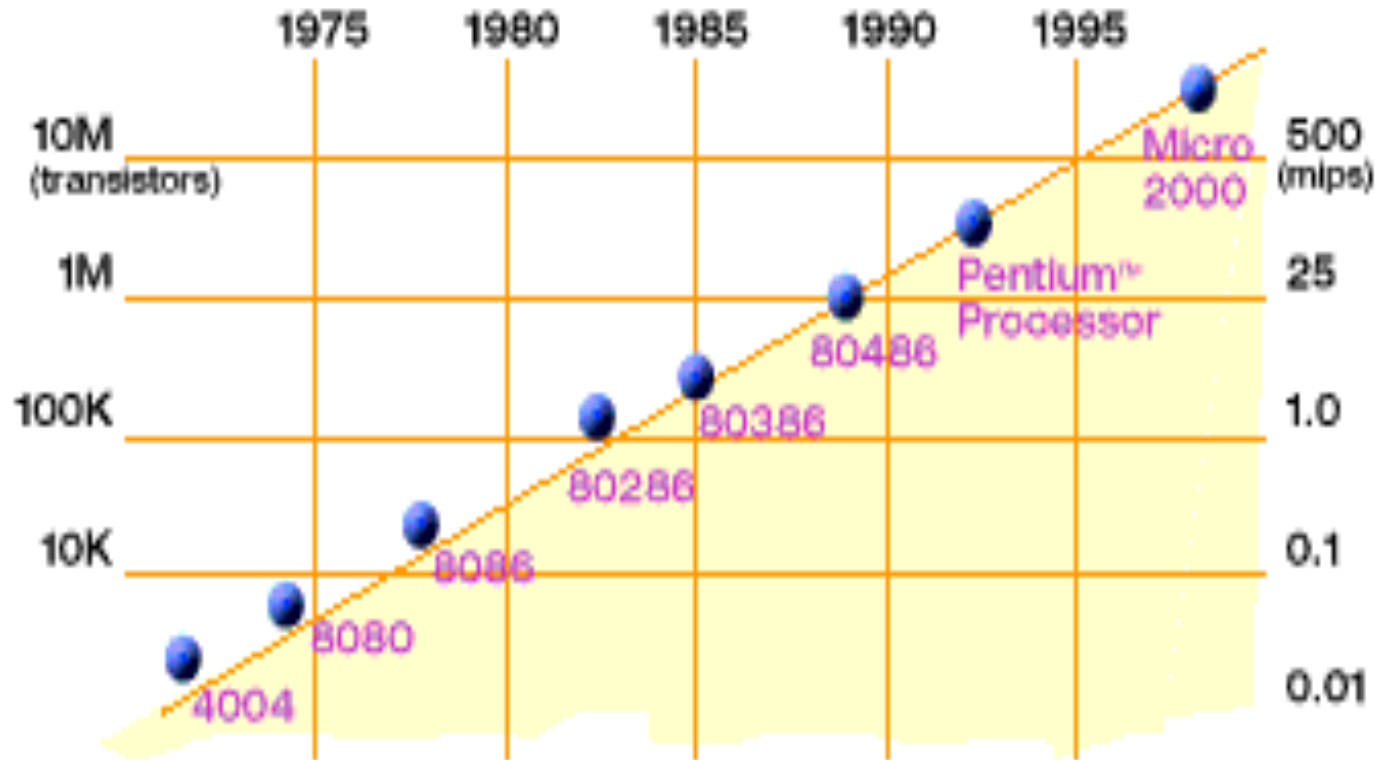
- Our plan ... Active learning!
 - We will mix short lectures with short exercises.
 - I assume you all have setup your laptops to use a gnu compiler, OR you have access to a remote server with an OpenMP compiler.
- Please follow these simple rules
 - Do the exercises that we assign and then change things around and experiment.
 - Embrace active learning!
 - Don't cheat: Do Not look at the solutions before you complete an exercise ... even if you get really frustrated.

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```

Why do you need to understand parallel computing?

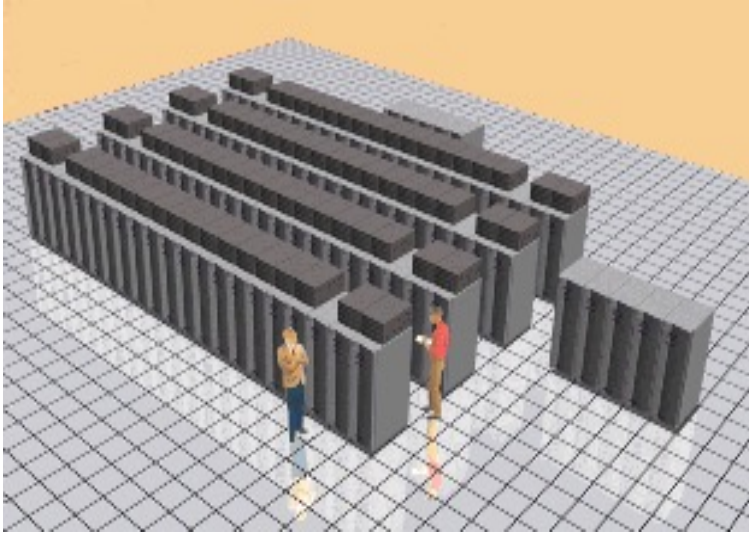
Moore's Law



- In 1965, Intel co-founder Gordon Moore predicted (from just 3 data points!) that semiconductor density would double every 18 months.
 - ***He was right!*** Over the last 50 years, transistor densities have increased as he predicted.

Moore's Law: A personal perspective

First TeraScale* computer: 1997



Intel's ASCI Option Red

Intel's ASCI Red Supercomputer

9000 CPUs

one megawatt of electricity.

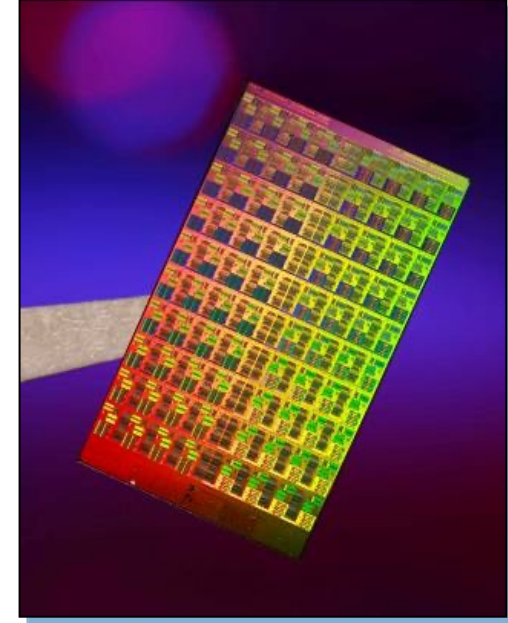
1600 square feet of floor space.

*Double Precision TFLOPS running MP-Linpack

A TeraFLOP in 1996: The ASCI TeraFLOP Supercomputer,
Proceedings of the International Parallel Processing
Symposium (1996), T.G. Mattson, D. Scott and S. Wheat.

10 years
later

First TeraScale% chip: 2007



Intel's 80 core teraScale Chip

1 CPU

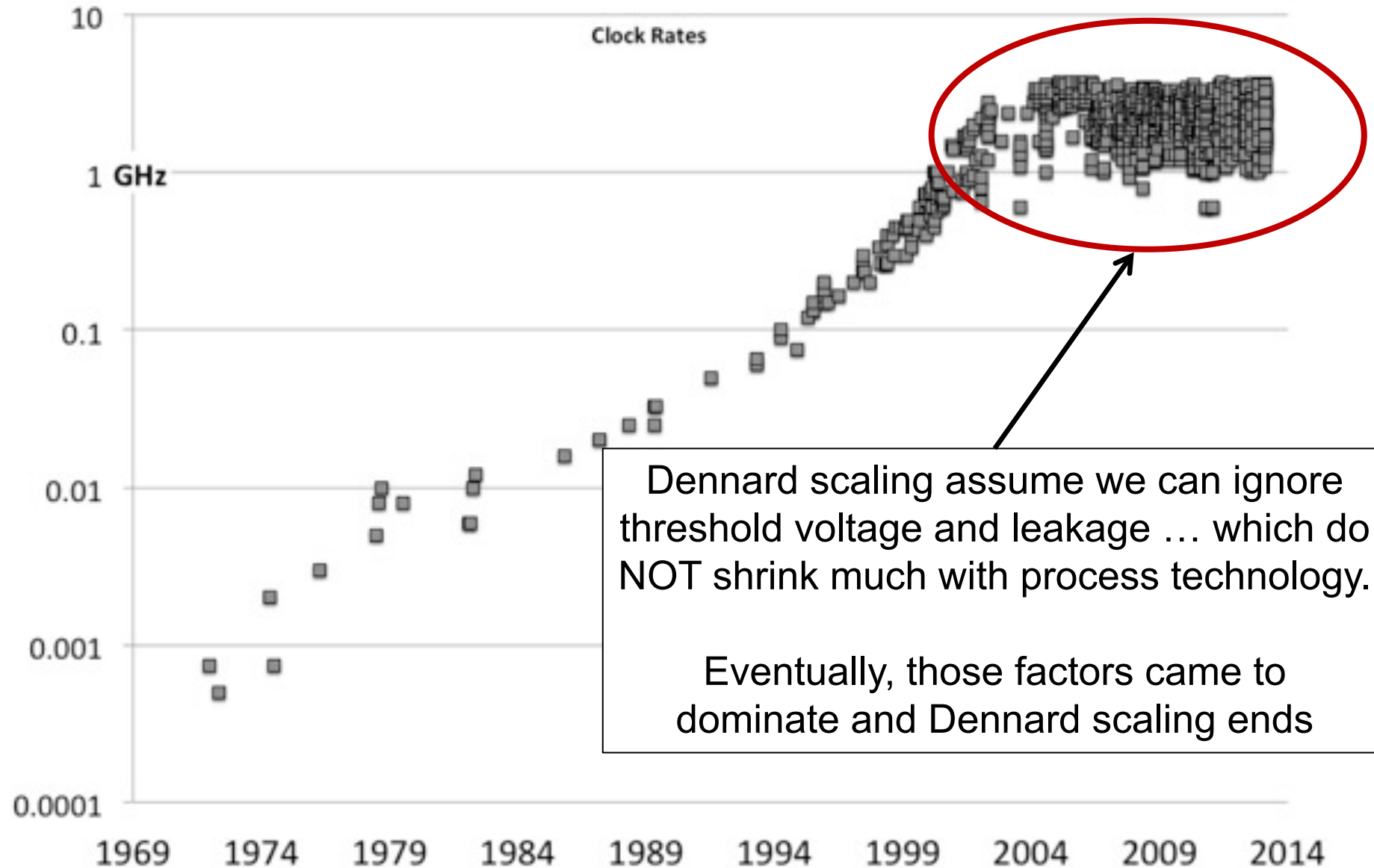
97 watt

275 mm²

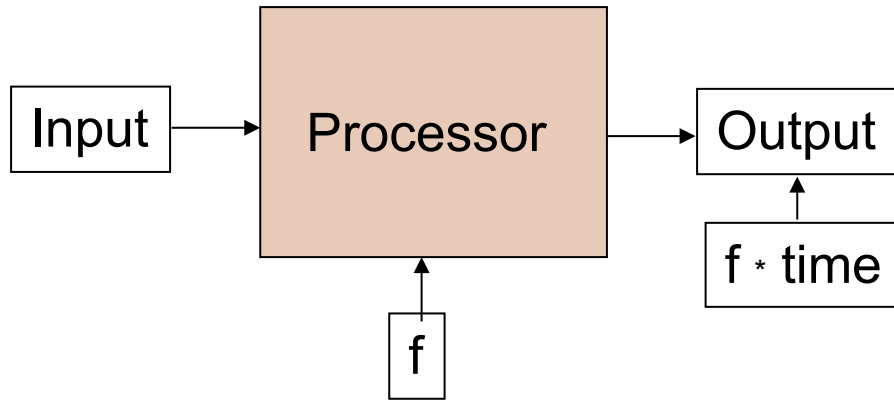
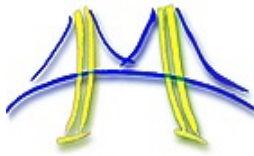
%Single Precision TFLOPS running stencil

Programming Intel's 80 core terascale processor
SC08, Austin Texas, Nov. 2008, Tim Mattson,
Rob van der Wijngaart, Michael Frumkin

CPU Frequency (GHz) over time (years)



Consider power in a chip ...



Capacitance = C
Voltage = V
Frequency = f
Power = CV^2f

C = capacitance ... it measures the ability of a circuit to store energy:

$$C = q/V \rightarrow q = CV$$

Work is pushing something (charge or q) across a “distance” ... in electrostatic terms pushing q from 0 to V:

$$V * q = W.$$

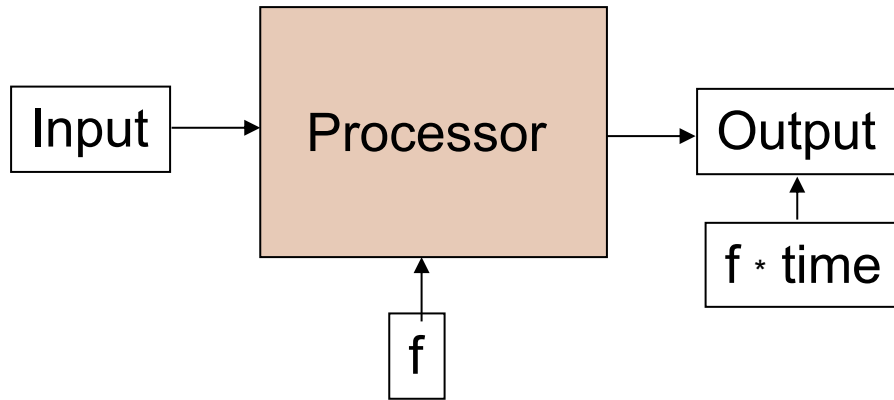
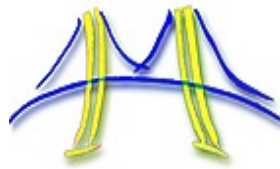
But for a circuit $q = CV$ so

$$W = CV^2$$

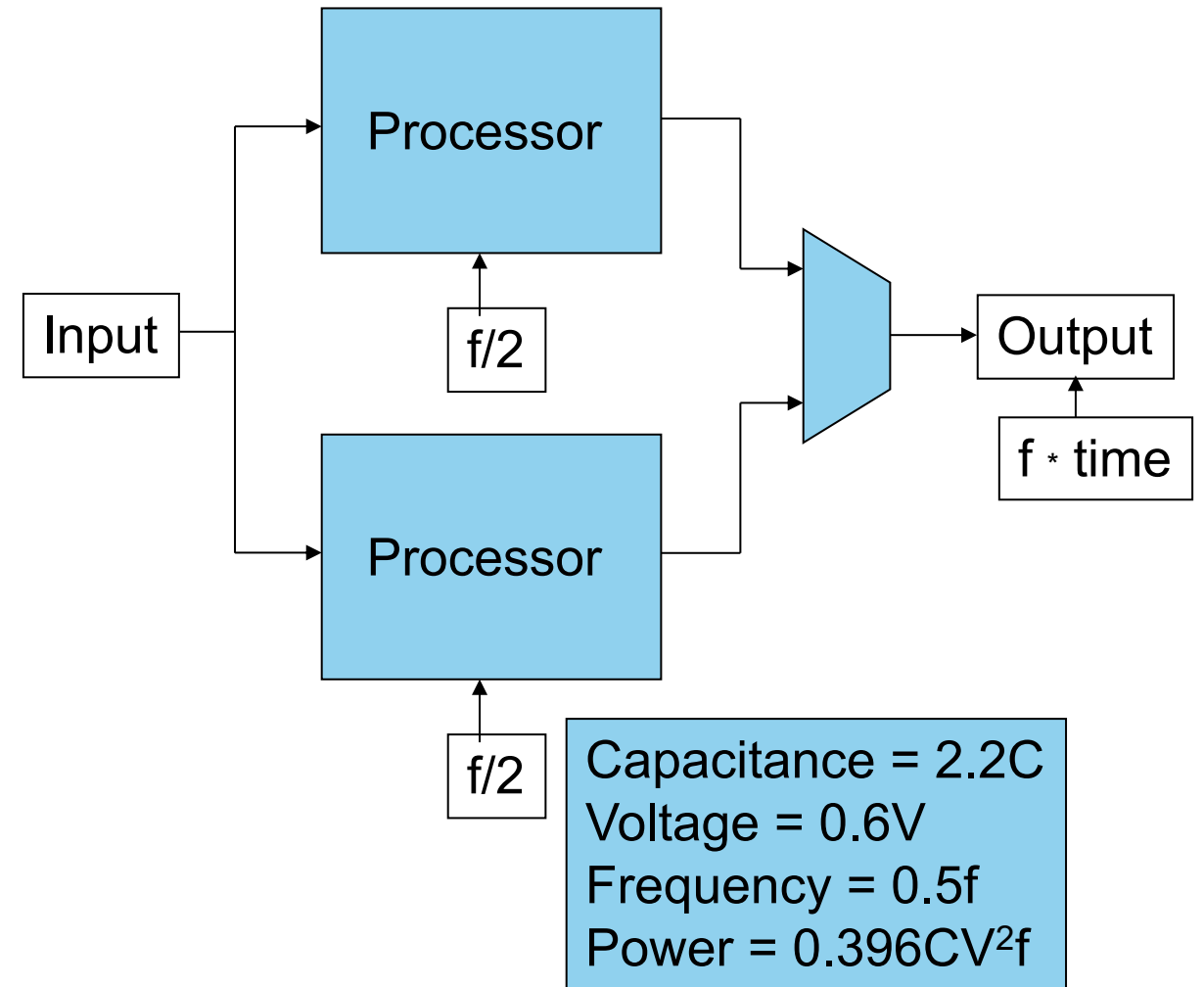
power is work over time ... or how many times per second we oscillate the circuit

$$\text{Power} = W * F \rightarrow \text{Power} = CV^2f$$

... Reduce power by adding cores



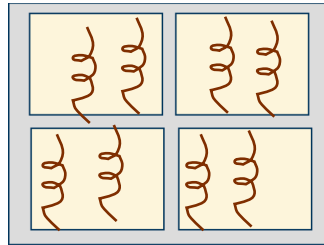
Capacitance = C
Voltage = V
Frequency = f
Power = CV^2f



Capacitance = $2.2C$
Voltage = $0.6V$
Frequency = $0.5f$
Power = $0.396CV^2f$

For hardware ... parallelism is the path to performance

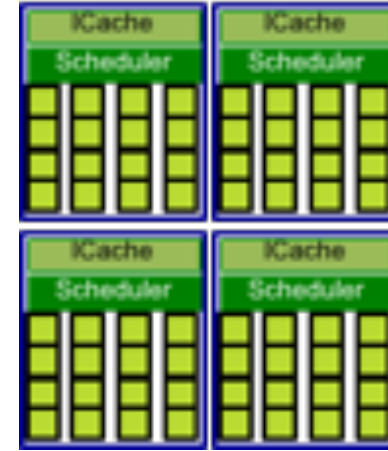
All hardware vendors are in the game ... parallelism is ubiquitous so if you care about getting the most from your hardware, you will need to create parallel software.



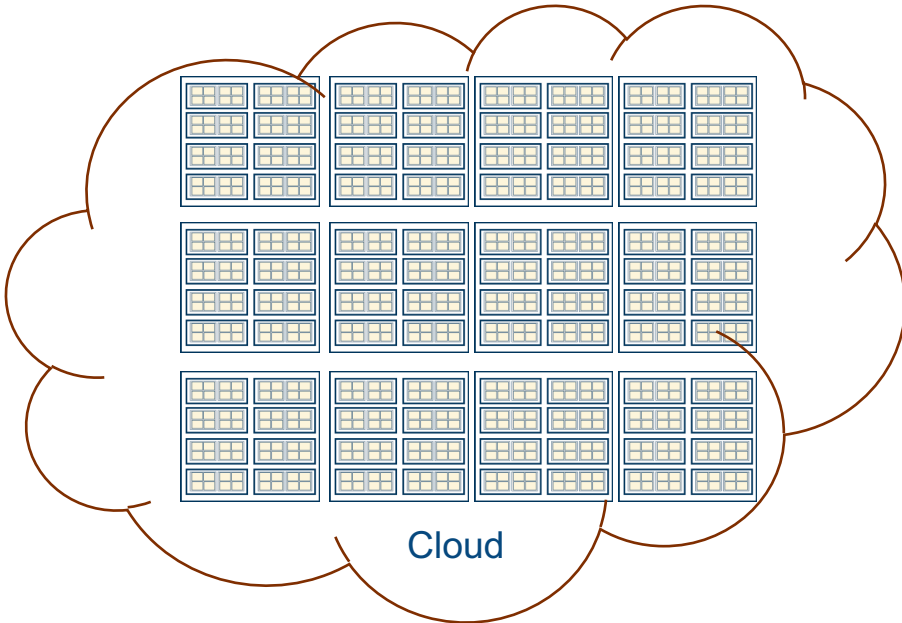
CPU



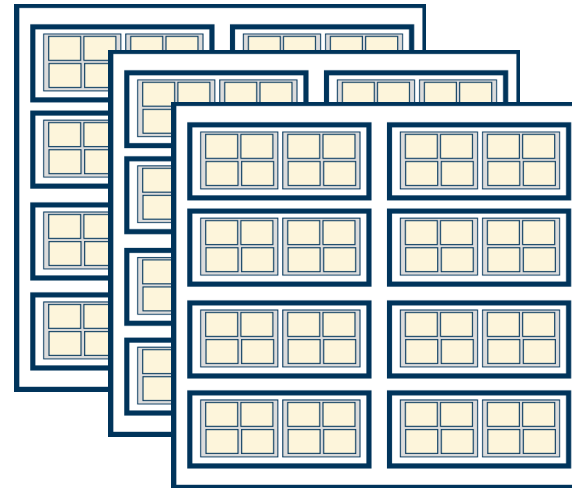
SIMD/Vector



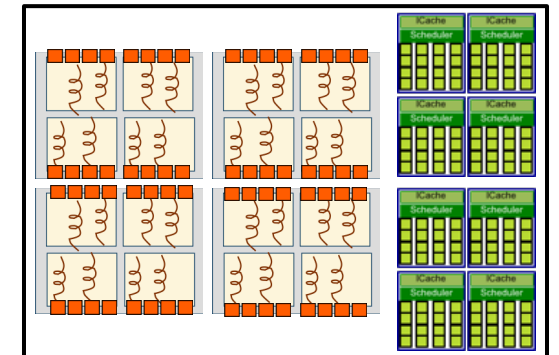
GPU



Cloud



Cluster



Heterogeneous node

The best way to master parallel computing ...

**start with a simple approach to parallelism and build
an intellectual foundation by writing parallel code.**

... and the simplest API for parallelism is?

OpenMP®

- ➡ • Introduction to OpenMP
 - Creating Threads
 - Synchronization ... simple collective sync
 - Parallel Loops
 - Data Environment
 - Synchronization ... the general case (including Memory Models)
 - Irregular Parallelism and Tasks
 - OpenMP for real hardware (NUMA)
 - Recap

OpenMP* Overview

C\$OMP FLUSH

#pragma omp critical

#pragma omp single

C\$OMP THREADPRIVATE (/ABC/)

C\$OMP ATOMIC

CALL OMP_SET_NUM_THREADS(10)

OpenMP: An API for Writing Parallel Applications

- A set of compiler directives and library routines for parallel application programmers
- Originally ... Greatly simplifies writing multithreaded programs in Fortran, C and C++
- Later versions ... supports non-uniform memories, vectorization and GPU programming

#pragma omp parallel for private(A, B)

C\$OMP PARALLEL REDUCTION (+: A, B)

C\$OMP PARALLEL COPYIN (/blk/)

C\$OMP DO lastprivate(XX)

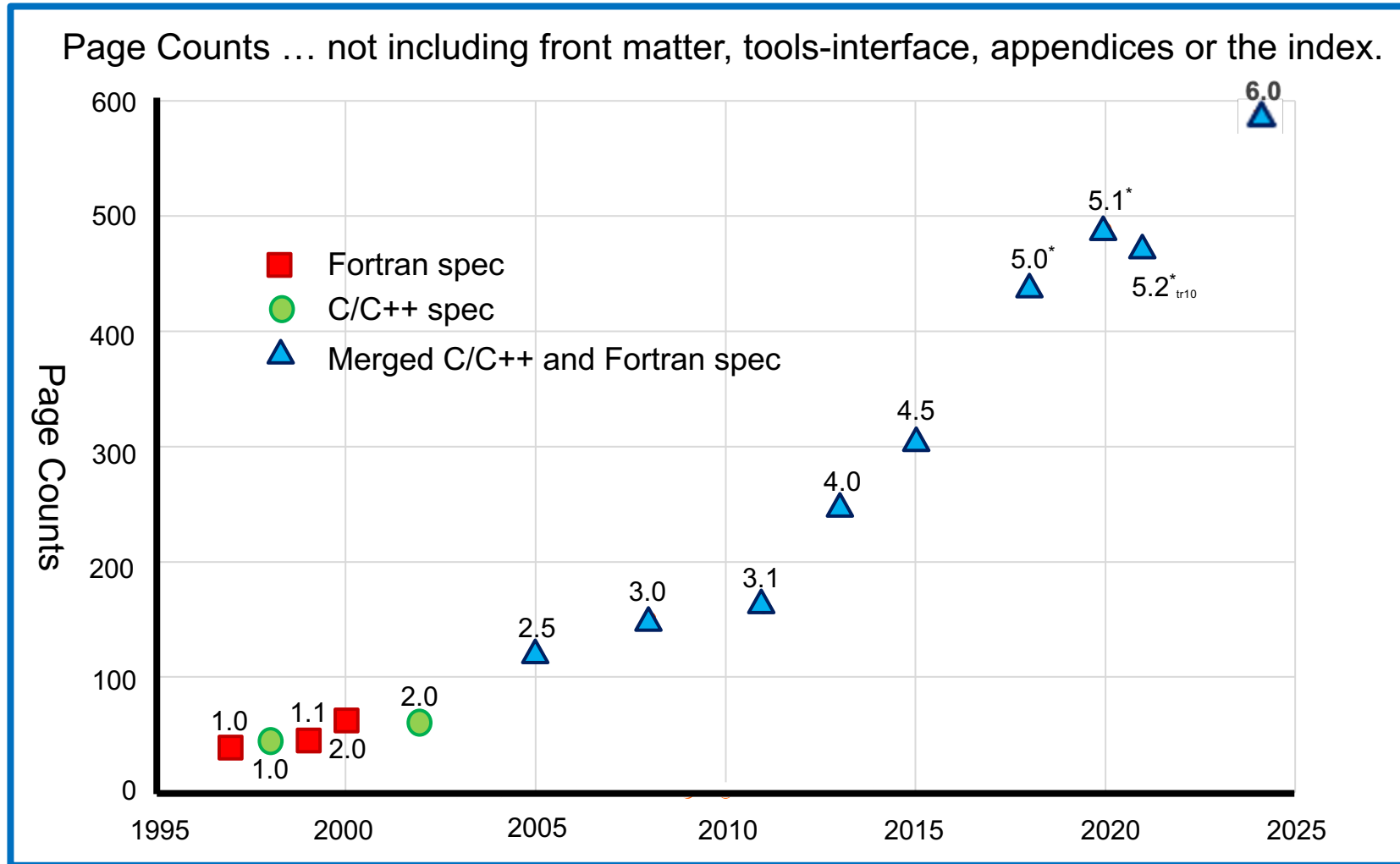
#pragma omp atomic seq_cst

Nthrds = OMP_GET_NUM_PROCS()

omp_set_lock(lck)

The Growth of Complexity in OpenMP

Our goal in 1997 ... A simple interface for application programmers

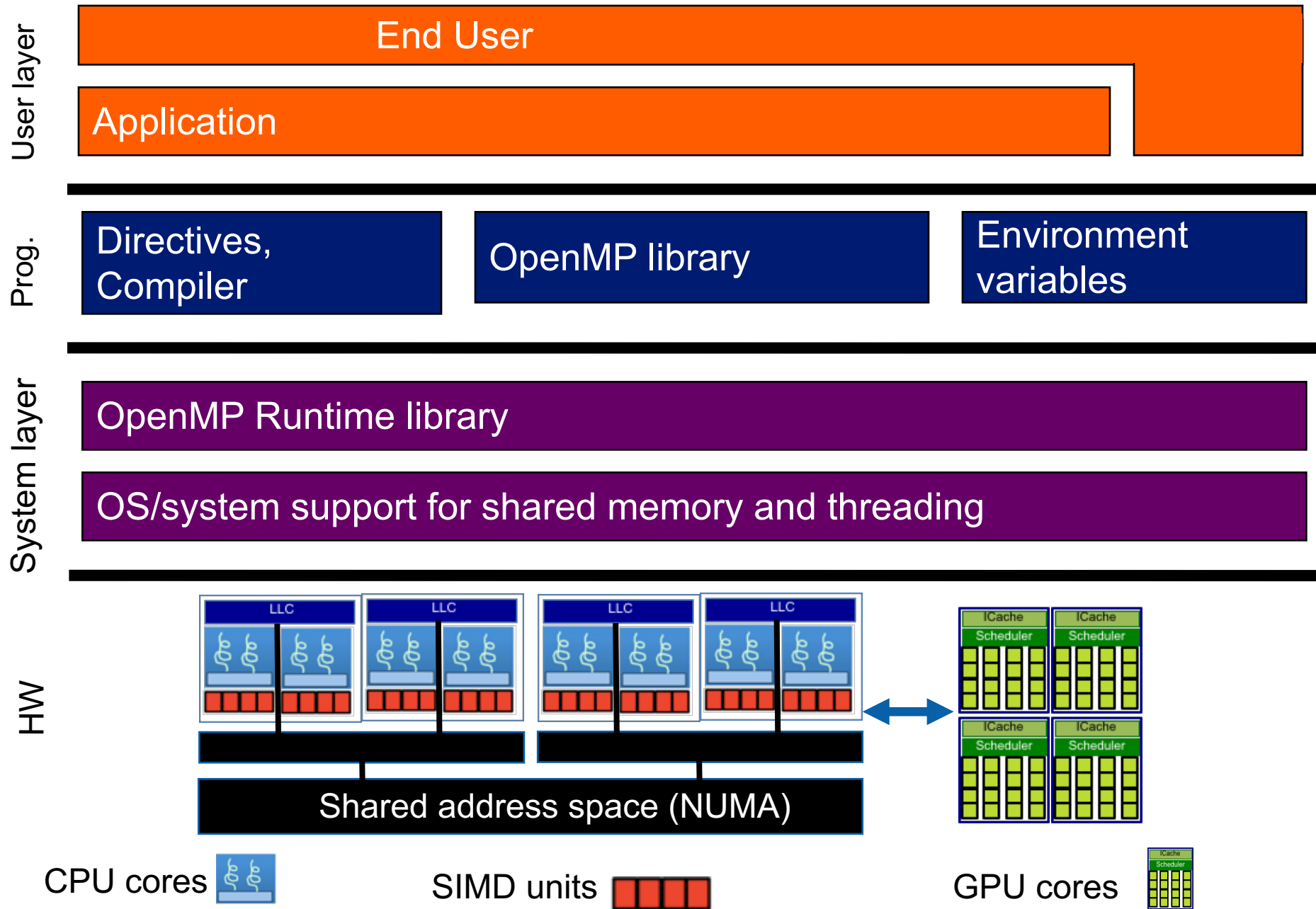


The OpenMP specification is so long and complex that few (if any) humans understand the full document

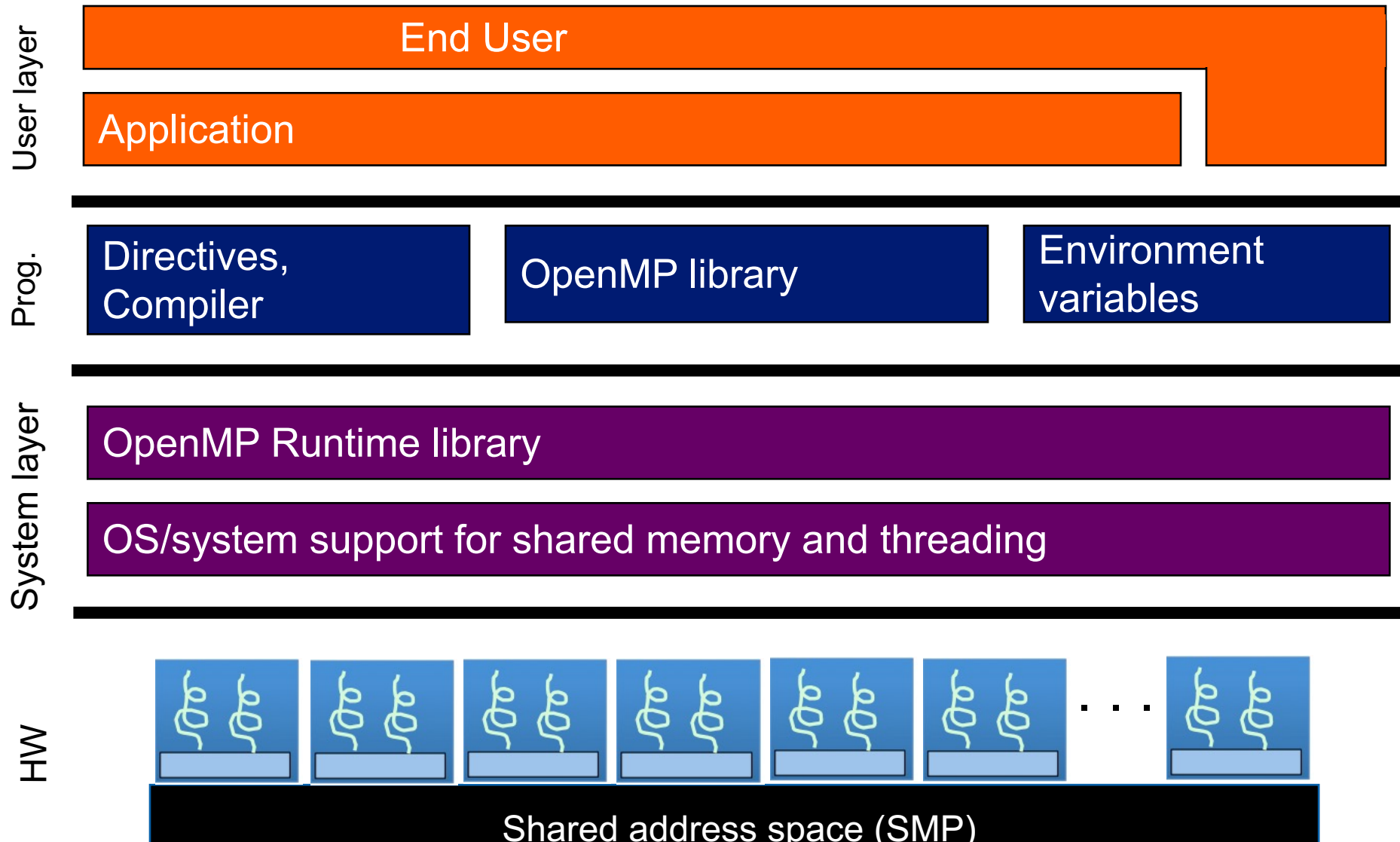
The OpenMP Common Core: Most OpenMP programs only use these 21 items

OpenMP pragma, function, or clause	Concepts
#pragma omp parallel	Parallel region, teams of threads, structured block, interleaved execution across threads.
void omp_set_thread_num() int omp_get_thread_num() int omp_get_num_threads()	Default number of threads and internal control variables. SPMD pattern: Create threads with a parallel region and split up the work using the number of threads and the thread ID.
double omp_get_wtime()	Speedup and Amdahl's law. False sharing and other performance issues.
setenv OMP_NUM_THREADS N	Setting the internal control variable for the default number of threads with an environment variable
#pragma omp barrier #pragma omp critical	Synchronization and race conditions. Revisit interleaved execution.
#pragma omp for #pragma omp parallel for	Worksharing, parallel loops, loop carried dependencies.
reduction(op:list)	Reductions of values across a team of threads.
schedule (static [,chunk]) schedule(dynamic [,chunk])	Loop schedules, loop overheads, and load balance.
shared(list), private(list), firstprivate(list)	Data environment.
default(none)	Force explicit definition of each variable's storage attribute
nowait	Disabling implied barriers on workshare constructs, the high cost of barriers, and the flush concept (but not the flush directive).
#pragma omp single	Workshare with a single thread.
#pragma omp task #pragma omp taskwait	Tasks including the data environment for tasks.

OpenMP Basic Definitions: Basic Solution Stack



OpenMP Basic Definitions: Basic Solution Stack



For the OpenMP Common Core, we focus on Symmetric Multiprocessor Case
i.e., lots of threads with “equal cost access” to memory

OpenMP Basic Syntax

- Most of OpenMP happens through compiler directives.

C and C++	Fortran
Compiler directives	
<i>#pragma omp construct [clause [clause]...]</i>	<i>!\$OMP construct [clause [clause] ...]</i>
Example	
<i>#pragma omp parallel private(x)</i> <i>{</i> <i>}</i>	<i>!\$OMP PARALLEL PRIVATE(X)</i> <i>!\$OMP END PARALLEL</i>
Function prototypes and types:	
<i>#include <omp.h></i>	<i>use OMP_LIB</i>

- Most OpenMP constructs apply to a “structured block”.
 - **Structured block**: a block of one or more statements with one point of entry at the top and one point of exit at the bottom.
 - It’s OK to have an exit() within the structured block.

Exercise, Part A: Hello World

Verify that your environment works

- Write a program that prints “hello world”. Put it in a file called hello.c

```
#include<stdio.h>
int main()
{

    printf(" hello ");
    printf(" world \n");

}
```

```
% vi hello.c
% gcc hello.c
% ./a.out
hello
world
```

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Exercise, Part B: Hello World

Verify that your OpenMP environment works

- Write a multithreaded program that prints “hello world”.

```
#include <omp.h>
#include <stdio.h>
int main()
{
    #pragma omp parallel
    {

        printf(" hello ");
        printf(" world \n");

    }
}
```

Switches for compiling and linking

gcc -fopenmp	Gnu (Linux, OSX)
cc -qopenmp	Intel (Linux@NERSC)
icc -fopenmp	Intel (Linux, OSX)

Solution

A Multi-Threaded “Hello World” Program

- Write a multithreaded program where each thread prints “hello world”.

```
#include <omp.h>
#include <stdio.h>
int main()
{
    #pragma omp parallel
    {
        printf(" hello ");
        printf(" world \n");
    }
}
```

OpenMP include file

Parallel region with default number of threads

End of the Parallel region

Sample Output:

```
hello hello world
world
hello  hello world
world
```

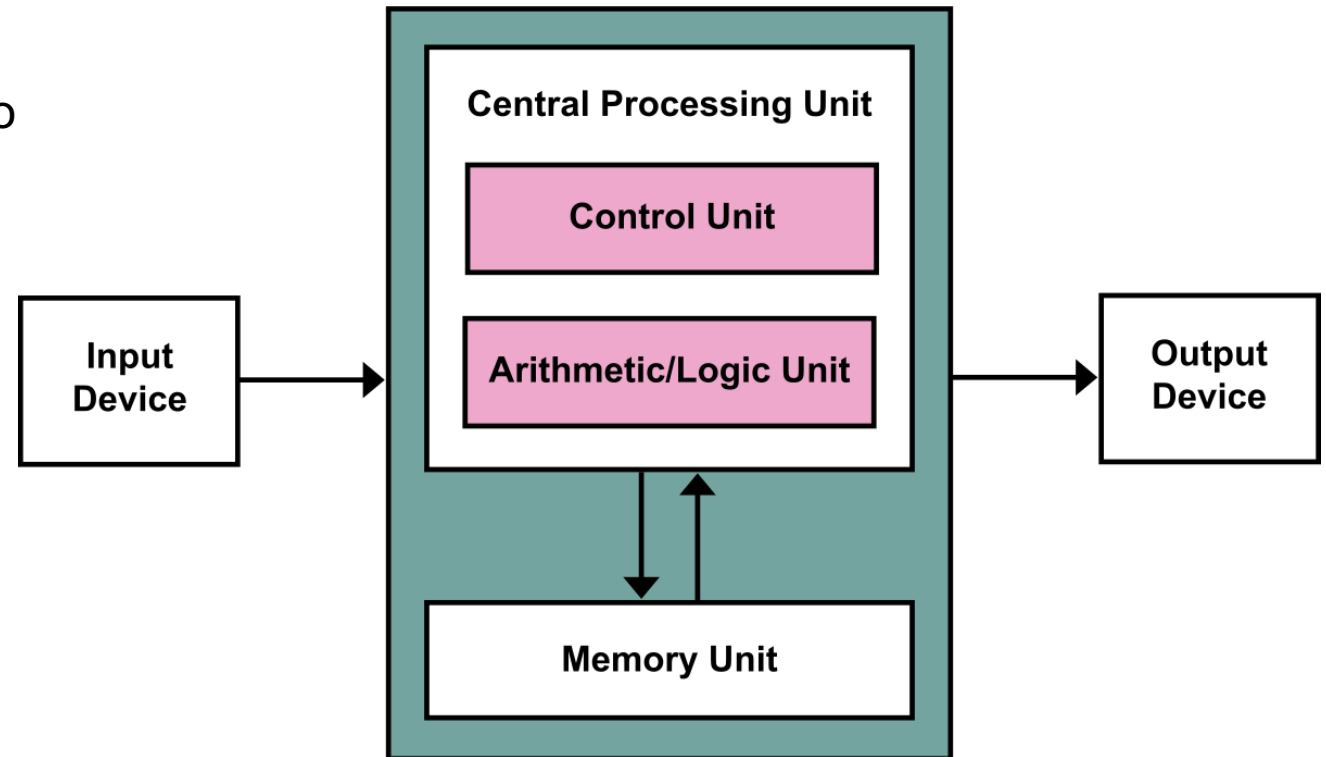
The statements are interleaved based on how the operating schedules the threads

A brief digression on the terminology of parallel computing

Let's agree on a few definitions:

- **Computer:**

- A machine that transforms *input values* into *output values*.
- Typically, a computer consists of Control, Arithmetic/Logic, and Memory units.
- The transformation is defined by a stored **program** (von Neumann architecture).



- **Task:**

- A sequence of instructions plus a data environment. A program is composed of one or more tasks.

- **Active task:**

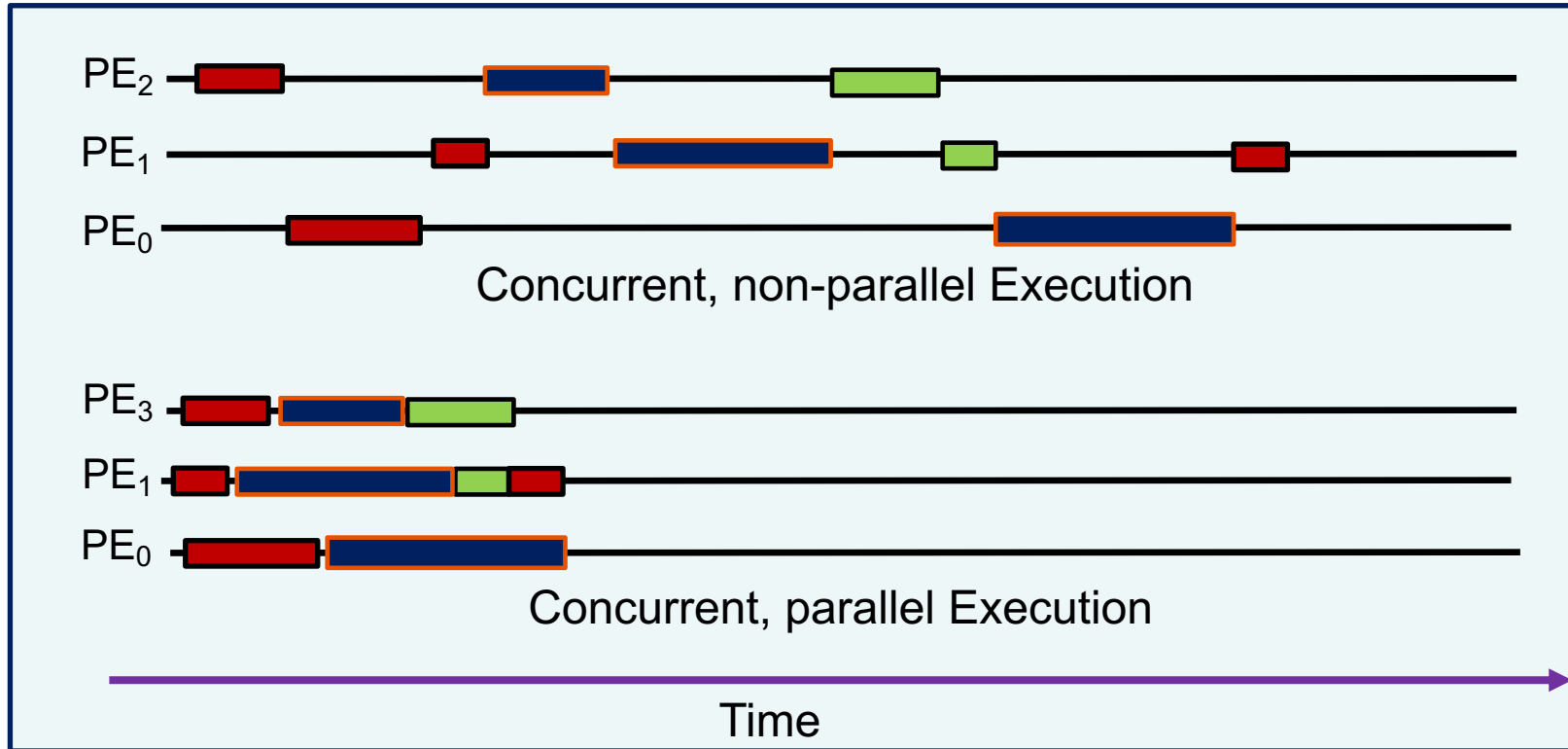
- A task that is available to be scheduled for execution. When the task is moving through its sequence of instructions, we say it is making **forward progress**

- **Fair scheduling:**

- When a scheduler gives each active task an equal *opportunity* for execution.

Concurrency vs. Parallelism

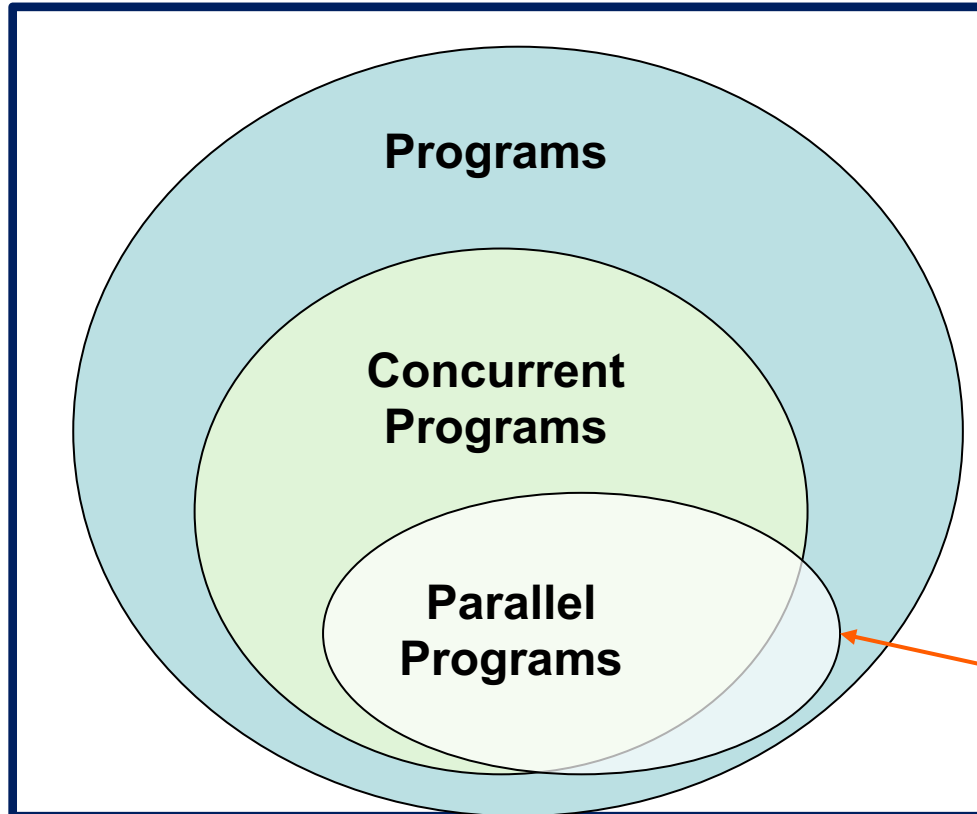
- Two important definitions:
 - Concurrency: A condition of a system in which multiple tasks are active and unordered. If **scheduled fairly**, they can be described as logically making **forward progress** at the same time.
 - Parallelism: A condition of a system in which multiple tasks are actually making **forward progress** at the same time.



PE = Processing Element

Concurrency vs. Parallelism

- Two important definitions:
 - Concurrency: A condition of a system in which multiple tasks are active and unordered. If **scheduled fairly**, they can be described as **logically** making **forward progress** at the same time.
 - Parallelism: A condition of a system in which multiple tasks are **actually** making **forward progress** at the same time.



In most cases, parallel programs exploit concurrency in a problem to run tasks on multiple processing elements

We use Parallelism to:

- Do more work in less time
- Work with larger problems

If tasks execute in “lock step” they are not concurrent, but they are still parallel.
Example ... a SIMD unit.

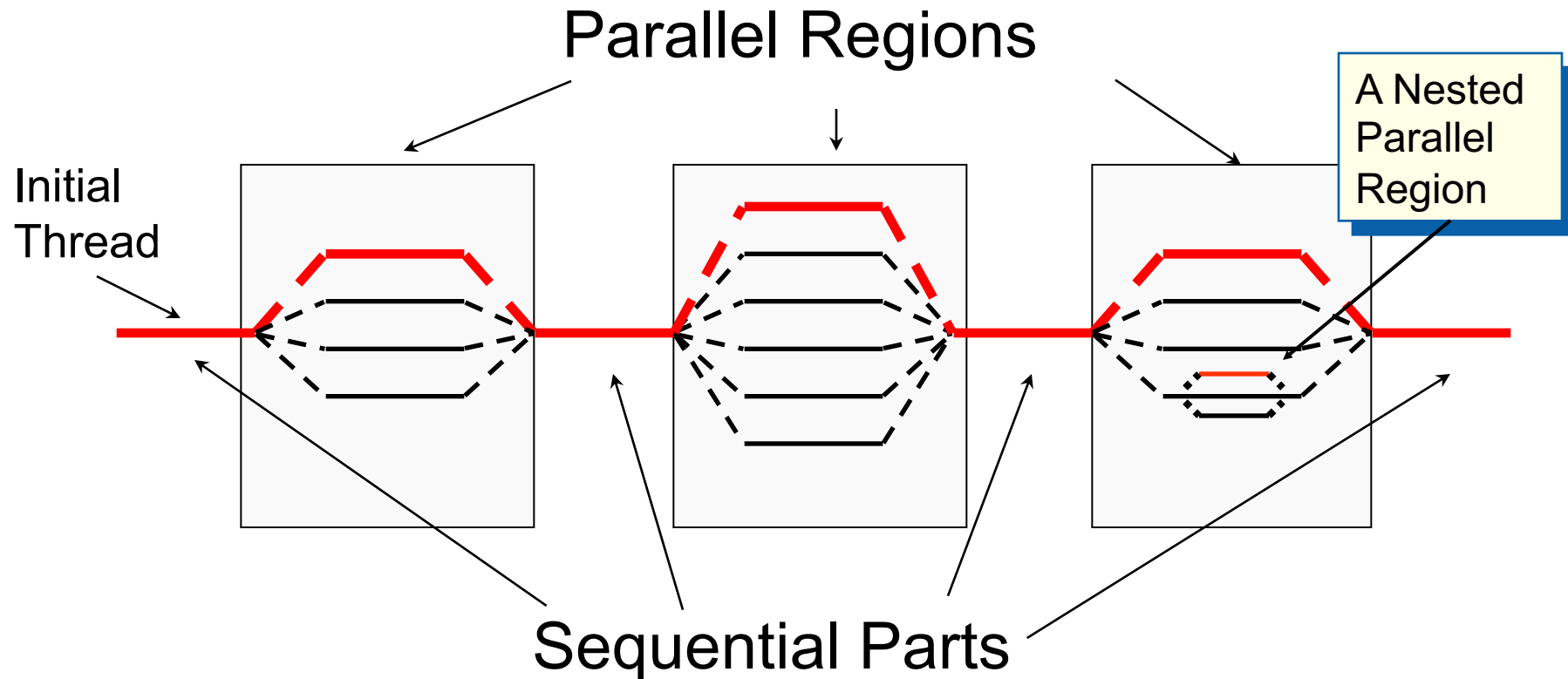
- Introduction to OpenMP

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OpenMP Execution model:

Fork-Join Parallelism:

- ◆ Initial thread spawns a team of threads as needed.
- ◆ Parallelism added incrementally until performance goals are met, i.e., the sequential program evolves into a parallel program.



Thread Creation: Parallel Regions

- You create threads in OpenMP with the parallel construct.
- For example, to create a 4 thread Parallel region:

Each thread executes a copy of the code within the structured block

```
double A[1000];  
omp_set_num_threads(4);  
#pragma omp parallel  
{  
    int ID = omp_get_thread_num();  
    pooh(ID,A);  
}
```

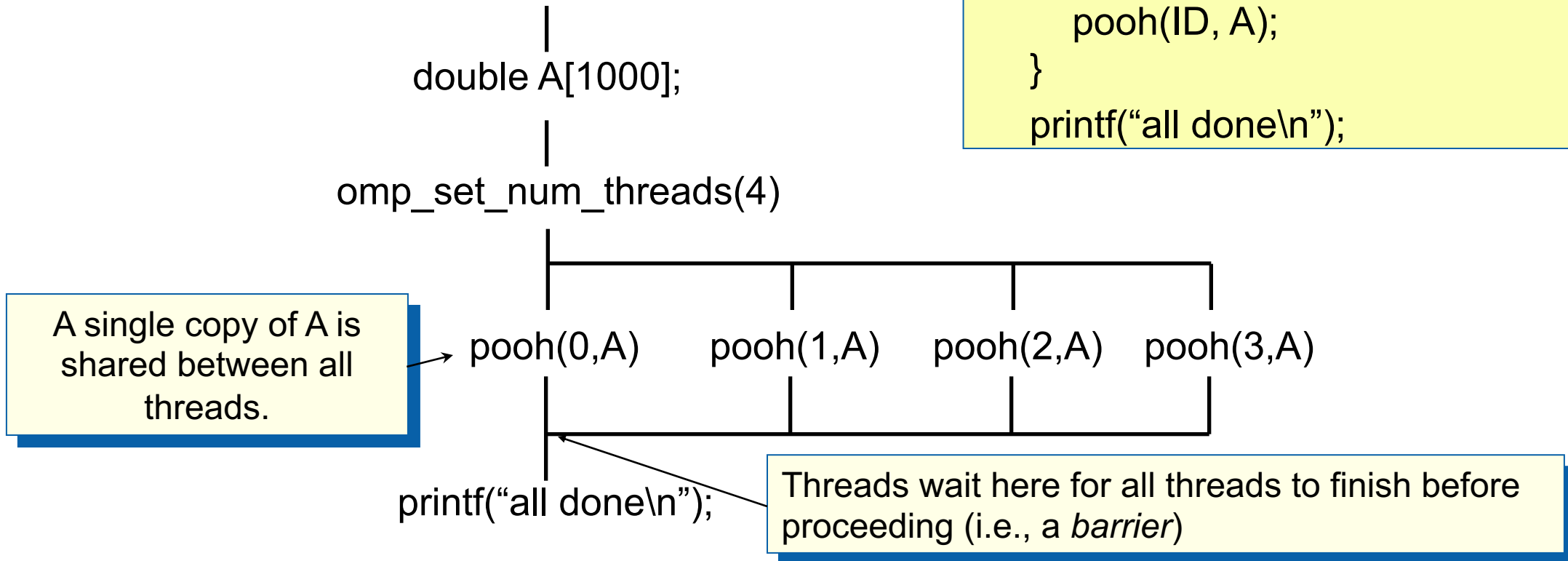
Runtime function to request a certain number of threads

Runtime function returning a thread ID

- Each thread calls pooh(ID,A) for ID = 0 to 3

Thread Creation: Parallel Regions Example

- Each thread executes the same code redundantly.



```
double A[1000];
omp_set_num_threads(4);
#pragma omp parallel
{
    int ID = omp_get_thread_num();
    pooh(ID, A);
}
printf("all done\n");
```

Thread creation: How many threads did you actually get?

- Request a number of threads with `omp_set_num_threads()`
- The number requested may not be the number you actually get.
 - An implementation may silently give you fewer threads than you requested.
 - Once a team of threads has launched, it will not be reduced.

Each thread executes a copy of the code within the structured block

```
double A[1000];  
omp_set_num_threads(4);  
#pragma omp parallel  
{  
    int ID = omp_get_thread_num();  
    int nthrds = omp_get_num_threads();  
    pooh(ID,A);  
}
```

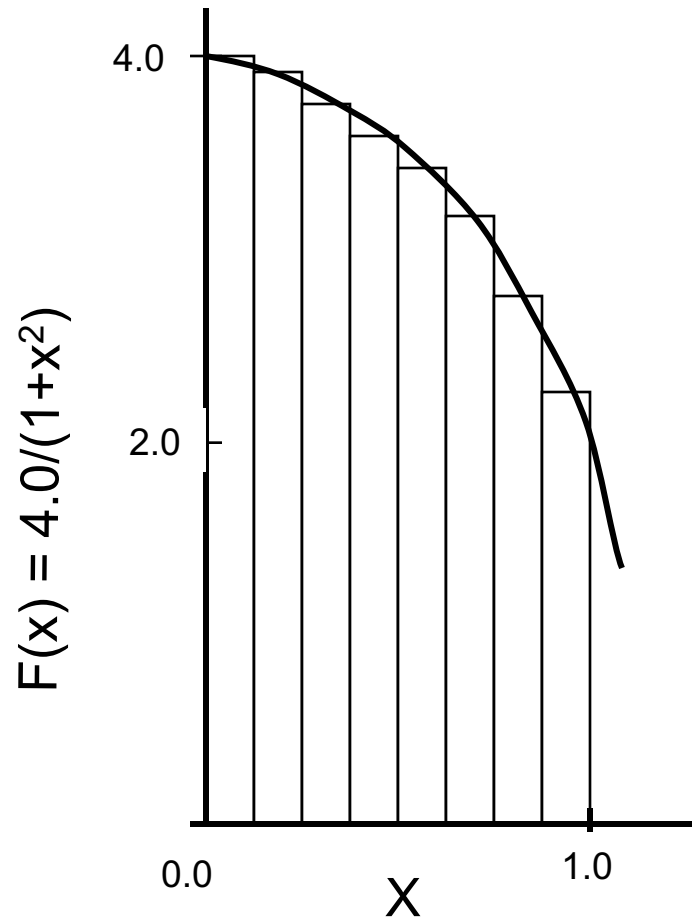
Runtime function to request a certain number of threads

Runtime function to return actual number of threads in the team

- Each thread calls `pooh(ID,A)` for `ID = 0` to `nthrds-1`

An Interesting Problem to Play With

Numerical Integration



Mathematically, we know that:

$$\int_0^1 \frac{4.0}{(1+x^2)} dx = \pi$$

We can approximate the integral as a sum of N rectangles:

$$\sum_{i=0}^N F(x_i) \Delta x = \Delta x \sum_{i=0}^N F(x_i) \approx \pi$$

Where each rectangle has width Δx and height $F(x_i)$ at the middle of interval i .

Serial PI Program

```
static long num_steps = 100000;  
double step;  
int main ()  
{  
    double x, pi, sum = 0.0;  
  
    step = 1.0/(double) num_steps;  
  
    for (int i=0;i< num_steps; i++){  
        x = (i+0.5)*step;  
        sum = sum + 4.0/(1.0+x*x);  
    }  
    pi = step * sum;  
}
```

Serial PI Program

```
#include <omp.h>
static long num_steps = 100000;
double step;
int main ()
{
    double x, pi, sum = 0.0;

    step = 1.0/(double) num_steps;
double tdata = omp_get_wtime();
    for (int i=0;i< num_steps; i++){
        x = (i+0.5)*step;
        sum = sum + 4.0/(1.0+x*x);
    }
    pi = step * sum;
tdata = omp_get_wtime() - tdata;
    printf(" pi = %f in %f secs\n",pi, tdata);
}
```

The library routine
get_omp_wtime()
is used to find the
elapsed “wall
time” for blocks of
code

Exercise: the Parallel Pi Program

- Create a parallel version of the pi program using a parallel construct:
 `#pragma omp parallel`
- Pay close attention to shared versus private variables.
- In addition to a parallel construct, you will need the runtime library routines

– `int omp_get_num_threads();`

Number of threads in the team

– `int omp_get_thread_num();`

Thread ID or rank

– `double omp_get_wtime();`

Time in seconds since a fixed point in the past

– `omp_set_num_threads();`

Request a number of threads in the team

Code for this program is in the file pi.c.

Save a copy of the file pi.c so you can go back to the original code for latter exercise.

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Hints: the Parallel Pi Program

- Use a parallel construct:

```
#pragma omp parallel
```

- The challenge is to:
 - divide loop iterations between threads (use the thread ID and the number of threads).
 - Create an accumulator for each thread to hold partial sums that you can later combine to generate the global sum.
- In addition to a parallel construct, you will need the runtime library routines
 - `int omp_set_num_threads();`
 - `int omp_get_num_threads();`
 - `int omp_get_thread_num();`
 - `double omp_get_wtime();`

Example: A simple SPMD* pi program

```
#include <omp.h>
```

```
static long num_steps = 100000;    double step;
```

```
#define NUM_THREADS 2
```

```
void main ()
```

```
{  int i, nthreads; double pi, sum[NUM_THREADS];
```

```
    step = 1.0/(double) num_steps;
```

```
    omp_set_num_threads(NUM_THREADS);
```

```
    #pragma omp parallel
```

```
{
```

```
    int i, id,numthrds;
```

```
    double x;
```

```
    id = omp_get_thread_num();
```

```
    numthrds = omp_get_num_threads();
```

```
    if (id == 0) nthreads = numthrds;
```

```
    for (i=id, sum[id]=0.0;i< num_steps; i=i+numthrds) {
```

```
        x = (i+0.5)*step;
```

```
        sum[id] += 4.0/(1.0+x*x);
```

```
    }
```

```
}
```

```
for(i=0, pi=0.0;i<nthreads;i++) pi += sum[i] * step;
```

```
}
```

Promote scalar to an array dimensioned by number of threads to avoid race condition.

Only one thread should copy the number of threads to the global value to make sure multiple threads writing to the same address don't conflict.

This is a common trick in SPMD programs to create a **cyclic distribution** of loop iterations

*SPMD: Single Program Multiple Data

Example: A simple SPMD pi program ... an alternative solution

```
#include <omp.h>
static long num_steps = 100000;      double step;
#define NUM_THREADS 2
void main ()
{  int i, nthreads; double pi, sum[NUM_THREADS];
  step = 1.0/(double) num_steps;
  omp_set_num_threads(NUM_THREADS);
  #pragma omp parallel
  {
    int i, id, numthrds, istart, iend;
    double x;
    id = omp_get_thread_num();
    numthrds = omp_get_num_threads();
    istart = id*(num_steps/numthrds);    iend=(id+1)*(num_steps/numthrds);
    if(id == (numthrds-1)) iend = num_steps;
    if (id == 0)  nthreads = numthrds;
    for (i=istart, sum[id]=0.0; i< iend; i++) {
      x = (i+0.5)*step;
      sum[id] += 4.0/(1.0+x*x);
    }
  }
  for(i=0, pi=0.0; i<nthreads; i++) pi += sum[i] * step;
}
```

This is a common trick in SPMD algorithms ...
it's a **blocked distribution** with one block per thread.

Results*

- Original Serial pi program with 100000000 steps ran in 1.83 seconds.

```
#include <omp.h>
static long num_steps = 100000;      double step;
#define NUM_THREADS 2
void main ()
{  int i, nthreads; double pi, sum[NUM_THREADS];
  step = 1.0/(double) num_steps;
  omp_set_num_threads(NUM_THREADS);
  #pragma omp parallel
  {
    int i, id, nthrds;
    double x;
    id = omp_get_thread_num();
    nthrds = omp_get_num_threads();
    if (id == 0) nthreads = nthrds;
    for (i=id, sum[id]=0.0; i< num_steps; i=i+nthrds) {
      x = (i+0.5)*step;
      sum[id] += 4.0/(1.0+x*x);
    }
  }
  for(i=0, pi=0.0; i<nthreads; i++) pi += sum[i] * step;
}
```

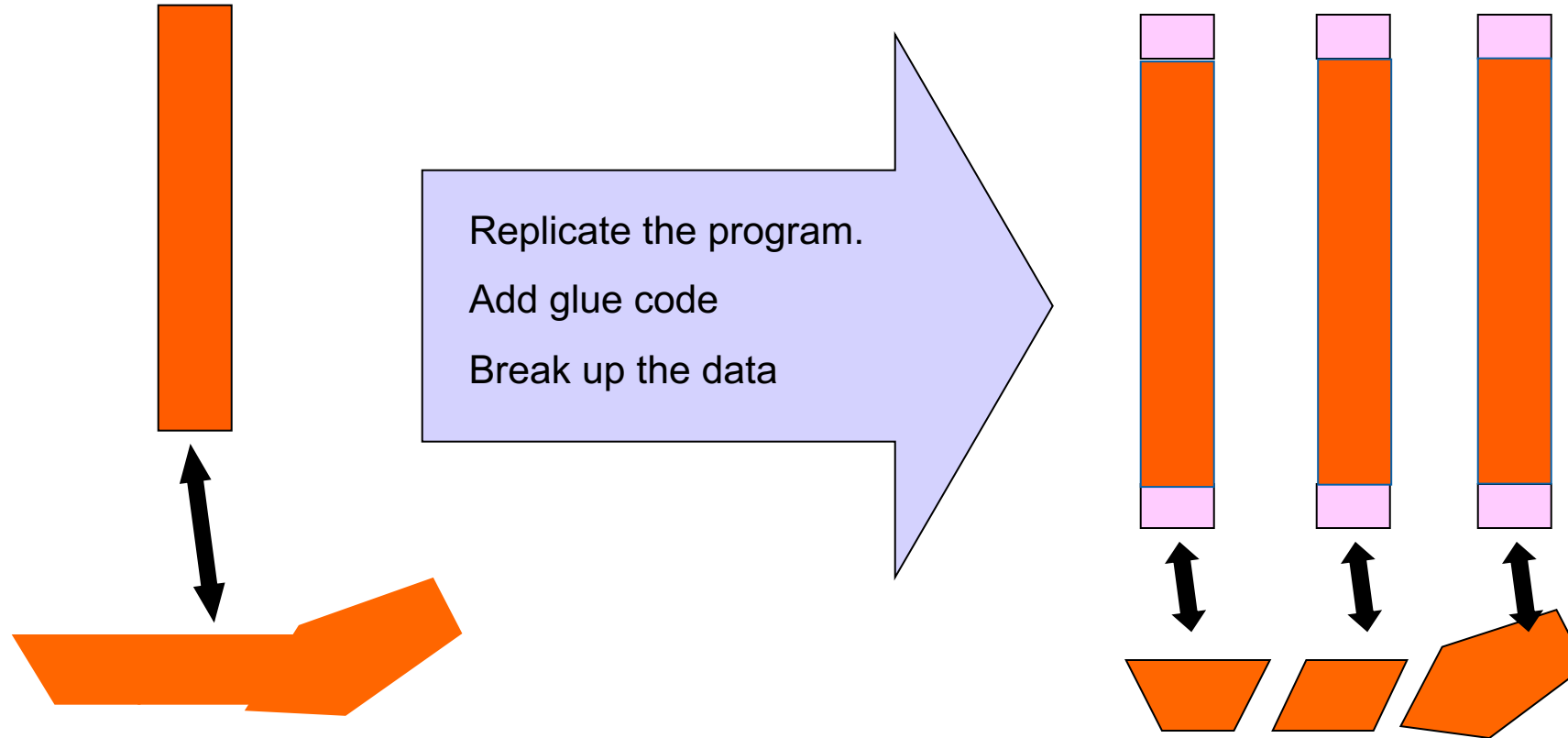
threads	1 st SPMD*
1	1.86
2	1.03
3	1.08
4	0.97

Intel compiler (icpc) with default optimization level (O2) on Apple OS X 10.7.3 with a dual core (four HW thread) Intel® Core™ i5 processor at 1.7 Ghz and 4 Gbyte DDR3 memory at 1.333 Ghz.

*SPMD: Single Program Multiple Data

SPMD: Single Program Multiple Data

- Run the same program on P processing elements where P can be arbitrarily large.



- Use the rank ... an ID ranging from 0 to $(P-1)$... to select between a set of tasks and to manage any shared data structures.

MPI programs almost always use this pattern ... it is probably the most commonly used pattern in the history of parallel programming.

How do we describe performance in parallel programs

Consider performance of parallel programs

Compute N independent tasks on one processor

Load Data

Compute T_1

...

Compute T_N

Consume Results

$$\text{Time}_{\text{seq}}(1) = T_{\text{load}} + N * T_{\text{task}} + T_{\text{consume}}$$

Compute N independent tasks with P processors

Load Data

Compute T_1

...

Compute T_N

Consume Results

$$\text{Time}_{\text{par}}(P) = T_{\text{load}} + (N/P) * T_{\text{task}} + T_{\text{consume}}$$

Ideally Cut
runtime by $\sim 1/P$

*(Note: Parallelism
only speeds-up the
concurrent part)*

Talking about performance

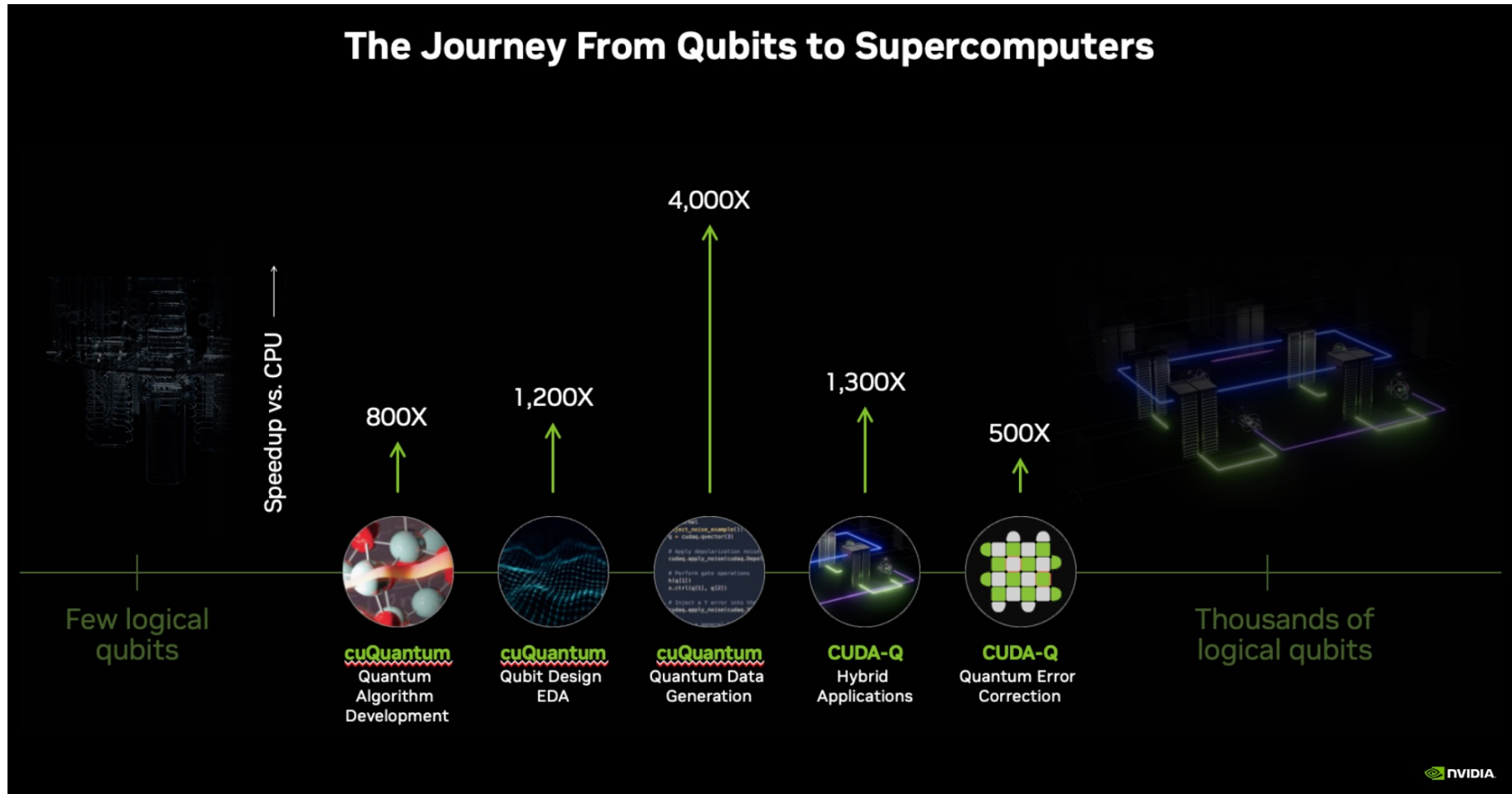
- Speedup: the increased performance from running on P processors.
- Perfect Linear Speedup: happens when no parallel overhead and algorithm is 100% parallel.
- Efficiency: How well does your observed speedup compare to the ideal case?

$$S(P) = \frac{Time_{seq}(1)}{Time_{par}(P)}$$

$$S(P) = P$$

$$\varepsilon(P) = \frac{S(P)}{P}$$

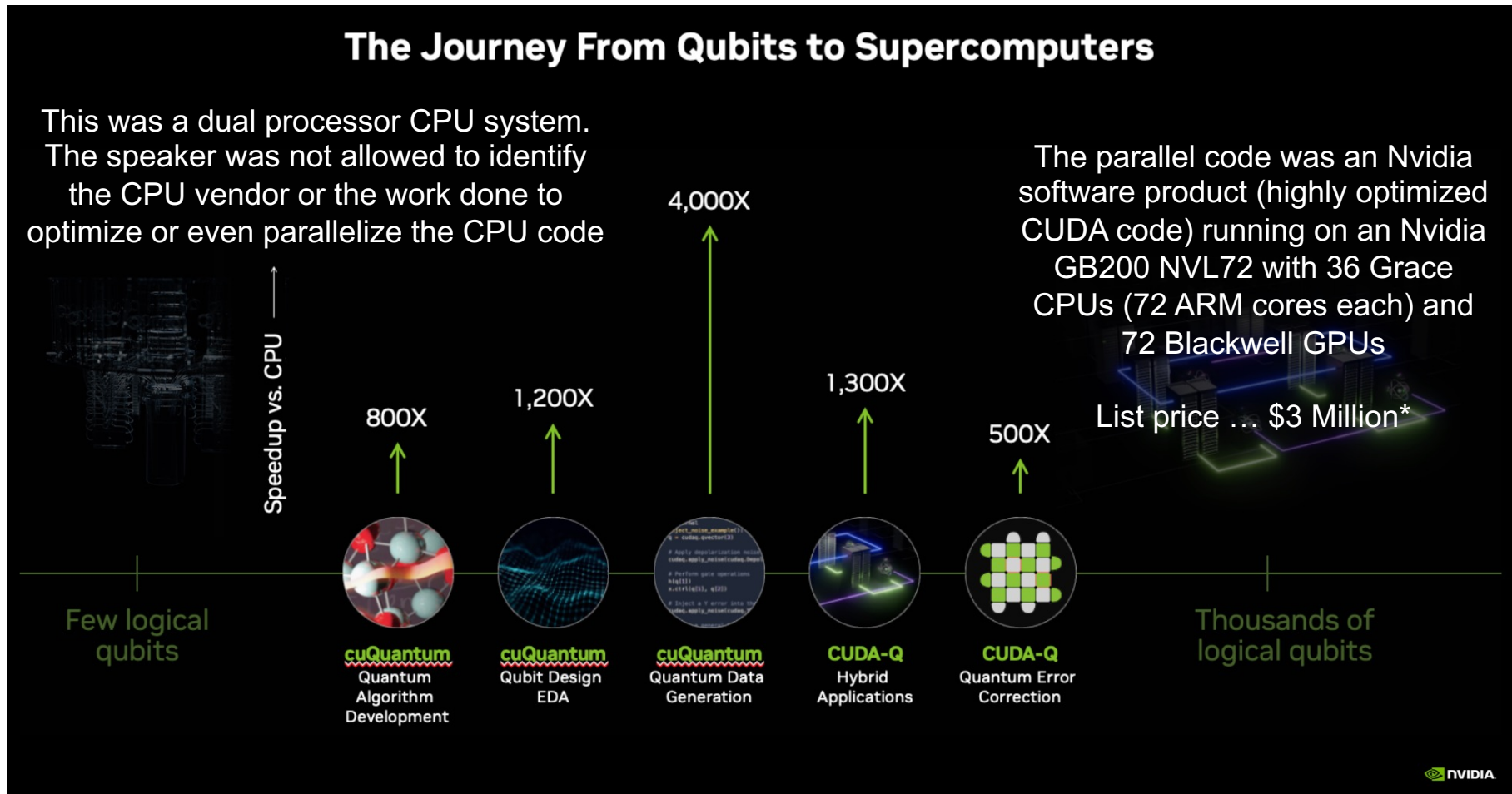
Speedups as disinformation



A slide from an Nvidia talk at ATPESC'25 about their quantum computing product

Reported speedups without defining what they were comparing against ... included a verbal comment that this showed if you really care about performance, you must use a GPU

Speedups as disinformation



These Speedups are blatant disinformation... it's demeaning to show such data.

The sad thing is, the Nvidia product is **excellent**. In a fair comparison, they'd still come out on top.
Why they need to resort to such misleading statements is baffling.

*<https://www.datacenterdynamics.com/en/news/lambda-partners-with-pegatron-to-deploy-nvidia-gb200-nvl72-rack/>

Amdahl's Law

- What is the maximum speedup you can expect from a parallel program?
- Approximate the runtime as a part that can be sped up with additional processors and a part that is fundamentally serial.

$$Time_{par}(P) = (serial_fraction + \frac{parallel_fraction}{P}) * Time_{seq}$$

- If the serial fraction is α and the parallel fraction is $(1 - \alpha)$ then the speedup is:

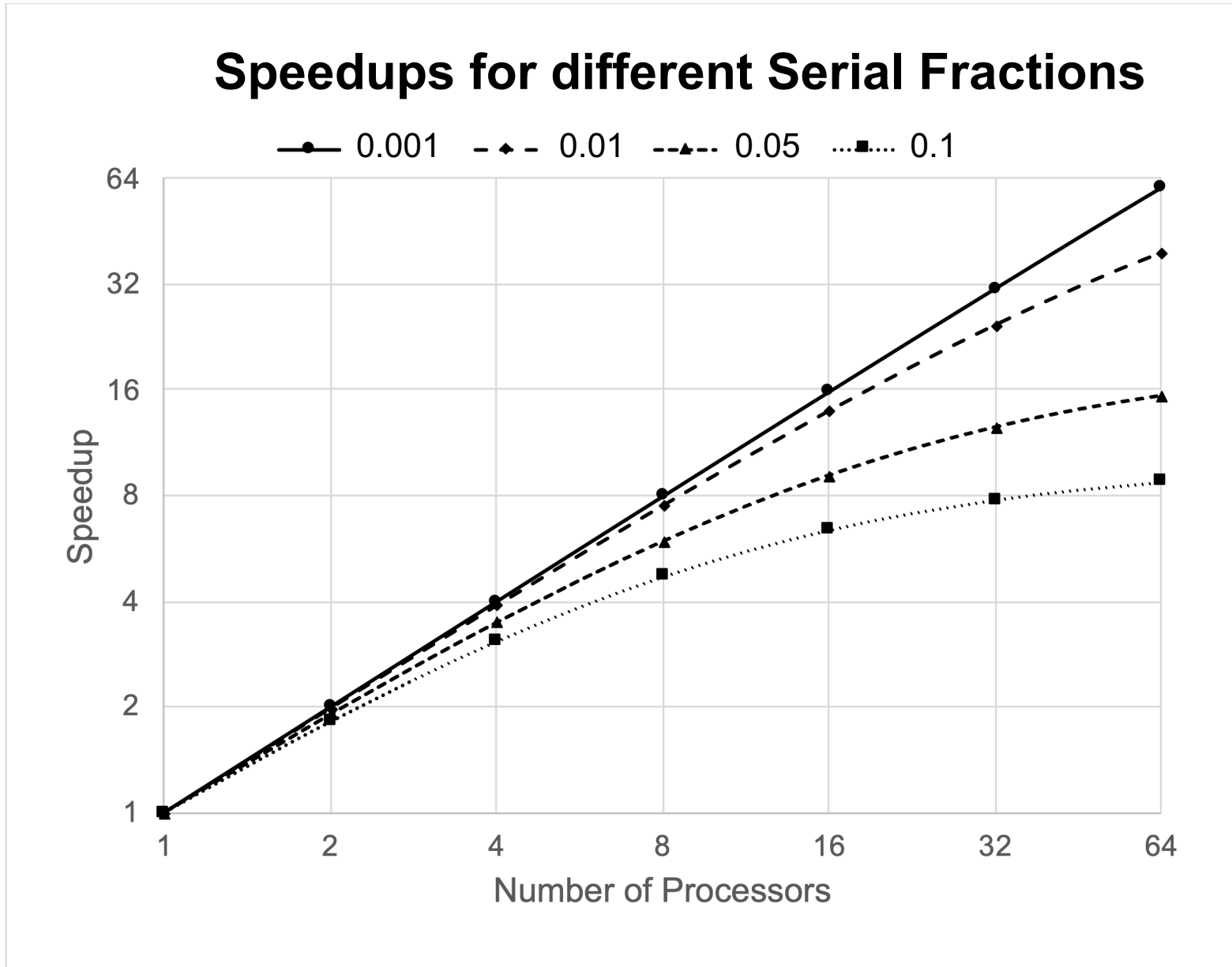
$$S(P) = \frac{Time_{seq}}{Time_{par}(P)} = \frac{Time_{seq}}{(\alpha + \frac{1 - \alpha}{P}) * Time_{seq}} = \frac{1}{\alpha + \frac{1 - \alpha}{P}}$$

- If you had an unlimited number of processors: $P \rightarrow \infty$

- The maximum possible speedup is: $S = \frac{1}{\alpha}$ ←

Amdahl's Law

Amdahl's Law ... It's not just about the maximum speedup



$$S(P, \alpha) = \frac{1}{\alpha - \frac{1 - \alpha}{P}}$$

So now you should understand my silly introduction slide.

Introduction

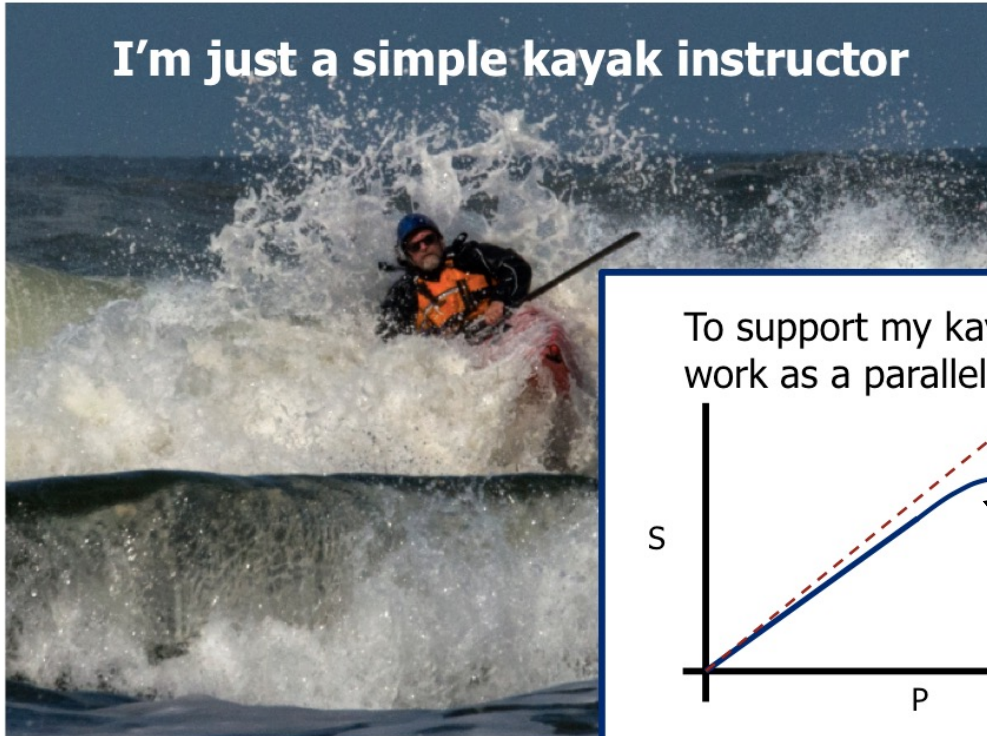
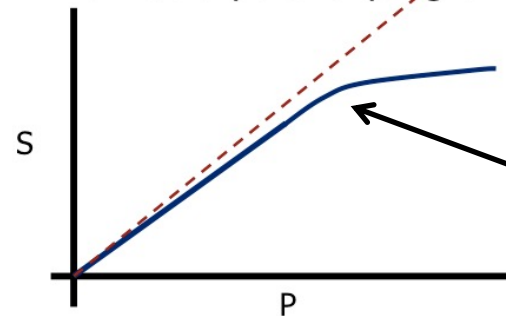


Photo © by Greg Clopton, 2014

To support my kayaking habit I work as a parallel programmer



Which means I know how to turn math into lines on a speedup plot

We measure our success as parallel programmers by how close we come to ideal linear speedup.

A good parallel programmer always figures out when you fall off the linear speedup curve and why that has occurred.

**.... Now that we know how to describe
performance for parallel computations,
lets get back to OpenMP**

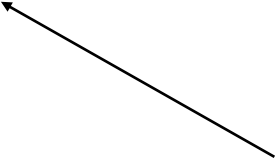
Internal control variables and how to control the number of threads in a team

- We've used the following construct to control the number of threads. (e.g. to request 12 threads):
 - `omp_set_num_threads(12)`
- What does `omp_set_num_threads()` actually do?
 - It **resets** an “**internal control variable**” the system queries to select the default number of threads to request on subsequent parallel constructs.
- Is there an easier way to change this internal control variable ... perhaps one that doesn't require re-compilation? Yes.
 - When an OpenMP program starts up, it queries an environment variable `OMP_NUM_THREADS` and sets the appropriate internal control variable to the value of **`OMP_NUM_THREADS`**
 - For example, to set the initial, default number of threads to request in OpenMP from my apple laptop
 - > **`export OMP_NUM_THREADS=12`**

Exercise

- Go back to your parallel pi program and explore how well it scales with the number of threads.
- Can you explain your performance with Amdahl's law? If not what else might be going on?

- `int omp_get_num_threads();`
- `int omp_get_thread_num();`
- `double omp_get_wtime();`
- `omp_set_num_threads();`
- `export OMP_NUM_THREADS = N`



An environment variable
to set the default number
of threads to request to N

Results*

- Original Serial pi program with 100000000 steps ran in 1.83 seconds.

```
#include <omp.h>
static long num_steps = 100000;      double step;
#define NUM_THREADS 2
void main ()
{  int i, nthreads; double pi, sum[NUM_THREADS];
  step = 1.0/(double) num_steps;
  omp_set_num_threads(NUM_THREADS);
  #pragma omp parallel
  {
    int i, id, nthrds;
    double x;
    id = omp_get_thread_num();
    nthrds = omp_get_num_threads();
    if (id == 0) nthreads = nthrds;
    for (i=id, sum[id]=0.0; i< num_steps; i=i+nthrds) {
      x = (i+0.5)*step;
      sum[id] += 4.0/(1.0+x*x);
    }
  }
  for(i=0, pi=0.0; i<nthreads; i++) pi += sum[i] * step;
}
```

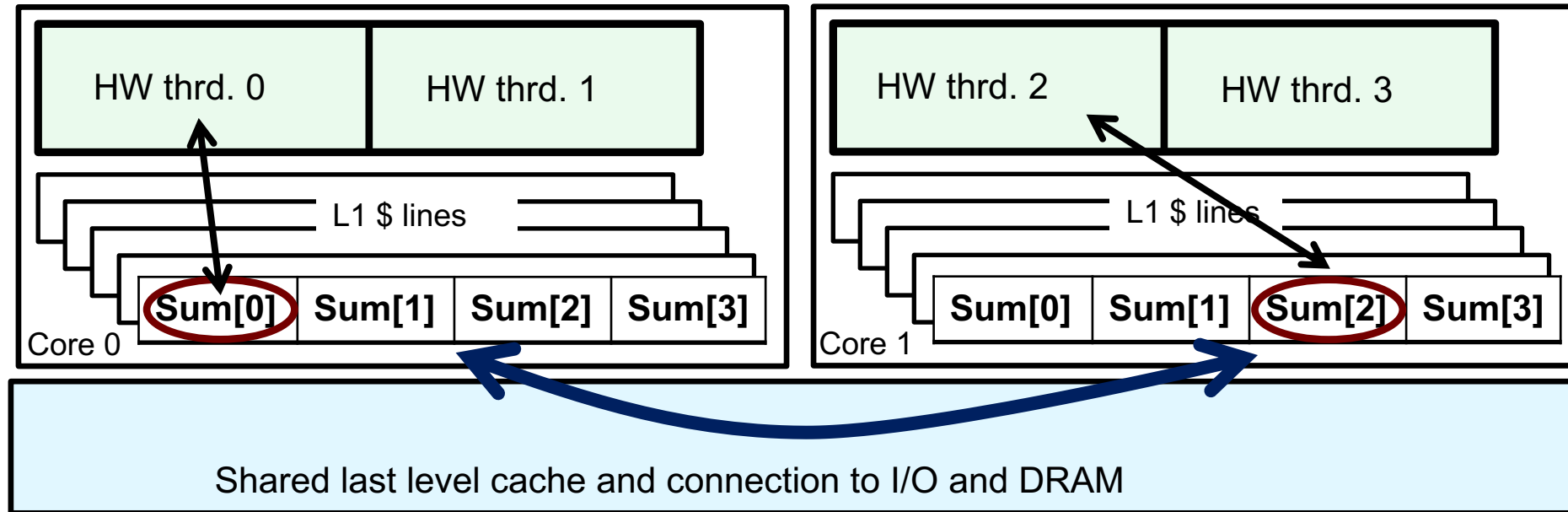
threads	1 st SPMD*
1	1.86
2	1.03
3	1.08
4	0.97

Intel compiler (icpc) with default optimization level (O2) on Apple OS X 10.7.3 with a dual core (four HW thread) Intel® Core™ i5 processor at 1.7 Ghz and 4 Gbyte DDR3 memory at 1.333 Ghz.

*SPMD: Single Program Multiple Data

Why Such Poor Scaling? False Sharing

- If independent data elements happen to sit on the same cache line, each update will cause the cache lines to “slosh back and forth” between threads ... This is called **“false sharing”**.



- If you promote scalars to an array to support creation of an SPMD program, the array elements are contiguous in memory and hence share cache lines ... Results in poor scalability.
- Solution: Pad arrays so elements you use are on distinct cache lines.

Example: Eliminate false sharing by padding the sum array

```
#include <omp.h>
static long num_steps = 100000;      double step;
#define NUM_THREADS 2
#define PAD 8      // assume 64 byte L1 cache line size
void main ()
{  int i, nthreads; double pi, sum[NUM_THREADS][PAD] ;
  step = 1.0/(double) num_steps;
  omp_set_num_threads(NUM_THREADS);
  #pragma omp parallel
  {
    int i, id, nthrds;
    double x;
    id = omp_get_thread_num();
    nthrds = omp_get_num_threads();
    if (id == 0) nthreads = nthrds;
    for (i=id, sum[id]=0.0; i< num_steps; i=i+nthrds) {
      x = (i+0.5)*step;
      sum[id][0] += 4.0/(1.0+x*x);
    }
  }
  for(i=0, pi=0.0; i<nthreads; i++) pi += sum[i][0] * step;
}
```

Pad the array so each
sum value is in a
different cache line



Results*: PI Program, Padded Accumulator

- Original Serial pi program with 100000000 steps ran in 1.83 seconds.

```
#include <omp.h>
static long num_steps = 100000;      double step;
#define NUM_THREADS 2
#define PAD 8      // assume 64 byte L1 cache line size
void main ()
{  int i, nthreads; double pi, sum[NUM_THREADS][PAD] ;
  step = 1.0/(double) num_steps;
  omp_set_num_threads(NUM_THREADS);
  #pragma omp parallel
  {
    int i, id,nthrds;
    double x;
    id = omp_get_thread_num();
    nthrds = omp_get_num_threads();
    if (id == 0)  nthreads = nthrds;
    for (i=id, sum[id]=0.0;i< num_steps; i=i+nthrds) {
      x = (i+0.5)*step;
      sum[id][0] += 4.0/(1.0+x*x);
    }
  }
  for(i=0, pi=0.0;i<nthreads;i++)pi += sum[i][0] * step;
}
```

threads	1 st SPMD	1 st SPMD padded
1	1.86	1.86
2	1.03	1.01
3	1.08	0.69
4	0.97	0.53

*Intel compiler (icpc) with default optimization level (O2) on Apple OS X 10.7.3 with a dual core (four HW thread) Intel® Core™ i5 processor at 1.7 Ghz and 4 Gbyte DDR3 memory at 1.333 Ghz.

- Introduction to OpenMP
- Creating Threads
- ➔ • Synchronization ... simple collective sync
- Parallel Loops
- Data Environment
- Synchronization ... the general case (including Memory Models)
- Irregular Parallelism and Tasks
- OpenMP for real hardware (NUMA)
- Recap

Synchronization

Synchronization is used to impose order constraints and to protect access to shared data

- High level synchronization included in the common core:
 - critical
 - barrier
- Other, more advanced, synchronization operations:
 - atomic
 - flush
 - locks (both simple and nested)

Synchronization: critical

- Mutual exclusion: Only one thread at a time can enter a **critical** region.

Threads wait their turn
– only one thread at a
time calls consume()

```
float res;  
  
#pragma omp parallel  
{  float B;  int i, id, nthrds;  
    id = omp_get_thread_num();  
    nthrds = omp_get_num_threads();  
    B = big_SPMD_job(id, nthrds);  
    #pragma omp critical  
        res += consume (B);  
}
```

Synchronization: barrier

- Barrier: a point in a program all threads must reach before any threads are allowed to proceed.
- It is a “stand alone” pragma meaning it is not associated with user code ... it is an executable statement.

```
double Arr[8], Brr[8];      int numthrds;

omp_set_num_threads(8)

#pragma omp parallel
{  int id, nthrds;

    id = omp_get_thread_num();
    nthrds = omp_get_num_threads();
    if (id==0) numthrds = nthrds;

    Arr[id] = big_ugly_calc(id, nthrds);

    #pragma omp barrier
    Brr[id] = really_big_and_ugly(id, nthrds, Arr);
}
```

Threads wait until all
threads hit the barrier.
Then they can go on.



Exercise

- In your first Pi program, you probably used an array to create space for each thread to store its partial sum.
- If array elements happen to share a cache line, this leads to false sharing.
 - Non-shared data in the same cache line so each update invalidates the cache line ... in essence “sloshing independent data” back and forth between threads.
- Modify your “pi program” to avoid false sharing due to the partial sum array.

```
int omp_get_num_threads();
int omp_get_thread_num();
double omp_get_wtime();
omp_set_num_threads();
#pragma omp parallel
#pragma omp critical
```

PI Program with False Sharing

- Original Serial pi program with 100000000 steps ran in 1.83 seconds.

```
#include <omp.h>
static long num_steps = 100000;      double step;
#define NUM_THREADS 2
void main ()
{  int i, nthreads; double pi, sum[NUM_THREADS];
  step = 1.0/(double) num_steps;
  omp_set_num_threads(NUM_THREADS);
  #pragma omp parallel
  {
    int i, id, nthrds;
    double x;
    id = omp_get_thread_num();
    nthrds = omp_get_num_threads();
    if (id == 0) nthreads = nthrds;
    for (i=id, sum[id]=0.0; i< num_steps; i=i+nthrds) {
      x = (i+0.5)*step;
      sum[id] += 4.0/(1.0+x*x);
    }
  }
  for(i=0, pi=0.0; i<nthreads; i++) pi += sum[i] * step;
}
```

Recall that promoting sum to an array made the coding easy, but led to false sharing and poor performance.

threads	1 st SPMD
1	1.86
2	1.03
3	1.08
4	0.97

*Intel compiler (icpc) with no optimization on Apple OS X 10.7.3 with a dual core (four HW thread) Intel® Core™ i5 processor at 1.7 Ghz and 4 Gbyte DDR3 memory at 1.333 Ghz.

Example: Using a critical section to remove impact of false sharing

```
#include <omp.h>
static long num_steps = 100000;      double step;
#define NUM_THREADS 2
void main ()
{ int nthreads; double pi=0.0;      step = 1.0/(double) num_steps;
  omp_set_num_threads(NUM_THREADS);
  #pragma omp parallel
  {
    int i, id, nthrds;  double x, sum;
    id = omp_get_thread_num();
    nthrds = omp_get_num_threads();
    if (id == 0) nthreads = nthrds;
    for (i=id, sum=0.0; i< num_steps; i=i+nthrds) {
      x = (i+0.5)*step;
      sum += 4.0/(1.0+x*x);
    }
    #pragma omp critical
    pi += sum * step;
  }
}
```

Create a scalar local to each thread to accumulate partial sums.

No array, so no false sharing.

Sum goes “out of scope” beyond the parallel region ... so you must sum it in here. Must protect summation into pi in a critical region so updates don’t conflict

Results*: pi program critical section

- Original Serial pi program with 100000000 steps ran in 1.83 seconds.

```
#include <omp.h>
static long num_steps = 100000;      double step;
#define NUM_THREADS 2
void main ()
{ int nthreads; double pi=0.0;      step = 1.0/(double) num_steps;
  omp_set_num_threads(NUM_THREADS);
  #pragma omp parallel
  {
    int i, id, nthrds;  double x, sum;
    id = omp_get_thread_num();
    nthrds = omp_get_num_threads();
    if (id == 0) nthreads = nthrds;
    for (i=id, sum=0.0; i< num_steps; i=i+nthrds) {
      x = (i+0.5)*step;
      sum += 4.0/(1.0+x*x);
    }
    #pragma omp critical
      pi += sum * step;
  }
}
```

threads	1 st SPMD	1 st SPMD padded	SPMD critical
1	1.86	1.86	1.87
2	1.03	1.01	1.00
3	1.08	0.69	0.68
4	0.97	0.53	0.53

*Intel compiler (icpc) with no optimization on Apple OS X 10.7.3 with a dual core (four HW thread) Intel® Core™ i5 processor at 1.7 Ghz and 4 Gbyte DDR3 memory at 1.333 Ghz.

Example: Using a critical section to remove impact of false sharing

```
#include <omp.h>
static long num_steps = 100000;      double step;
#define NUM_THREADS 2
void main ()
{ int nthreads; double pi=0.0;      step = 1.0/(double) num_steps;
  omp_set_num_threads(NUM_THREADS);
  #pragma omp parallel
  {
    int i, id, nthrds;  double x, sum;
    id = omp_get_thread_num();
    nthrds = omp_get_num_threads();
    if (id == 0) nthreads = nthrds;
    for (i=id, sum=0.0; i< num_steps; i=i+nthrds) {
      x = (i+0.5)*step;
      #pragma omp critical
      sum += 4.0/(1.0+x*x);
    }
  }
}
```

What would happen if you put the critical section inside the loop?

- Introduction to OpenMP
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- OpenMP for real hardware (NUMA)
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The Loop Worksharing Construct

- The loop worksharing construct splits up loop iterations among the threads in a team

```
#pragma omp parallel
{
  #pragma omp for
  for (I=0;I<N;I++){
    NEAT_STUFF(I);
  }
}
```

Loop construct name:

- C/C++: for
- Fortran: do

The loop control index I is made
“private” to each thread by default.

Threads wait here until all
threads are finished with the
parallel loop before any proceed
past the end of the loop

Loop Worksharing Construct

A motivating example

Sequential code

```
for(i=0;i<N;i++) { a[i] = a[i] + b[i];}
```

OpenMP parallel region
(SPMD Pattern)

```
#pragma omp parallel  
{  
    int id, i, Nthrds, istart, iend;  
    id = omp_get_thread_num();  
    Nthrds = omp_get_num_threads();  
    istart = id * N / Nthrds;  
    iend = (id+1) * (N / Nthrds);  
    if (id == Nthrds-1) iend = N;  
    for(i=istart;i<iend;i++) { a[i] = a[i] + b[i];}  
}
```

OpenMP parallel region and
a worksharing for construct

```
#pragma omp parallel  
#pragma omp for  
for(i=0;i<N;i++) { a[i] = a[i] + b[i];}
```

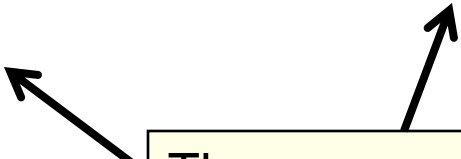
Combined Parallel/Worksharing Construct

- OpenMP shortcut: Put the “parallel” and the worksharing directive on the same line

```
double res[MAX]; int i;  
#pragma omp parallel  
{  
    #pragma omp for  
    for (i=0;i< MAX; i++) {  
        res[i] = huge();  
    }  
}
```

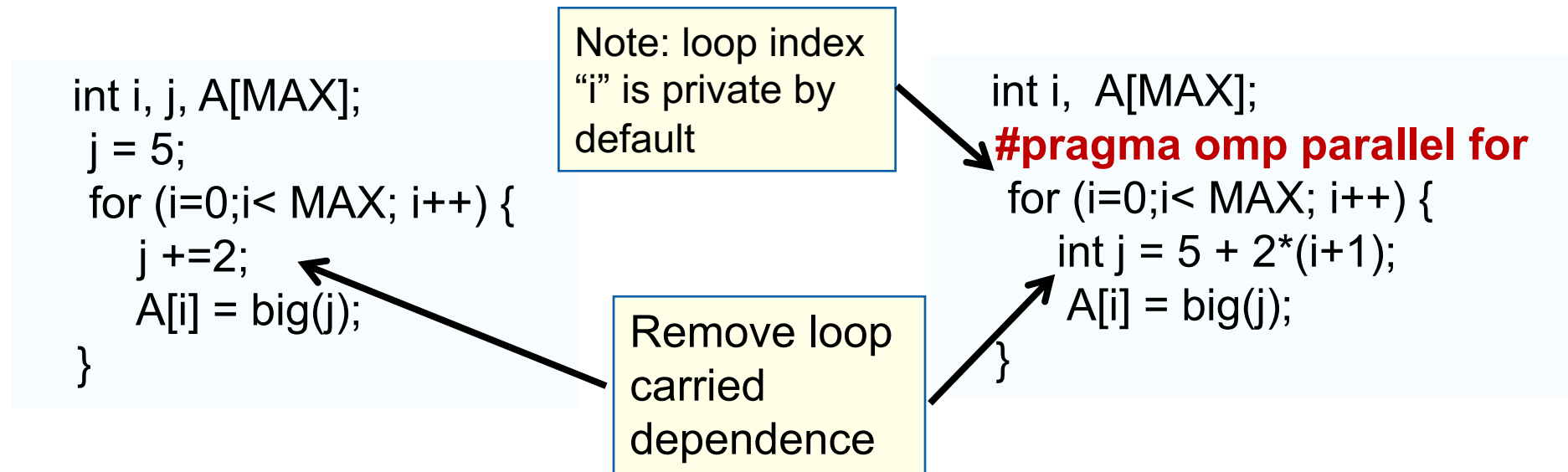
```
double res[MAX]; int i;  
#pragma omp parallel for  
    for (i=0;i< MAX; i++) {  
        res[i] = huge();  
    }
```

These are equivalent



Working with loops

- Basic approach
 - Find compute intensive loops
 - Make the loop iterations independent ... So they can safely execute in any order without loop-carried dependencies
 - Place the appropriate OpenMP directive and test



Reduction

- How do we handle this case?

```
double ave=0.0, A[MAX];  
int i;  
for (i=0;i< MAX; i++) {  
    ave + = A[i];  
}  
ave = ave/MAX;
```

- We are combining values into a single accumulation variable (ave) ... there is a true dependence between loop iterations that can't be trivially removed.
- This is a very common situation ... it is called a “reduction”.
- Support for reduction operations is included in most parallel programming environments.

Reduction

- OpenMP reduction clause:
reduction (op : list)
- Inside a parallel or a work-sharing construct:
 - A local copy of each list variable is made and initialized depending on the “op” (e.g. 0 for “+”).
 - Updates occur on the local copy.
 - Local copies are reduced into a single value and combined with the original global value.
- The variables in “list” must be shared in the enclosing parallel region.

```
double ave=0.0, A[MAX];  int i;  
#pragma omp parallel for reduction (+:ave)  
  for (i=0;i< MAX; i++) {  
    ave + = A[i];  
  }  
ave = ave/MAX;
```

OpenMP: Reduction operands/initial-values

- Many different associative operands can be used with reduction:
- Initial values are the ones that make sense mathematically.

Operator	Initial value
+	0
*	1
min	Largest pos. number
max	Most neg. number

C/C++ only	
Operator	Initial value
&	~0
 	0
^	0
&&	1
 	0

Fortran Only	
Operator	Initial value
.AND.	.true.
.OR.	.false.
.NEQV.	.false.
.IEOR.	0
.IOR.	0
.IAND.	All bits on
.EQV.	.true.

OpenMP includes user defined reductions and array-sections as reduction variables (we just don't cover those topics here)

Exercise: PI with loops

- Go back to the serial pi program and parallelize it with a loop construct
- Your goal is to minimize the number of changes made to the serial program.

```
#pragma omp parallel
#pragma omp for
#pragma omp parallel for
#pragma omp for reduction(op:list)
#pragma omp critical
int omp_get_num_threads();
int omp_get_thread_num();
double omp_get_wtime();
```

Example: PI with a loop and a reduction

```
#include <omp.h>
```

```
static long num_steps = 100000;    double step;
```

```
void main ()
```

```
{  int i;        double x, pi, sum = 0.0;
```

```
    step = 1.0/(double) num_steps;
```

```
    #pragma omp parallel
```

```
    {
```

```
        double x;
```

```
        #pragma omp for reduction(+:sum)
```

```
            for (i=0;i< num_steps; i++){
```

```
                x = (i+0.5)*step;
```

```
                sum = sum + 4.0/(1.0+x*x),
```

```
            }
```

```
    }
```

```
    pi = step * sum;
```

```
}
```

Create a team of threads ...
without a parallel construct, you'll
never have more than one thread

Create a scalar local to each thread to hold
value of x at the center of each interval

Break up loop iterations
and assign them to
threads ... setting up a
reduction into sum.
Note ... the loop index is
local to a thread by default.

Example: PI with a loop and a reduction

```
#include <omp.h>
static long num_steps = 100000;      double step;
void main ()
{
    double pi, sum = 0.0;
    step = 1.0/(double) num_steps;

    #pragma omp parallel for reduction(+:sum)
    for (int i=0;i< num_steps; i++){
        double x = (i+0.5)*step;
        sum = sum + 4.0/(1.0+x*x);
    }
    pi = step * sum;
}
```

Using modern C style, we put declarations close to where they are used ... which lets me use the parallel for construct.

Results*: PI with a loop and a reduction

- Original Serial pi program with 100000000 steps ran in 1.83 seconds.

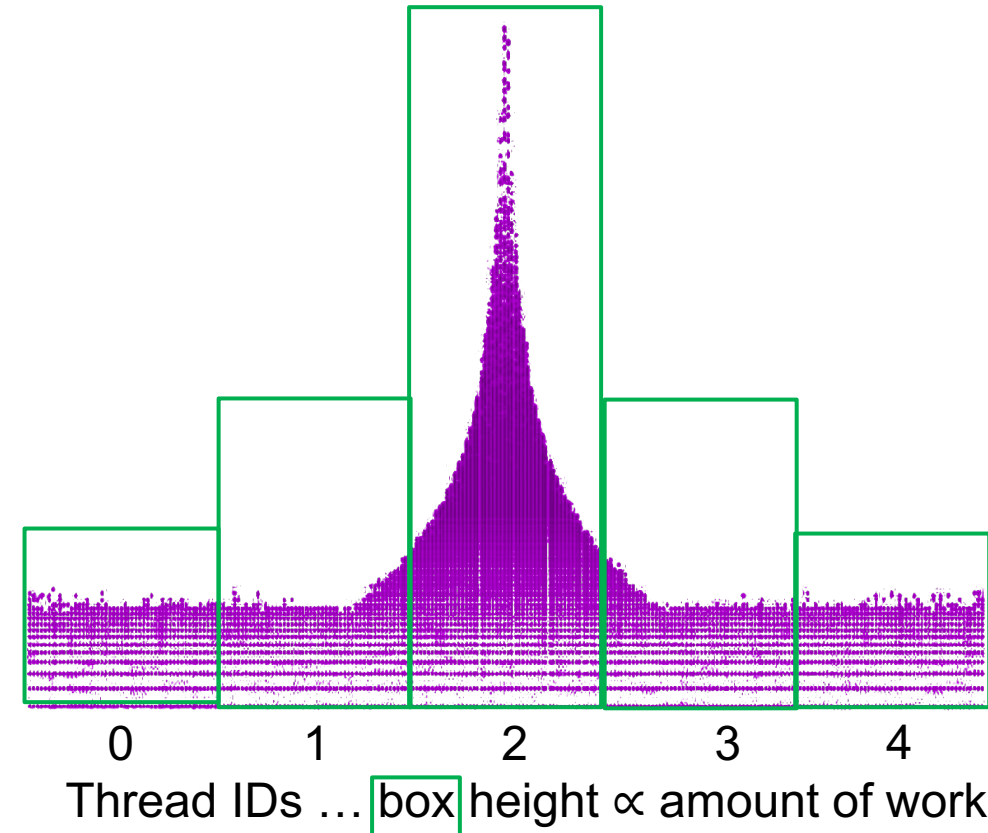
Example: Pi with a		threads	1 st SPMD	1 st SPMD padded	SPMD critical	PI Loop
<pre>#include <omp.h> static long num_steps = 100000000; void main () { int i; double x, pi, sum; step = 1.0/(double) num_steps; #pragma omp parallel { double x; #pragma omp for reduction(+:sum) for (i=0; i< num_steps; i++){ x = (i+0.5)*step; sum = sum + 4.0/(1.0+x*x); } } pi = step * sum; }</pre>		1	1.86	1.86	1.87	1.91
		2	1.03	1.01	1.00	1.02
		3	1.08	0.69	0.68	0.80
		4	0.97	0.53	0.53	0.68

*Intel compiler (icpc) with no optimization on Apple OS X 10.7.3 with a dual core (four HW thread) Intel® Core™ i5 processor at 1.7 Ghz and 4 Gbyte DDR3 memory at 1.333 Ghz.

**.... Let's pause a moment and consider
one of the fundamental issues EVERY
parallel programmer must grapple with**

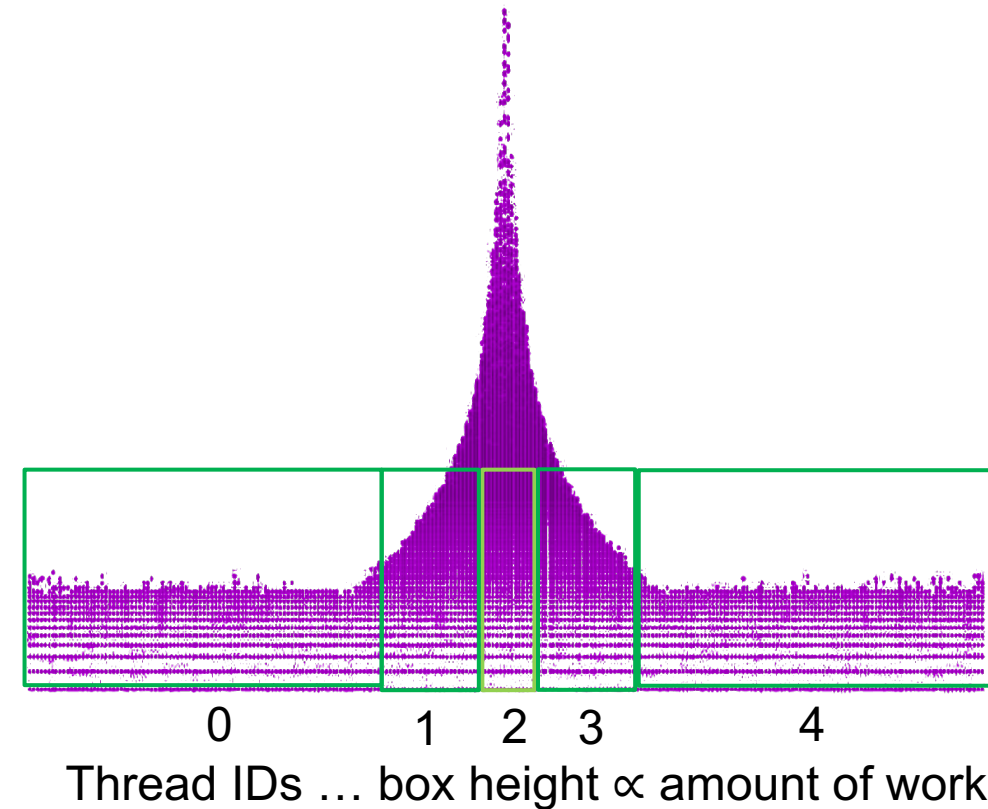
Load Balancing

- A parallel job isn't done until the last thread is finished
- Example: Partition a problem into equal sized chunks but for work that is unevenly distributed spatially.
 - Thread 2 has MUCH more work. The uneven distribution of work will limit performance.
- A key part of parallel programming is to design how you partition the work between threads so every thread has about the same amount of work. This topic is referred to as Load Balancing.



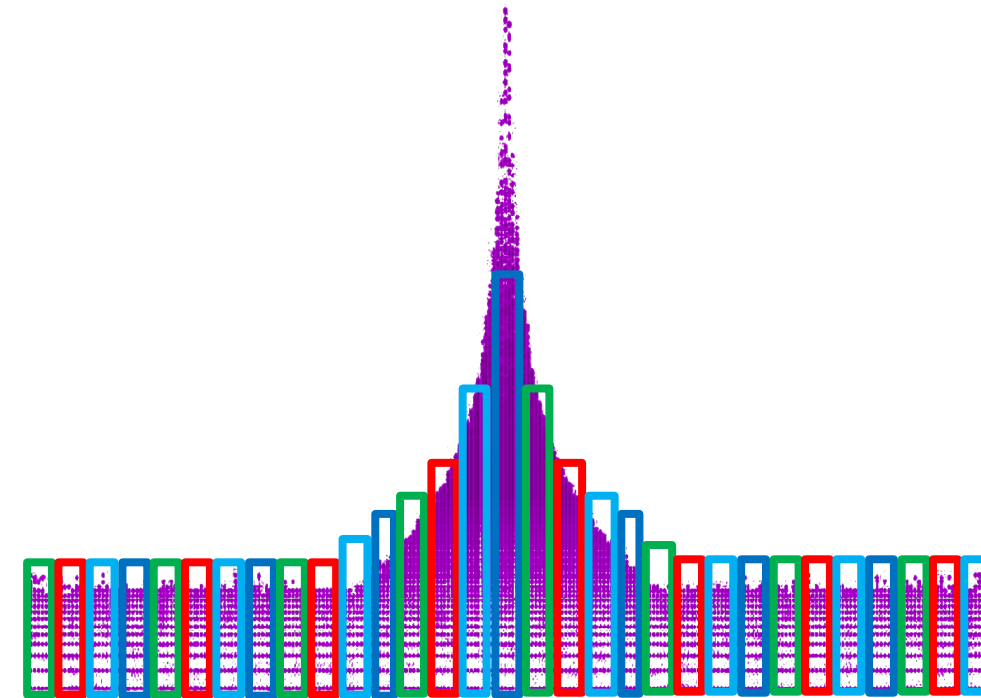
Load Balancing

- A parallel job isn't done until the last thread is finished
- The work in our problem is unevenly distributed spatially.
- A key part of parallel programming is to design how you partition the work between threads so every thread has about the same amount of work.
- This topic is referred to as Load Balancing.
- In this case we adjusted the size of each chunk to equalize the work assigned to each thread.
 - Getting the right sized chunks for a variable partitioning (as done here) can be really difficult.



Load Balancing

- A parallel job isn't done until the last thread is finished
- An easier path to Load Balancing.
 - Over-decompose the problem into small, fine-grained chunks
 - Spread the chunks out among the threads (in this case using a cyclic distribution)
 - The work is spread out and statistically, you are likely to get a good distribution of work

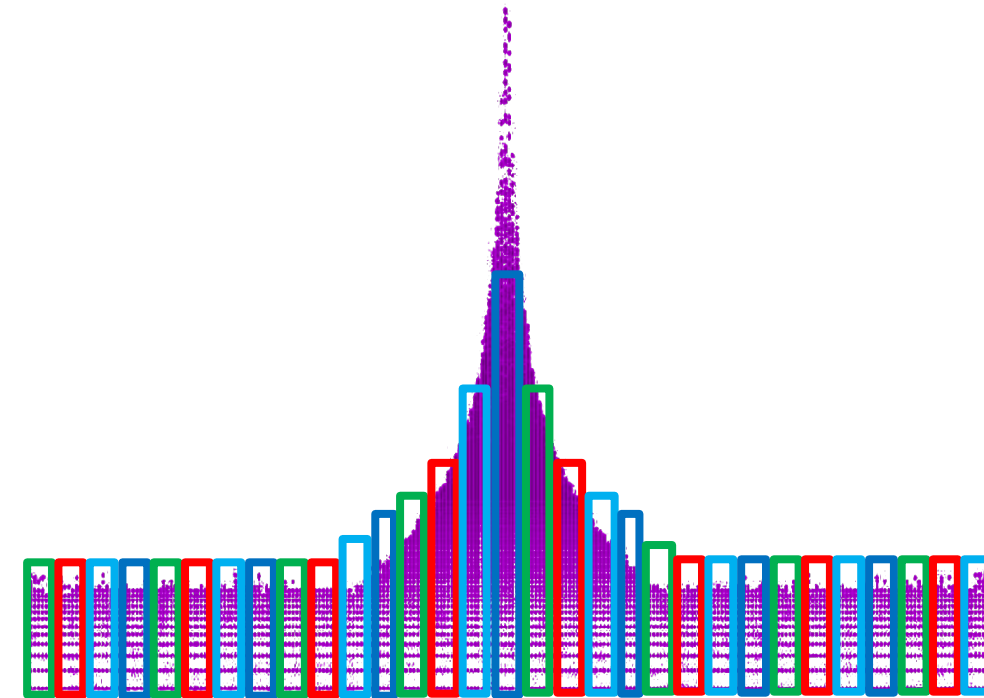


Colors mapped to 4 different Threads

- 0 
- 1 
- 2 
- 3 

Load Balancing

- A parallel job isn't done until the last thread is finished
- An easier path to Load Balancing.
 - Over-decompose the problem into small, fine-grained chunks
 - Spread the chunks out among the threads (in this case using a cyclic distribution)
 - The work is spread out and statistically, you are likely to get a good distribution of work
- Vocabulary review
 - **Load Balancing** ... giving each thread work sized so all threads take the same amount of time
 - **Partitioning** or **decomposition** ... breaking up the problem domain into partitions (or chunks) and assigning different partitions to different threads.
 - **Granularity** ... the size of the block of work. Fine grained (small chunks) vs coarse grained (large chunks)
 - **Over-decomposition** ... when you decompose your problem into partitions such that there are many more partitions than threads to do the work



Colors mapped to 4 different Threads

0	
1	
2	
3	

Loop Worksharing Constructs: The schedule clause

- The schedule clause affects how loop iterations are mapped onto threads
 - **schedule(static [,chunk])**
 - Deal-out blocks of iterations of size “chunk” to each thread.
 - **schedule(dynamic[,chunk])**
 - Each thread grabs “chunk” iterations off a queue until all iterations have been handled.
- Example:
 - #pragma omp for schedule(dynamic, 10)

Schedule Clause	When To Use
STATIC	Pre-determined and predictable by the programmer
DYNAMIC	Unpredictable, highly variable work per iteration

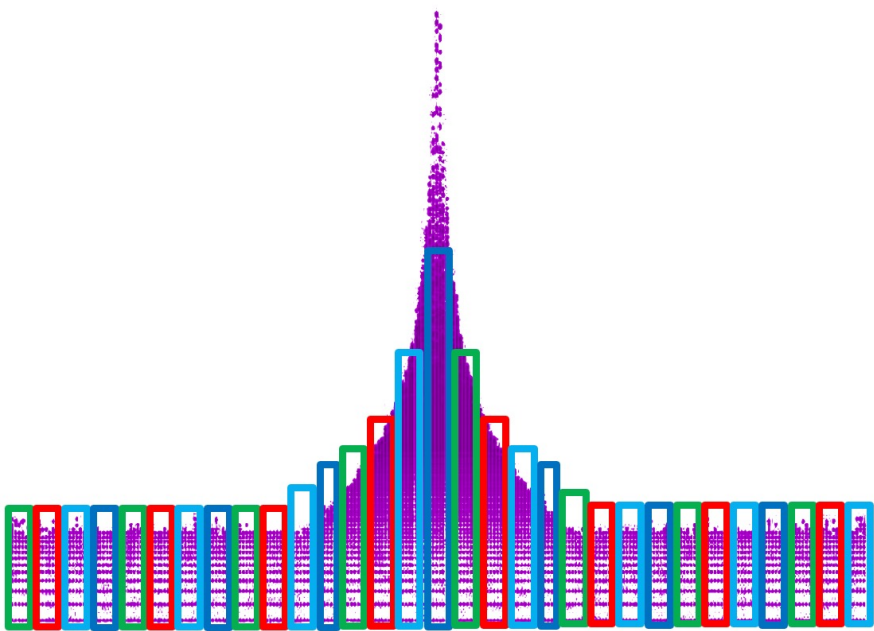
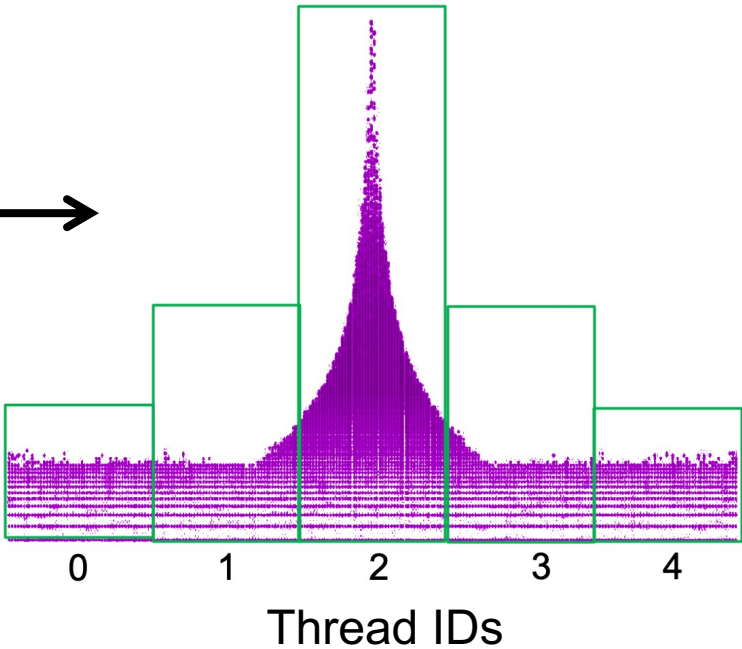
Least work at runtime :
scheduling done at
compile-time

Most work at runtime :
complex scheduling
logic used at run-time

Loop Worksharing Constructs: The schedule clause

- The schedule clause ... most common cases:

`#pragma omp parallel for schedule (static)` →



Colors mapped to 4 different Threads

- 0
- 1
- 2
- 3



`Int small = 8; // loop iterations, i.e., width of boxes in the figure`

`#pragma omp parallel for schedule (static, small)`

We'll finish with loops by looking one more time at synchronization overhead

The nowait clause

- Barriers are really expensive. You need to understand when they are implied and how to skip them when it's safe to do so.

```
double A[big], B[big], C[big];
```

```
#pragma omp parallel  
{
```

```
    int id=omp_get_thread_num();  
    A[id] = big_calc1(id);
```

```
#pragma omp barrier
```

```
#pragma omp for
```

```
    for(i=0;i<N;i++){C[i]=big_calc3(i,A);}
```

```
#pragma omp for nowait
```

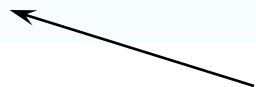
```
    for(i=0;i<N;i++){ B[i]=big_calc2(C, i); }  
    A[id] = big_calc4(id);
```

```
}
```

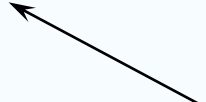
implicit barrier at the end of a for
worksharing construct



implicit barrier at the end
of a parallel region



no implicit barrier
due to nowait



- Introduction to OpenMP
- Creating Threads
- Synchronization ... simple collective sync
- Parallel Loops

-  • Data Environment
 - Synchronization ... the general case (including Memory Models)
 - Irregular Parallelism and Tasks
 - OpenMP for real hardware (NUMA)
 - Recap

Data Environment: Default storage attributes

- Shared memory programming model:
 - Most variables are shared by default
- Global variables are SHARED among threads
 - Fortran: COMMON blocks, SAVE variables, MODULE variables
 - C: File scope variables, static
 - Both: dynamically allocated memory (ALLOCATE, malloc, new)
- But not everything is shared...
 - Stack variables in subprograms(Fortran) or functions(C) called from parallel regions are PRIVATE
 - Automatic variables within a statement block are PRIVATE.

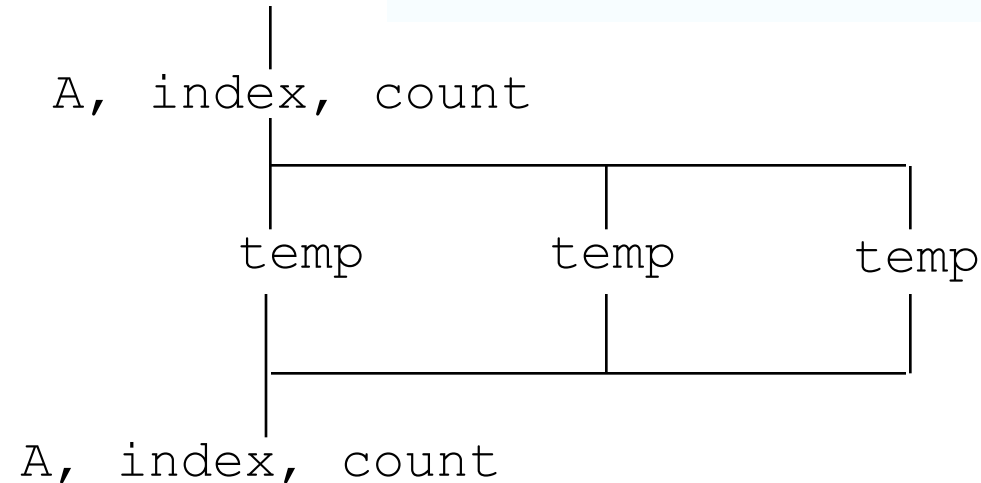
Data Sharing: Examples

```
double A[10];  
int main() {  
    int index[10];  
    #pragma omp parallel  
        work(index);  
    printf("%d\n", index[0]);  
}
```

A, index and count are shared by all threads.

temp is local to each thread

```
extern double A[10];  
void work(int *index) {  
    double temp[10];  
    static int count;  
    ...  
}
```



Data Sharing: Changing storage attributes

- One can selectively change storage attributes for constructs using the following clauses (note: *list* is a comma-separated list of variables)
 - **shared(list)**
 - **private(list)**
 - **firstprivate(list)**
- These can be used on **parallel** and **for** constructs ... other than **shared** which can only be used on a **parallel** construct
- Force the programmer to explicitly define storage attributes
 - **default (none)**

default() can only be used
on parallel constructs

Data Sharing: Private clause

- `private(var)` creates a new local copy of `var` for each thread.

```
int N = 1000;  
extern void init_arrays(int N, double *A, double *B, double *C);
```

```
void example () {  
    int i, j;  
    double A[N][N], B[N][N], C[N][N];  
    init_arrays(N, *A, *B, *C);  
  
    #pragma omp parallel for private(j)  
    for (i = 0; i < 1000; i++)  
        for( j = 0; j<1000; j++)  
            C[i][j] = A[i][j] + B[i][j];  
}
```

OpenMP makes the loop control index on the parallel loop (i) private by default ... but not for the second loop (j)

Data Sharing: Private clause

- `private(var)` creates a new local copy of `var` for each thread.
 - The value of the private copies is uninitialized
 - The value of the original variable is unchanged after the region

```
incr = 0;  
#pragma omp parallel for private(incr)  
for (i = 0; i <= MAX; i++) {  
    if ((i%2)==0) incr++;  
    A[i] = incr;  
}  
printf(" incr= %d\n", incr);
```

When you need to refer to the variable `incr` that exists prior to the construct, we call it the **original variable**.

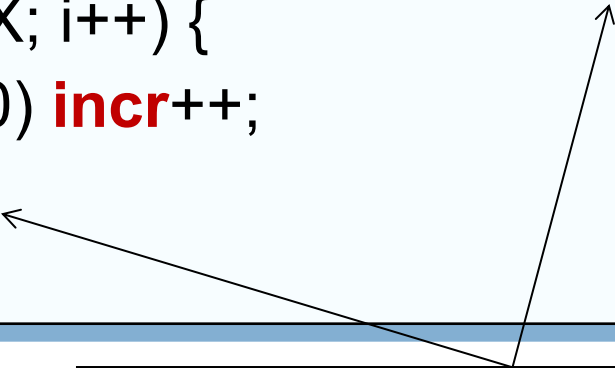
incr was not initialized

incr is 0 here

Firstprivate clause

- Variables initialized from a shared variable
- C++ objects are copy-constructed

```
incr = 0;  
#pragma omp parallel for firstprivate(incr)  
for (i = 0; i <= MAX; i++) {  
    if ((i%2)==0) incr++;  
    A[i] = incr;  
}
```



Each thread gets its own copy of incr with an initial value of 0

Data sharing:

A data environment test

- Consider this example of PRIVATE and FIRSTPRIVATE

```
variables: A = 1, B = 1, C = 1  
#pragma omp parallel private(B) firstprivate(C)
```

A → shared, 1, 1
B → private, undef, 1
C → private, 1, 1

- Are A,B,C private to each thread or shared inside the parallel region?
- What are their initial values inside and values after the parallel region?

Inside this parallel region ...

- “A” is shared by all threads; equals 1
- “B” and “C” are private to each thread.
 - B’s initial value is undefined
 - C’s initial value equals 1

Following the parallel region ...

- B and C revert to their original values of 1
- A is either 1 or the value it was set to inside the parallel region

Exercise: Mandelbrot set area

- The supplied program (mandel.c) computes the area of a Mandelbrot set.
- The program has been parallelized with OpenMP, but we were lazy and didn't do it right.
- Find and fix the errors.
- Once you have a working version, try to optimize the program.

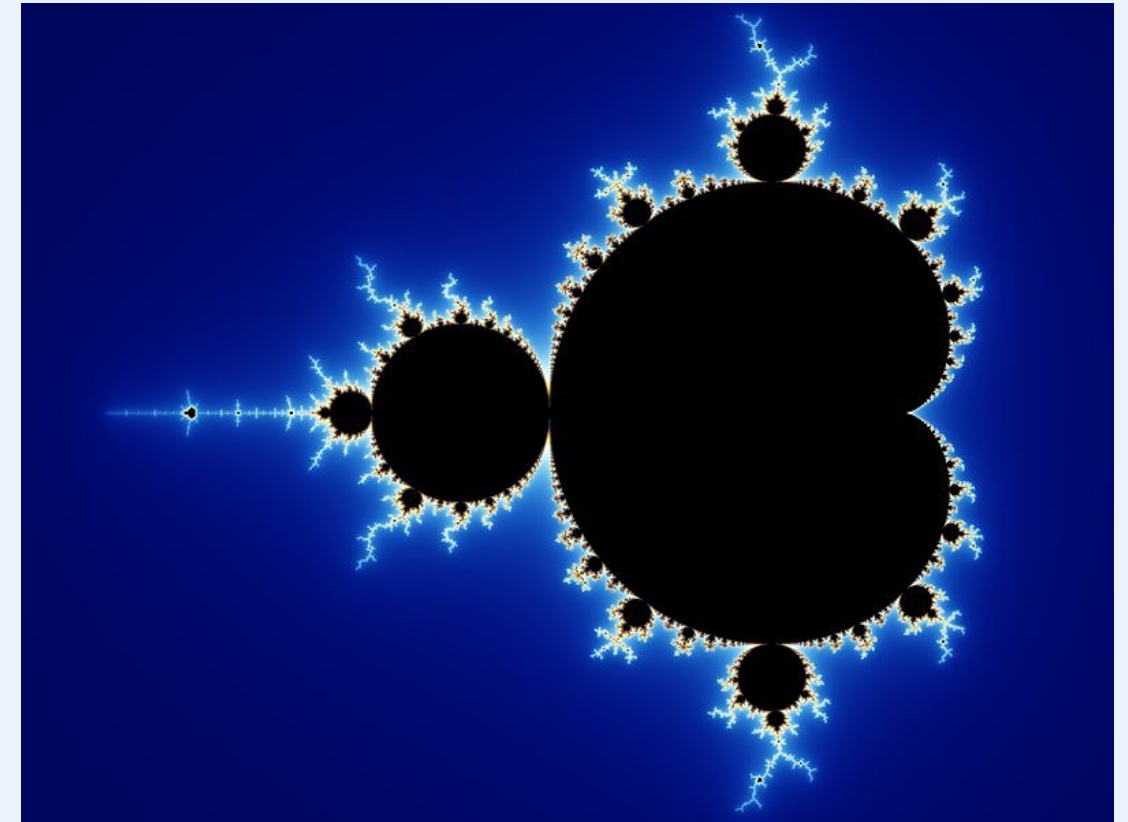


Image Source: Created by Wolfgang Beyer with the program Ultra Fractal 3. - Own work, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=321973>

The Mandelbrot set ... The points, c , for which the following iterative map converges

$$z_{n+1} = z_n^2 + c$$

With z_n and c as complex numbers and $z_0 = 0$.

The Mandelbrot Set Area Program (original code)

```
#include <omp.h>
# define NPOINTS 1000
# define MXITR 1000
void testpoint(double, double);
Int numoutside = 0;
int main(){
    int i, j;
    int num=0;
    double C_real, C_imag;
    double area, error, eps = 1.0e-5;
#pragma omp parallel for private(eps)
    for (i=0; i<NPOINTS; i++) {
        for (j=0; j<NPOINTS; j++) {
            C_real = -2.0+2.5*(double)(i)/(double)(NPOINTS)+eps;
            C_imag = 1.125*(double)(j)/(double)(NPOINTS)+eps;
            testpoint(C_real, C_imag);
        }
    }
    area=2.0*2.5*1.125*(double)(NPOINTS*NPOINTS-
numoutside)/(double)(NPOINTS*NPOINTS);
    error=area/(double)NPOINTS;
}
```

```
void testpoint(double C_real, double C_imag){
    double zr, zi;
    int iter;
    double temp;

    zr=C_real;    zi=C_imag;
    for (iter=0; iter<MXITR; iter++){
        temp = (zr*zr)-(zi*zi)+C_real;
        zi = zr*zi*2+C_imag;
        zr = temp;
        if ((zr*zr+zi*zi)>4.0) {
            numoutside++;
            break;
        }
    }
}
```

The Mandelbrot Set Area Program

```
#include <omp.h>
# define NPOINTS 1000
# define MXITR 1000
void testpoint(double, double);
Int numoutside = 0;
int main(){
    int i, j;
    int num=0;
    double C_real, C_imag;
    double area, error, eps = 1.0e-5;
    #pragma omp parallel for private(j, C_real, C_imag)
    for (i=0; i<NPOINTS; i++) {
        for (j=0; j<NPOINTS; j++) {
            C_real = -2.0+2.5*(double)(i)/(double)(NPOINTS)+eps;
            C_imag = 1.125*(double)(j)/(double)(NPOINTS)+eps;
            testpoint(C_real, C_imag);
        }
    }
    area=2.0*2.5*1.125*(double)(NPOINTS*NPOINTS-
numoutside)/(double)(NPOINTS*NPOINTS);
    error=area/(double)NPOINTS;
}
```

```
void testpoint(double C_real, double C_imag){
    double zr, zi;
    int iter;
    double temp;

    zr=C_real;    zi=C_imag;
    for (iter=0; iter<MXITR; iter++){
        temp = (zr*zr)-(zi*zi)+C_real;
        zi = zr*zi*2+C_imag;
        zr = temp;
        if ((zr*zr+zi*zi)>4.0) {
            #pragma omp critical
                numoutside++;
            break
        }
    }
}
```

- eps was not initialized
- Data race on j, C_real, and C_imag
- Protect updates of numoutside

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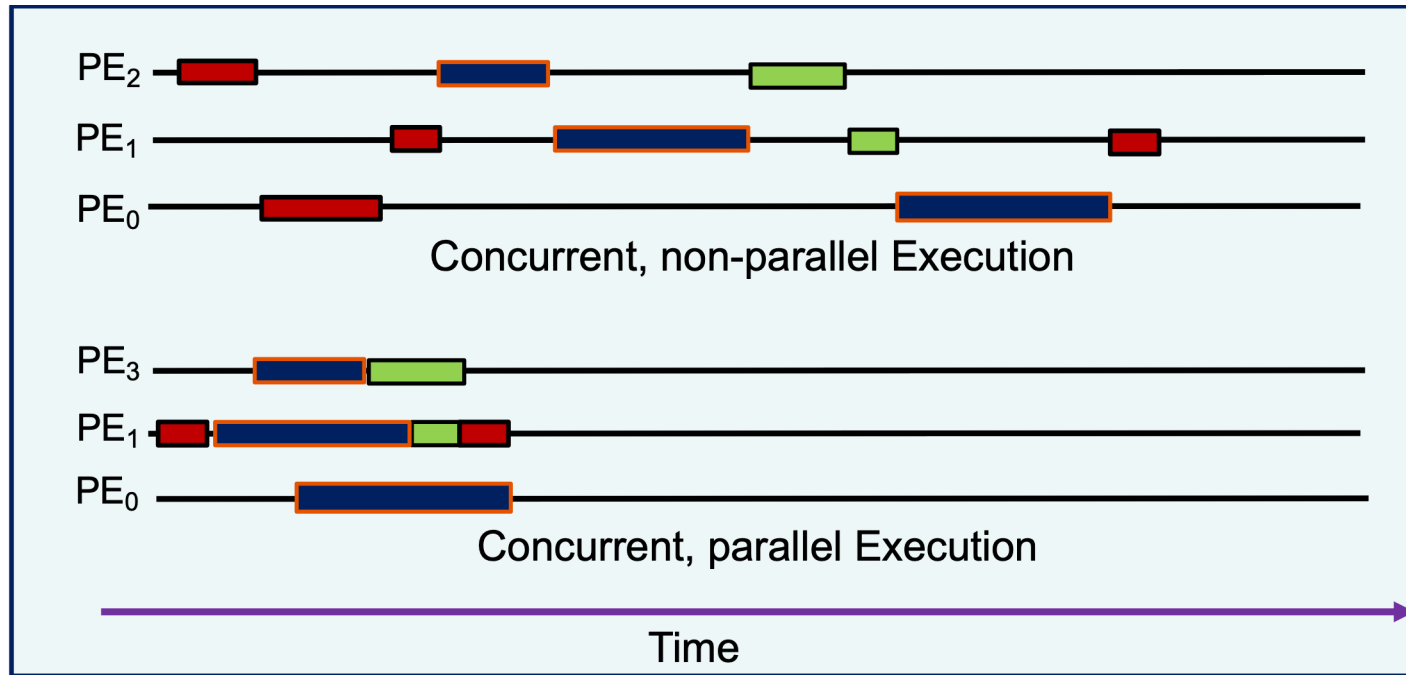
Synchronization

Synchronization is used to impose order constraints and to protect access to shared data

- High level synchronization included in the common core:
 - critical
 - barrier
- Other, more advanced, synchronization operations:
 - locks (both simple and nested)
 - atomic
 - flush

Concurrency vs. Parallelism

- Concurrency: A condition of a system in which multiple tasks are active and unordered. If **scheduled fairly**, they can be described as logically making **forward progress** at the same time.
- Parallelism: A condition of a system in which multiple tasks are actually making **forward progress** at the same time.



The fundamental execution model of Multithreading

A collection of active threads, scheduled fairly, that share an address space and execute concurrently.

Synchronization: Lock Routines

- Lock routines:
 - A lock is available if it is unset.
 - `omp_init_lock()`, `omp_set_lock()`,
`omp_unset_lock()`, `omp_test_lock()`, `omp_destroy_lock()`

A lock implies a memory fence ... hence a compiler cannot move memory operations around a lock.

Note: a thread always accesses the most recent copy of the lock (which when we discuss memory models, means you don't need to use a flush on the lock variable).

Locks with hints were added in OpenMP 4.5 to suggest a lock strategy based on intended use (e.g. contended, uncontended, speculative, unspeculative)

Synchronization: Simple Locks Example

- Count odds and evens in an input array(x) of N random values.

```
int i, ix, even_count = 0, odd_count = 0;
```

```
omp_lock_t odd_lck, even_lck;
```

```
omp_init_lock(&odd_lck);
```

```
omp_init_lock(&even_lck);
```

One lock per case ... even and odd

```
#pragma omp parallel for private(ix) shared(even_count, odd_count)
```

```
for(i=0; i<N; i++){
```

```
    ix = (int) x[i]; //truncate to int
```

```
    if(((int) x[i])%2 == 0) {
```

```
        omp_set_lock(&even_lck);
```

```
        even_count++;
```

```
        omp_unset_lock(&even_lck);
```

```
    }
```

```
    else{
```

```
        omp_set_lock(&odd_lck);
```

```
        odd_count++;
```

```
        omp_unset_lock(&odd_lck);
```

```
    }
```

```
}
```

```
omp_destroy_lock(&odd_lck);
```

```
omp_destroy_lock(&even_lck);
```

```
}
```

Enforce mutual exclusion updates,
but in parallel for each case.

Free-up storage when done.

Synchronization: Atomic

- Atomic provides mutual exclusion but only applies to the update of a memory location (the update of X in the following example)

```
#pragma omp parallel
{
    double B;
    B = DOIT();

    #pragma omp atomic
        X += big_ugly(B);
}
```

Synchronization: Atomic

- Atomic provides mutual exclusion but only applies to the update of a memory location (the update of X in the following example)

```
#pragma omp parallel
{
    double B, tmp;
    B = DOIT();
    tmp = big_ugly(B);
    #pragma omp atomic
    X += tmp;
}
```

Atomic only protects the read/update of X

The OpenMP 3.1 Atomics (1 of 2)

- Atomic was expanded to cover the full range of common scenarios where you need to protect a memory operation so it occurs atomically:

pragma omp atomic [read | write | update | capture]

- Atomic can protect loads

pragma omp atomic read

v = x;

- Atomic can protect stores

pragma omp atomic write

x = expr;

- Atomic can protect updates to a storage location (this is the default behavior ... i.e. when you don't provide a clause)

pragma omp atomic update

x++; or ++x; or x--; or --x; or

x binop= expr; or x = x binop expr;

This is the
original OpenMP
atomic

The OpenMP 3.1 Atomics (2 of 2)

- Atomic can protect the assignment of a value (its capture) AND an associated update operation:

`# pragma omp atomic capture`
`statement or structured block`

- Where the statement is one of the following forms:

`v = x++;` `v = ++x;` `v = x--;` `v = -x;` `v = x binop expr;`

- Where the structured block is one of the following forms:

`{v = x; x binop = expr;}`

`{x binop = expr; v = x;}`

`{v=x; x=x binop expr;}`

`{X = x binop expr; v = x;}`

`{v = x; x++;}`

`{v=x; ++x;}`

`{++x; v=x;}`

`{x++; v = x;}`

`{v = x; x--;}`

`{v= x; --x;}`

`{--x; v = x;}`

`{x--; v = x;}`

The capture semantics in atomic were added to map onto common hardware supported atomic operations and to support modern lock free algorithms

Atomics are far more important than just a lightweight way to enforce mutual exclusion.

Atomics are essential when you need to work with the details of the OpenMP Memory model.

And this comes up when you consider pairwise synchronization

Consider two threads: a producer/consumer pair

```
#include <stdio.h>
#include <omp.h>
#define COUNT 1000000
int main()
{
    int answer = 0, flag= 0,err=0;
    for (int i=0; i<COUNT; i++) {
        flag = 0; answer=0;
        #pragma omp parallel shared(flag,answer) num_threads(2)
        {
            int id = omp_get_thread_num();
            if (id == 0) {
                answer = 42;
                flag = 1;
            }
            else if (id == 1){
                while (flag == 0) { }
                if(answer!=42) err++;
            }
        }
    }
    return 0;
}
```

One thread **produces** a result
that a different thread **consumes**

Thread zero produces the answer and
then sets a flag to communicate the
answer to another thread

Thread one “spins” in a while loop
until the flag is non-zero which
indicates that answer is available.

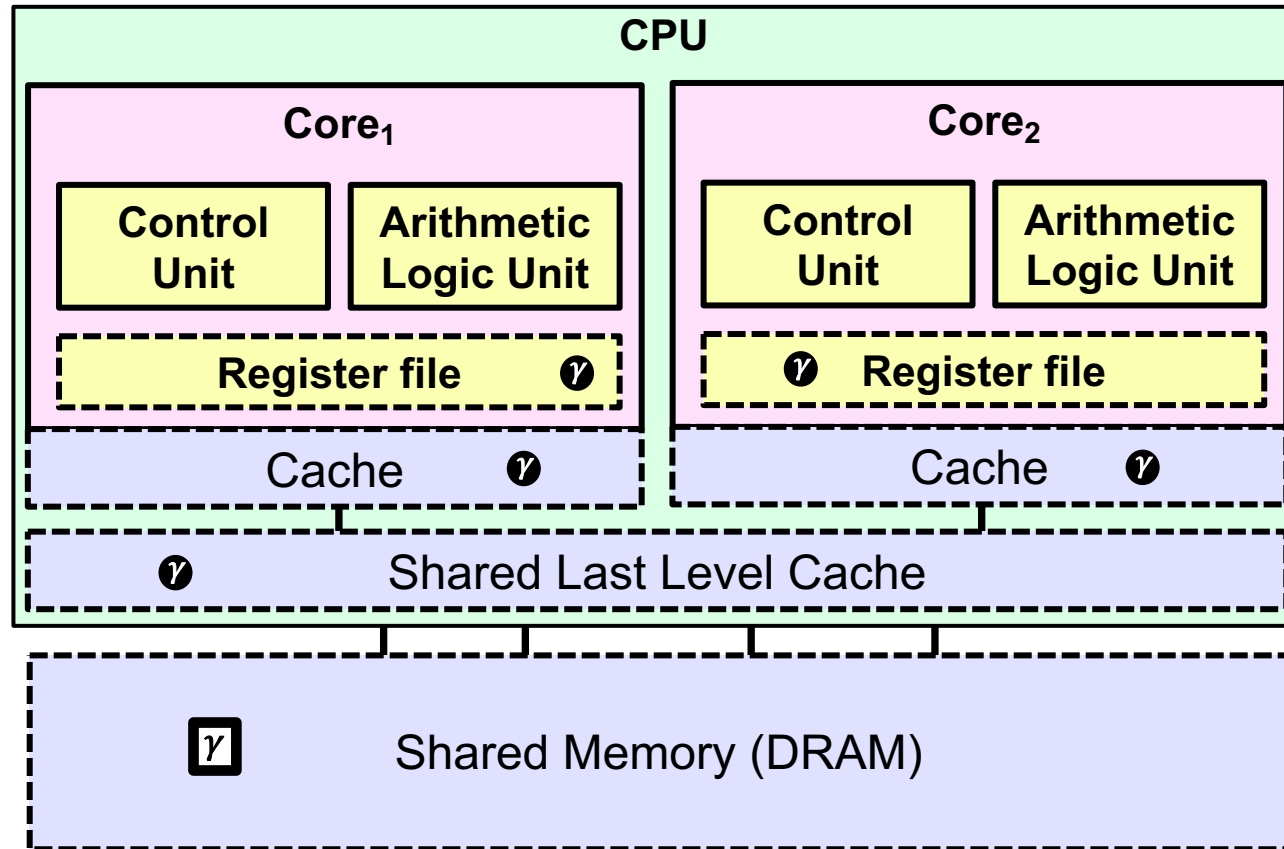
In the jargon of concurrent
programming, this is called
a “spin lock”

Put this in a file sync.c and compile as: **gcc -fopenmp -O3 sync.c**

The program went through a few loop iterations and then hangs Why?

Memory Models ...

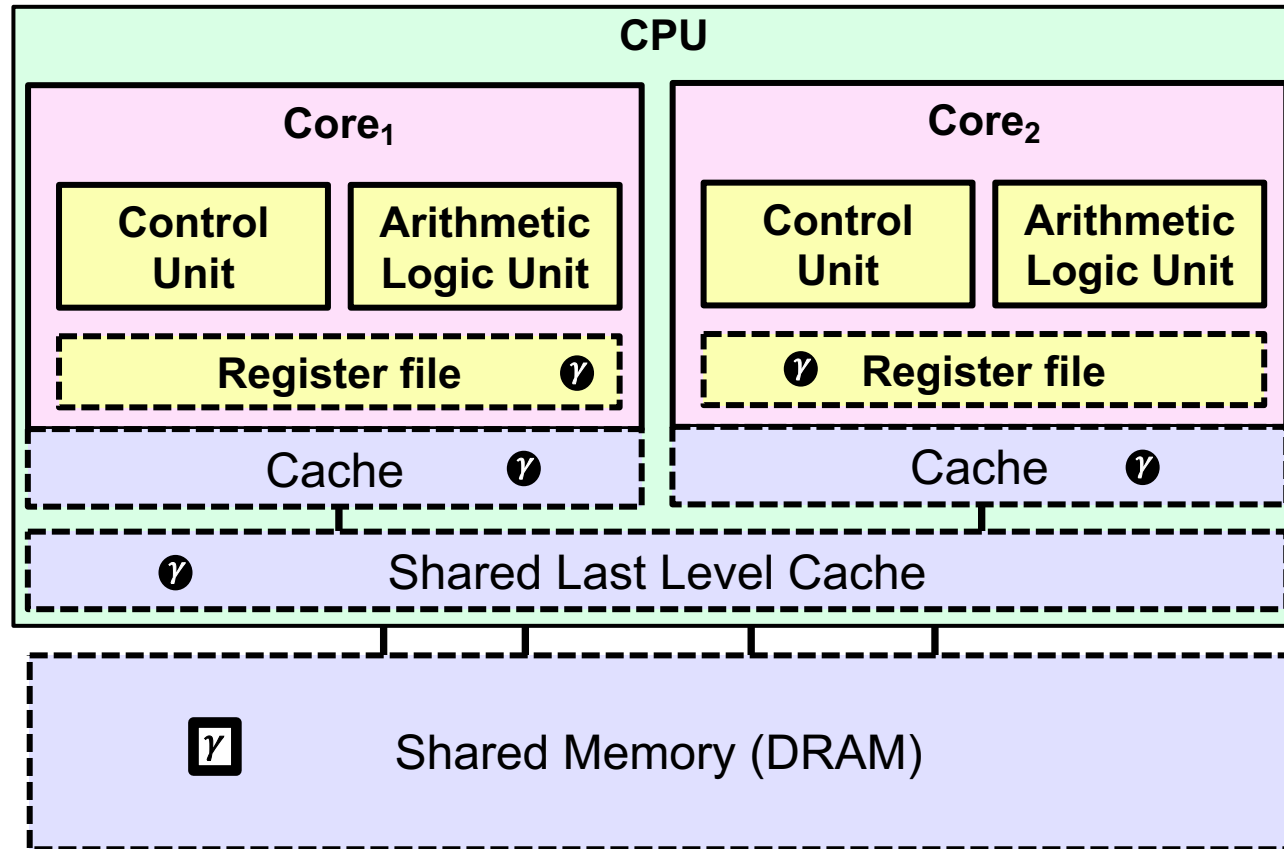
- A shared address space is a region of memory visible to the team of threads ... multiple threads can read and write variables in the shared address space.
- Multiple copies of a variable (such as γ) may be present in memory, at various levels of cache, or in registers and they may ALL have different values.



- Which value of γ is the one a thread should see at any point in a computation?

Memory Models ...

- A shared address space is a region of memory visible to the team of threads ... multiple threads can read and write variables in the shared address space.
- Multiple copies of a variable (such as γ) may be present in memory, at various levels of cache, or in registers and they may ALL have different values.



A memory consistency model (or “**memory model**” for short) provides the rules needed to answer this question.

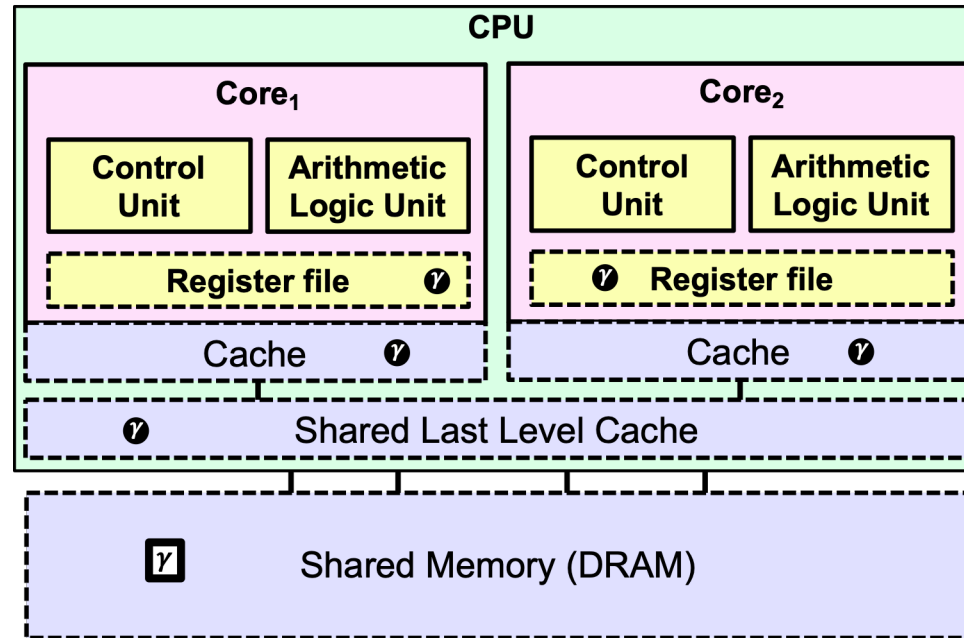
- Which value of γ is the one a thread should see at any point in a computation?

Memory Models ...

- The fundamental issue is how do the values of variables across the memory hierarchy interact with the statements executed by two or more threads?
- Two options:

1. Sequential Consistency

- Threads execute and the associated loads/stores appear in some order defined by the semantically allowed interleaving of program statements.
- **All threads see the same interleaved order of loads and stores**



2. Relaxed Consistency

- Threads execute and the associated loads/stores appear in some order defined by the semantically allowed interleaving of program statements.
- **Threads may see different orders of loads and stores**

Most (if not all) multithreading programming models assume **relaxed consistency**. Maintaining sequential consistency across the full program-execution adds too much synchronization overhead.

Why did this program fail?

Two issues:

(1) Can **flag** = 1 while **answer** = 0?

(2) Can thread 1 fail to see updates to **flag**?

```
#include <stdio.h>
#include <omp.h>
#define COUNT 1000000
int main()
{
    int answer = 0, flag= 0,err=0;
    for (int i=0; i<COUNT; i++) {
        flag = 0; answer=0;
        #pragma omp parallel shared(flag,answer) num_threads(2)
        {
            int id = omp_get_thread_num();
            if (id == 0) {
                answer = 42;
                flag = 1;
            }
            else if (id == 1){
                while (flag == 0) { }
                if(answer!=42) err++;
            }
        }
    }
    return 0;
}
```

The compiler can reorder statements, so **flag** is set to 1 before **answer** is set to **42**

Thread 1 can load **flag** from the register file. It may not even go to cache (let alone memory) to see an updated value.

Regardless of how the compiler orders stores to **answer** and **flag**, thread 1 may see a different order than thread 0

Why did this program fail?

Two issues:

(1) Can **flag** = 1 while **answer** = 0?

(2) Can thread 1 fail to see updates to **flag**?

```
#include <stdio.h>
#include <omp.h>
#define COUNT 1000000
int main()
{
    int answer = 0, flag= 0,err=0;
    for (int i=0; i<COUNT; i++) {
        flag = 0; answer=0;
        #pragma omp parallel shared(flag,answer) num_threads(2)
        {
            int id = omp_get_thread_num();
            if (id == 0) {
                answer = 42;
                flag = 1;
            }
            else if (id == 1){
                while (flag == 0) { }
                if(answer!=42) err++;
            }
        }
    }
    return 0;
}
```

The compiler can reorder statements, so **flag** is set to 1 before **answer** is set to **42**

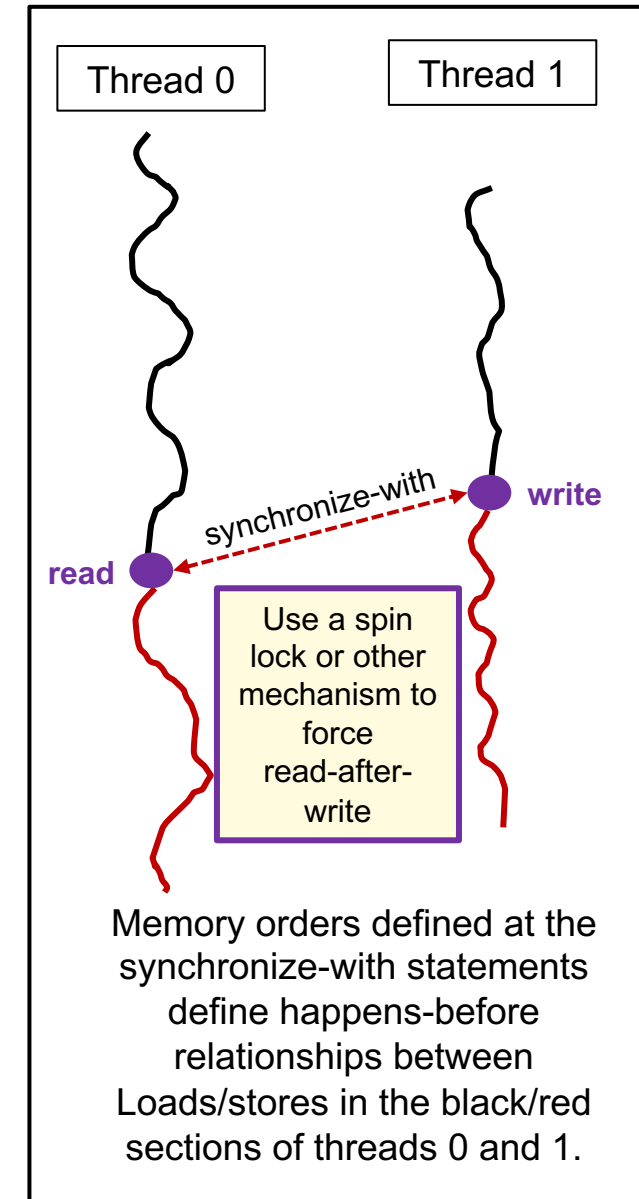
We need to enforce ordering constraints between the concurrent threads ... we need to consider the memory model and put the right synchronization constructs in place.

Thread 1 can load **flag** from the register file. It may not even go to cache (let alone memory) to see an updated value.

Regardless of how the compiler orders stores to **answer** and **flag**, thread 1 may see a different order than thread 0

Memory Models: *Happens-before* and *synchronized-with* relations

- Single thread execution:
 - Program order ... Loads and stores appear to occur in the order defined by the program's semantics. If you can't observe it, however, compilers can reorder instructions to maximize performance.
- Multithreaded execution ... concurrency in action
 - The compiler doesn't understand instruction-ordering across threads ... loads/stores to shared memory across threads can expose ambiguous orders of loads and stores
 - Instructions between threads are unordered except when specific ordering constraints are imposed, i.e., **synchronization**.
 - Synchronization lets us force that some instructions **happens-before** other instructions
- Two parts to synchronization:
 - A **synchronize-with** relationship exists at statements in 2 or more threads at which memory order constraints can be established.
 - **Memory order**: defines the view of loads/stores on either side of a synchronized-with operations.



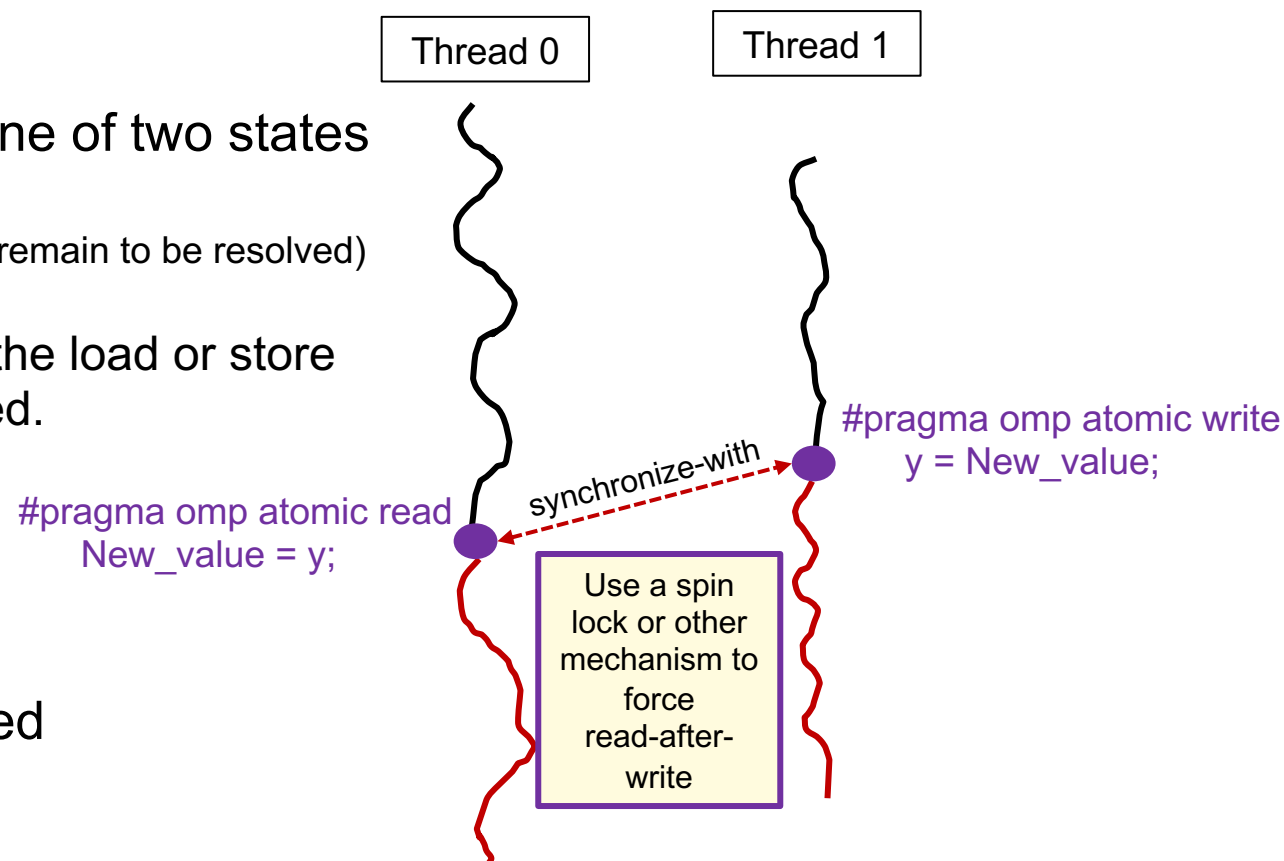
Atomic Operations and Synchronized-with

- An atomic operation can only be observed in one of two states
 - The operation has not happened yet
 - The operation has happened and is complete (no side-effects remain to be resolved)

- For example, on an atomic load or store operation, the load or store has happened and is complete, or it has not occurred.

- A **synchronized-with** relationship is established between a pair of atomic operations.

- The variables involved are visible to the programmer (such as with atomic constructs) or the variables are internal to a high level synchronization construct (barrier, critical, locks, etc).

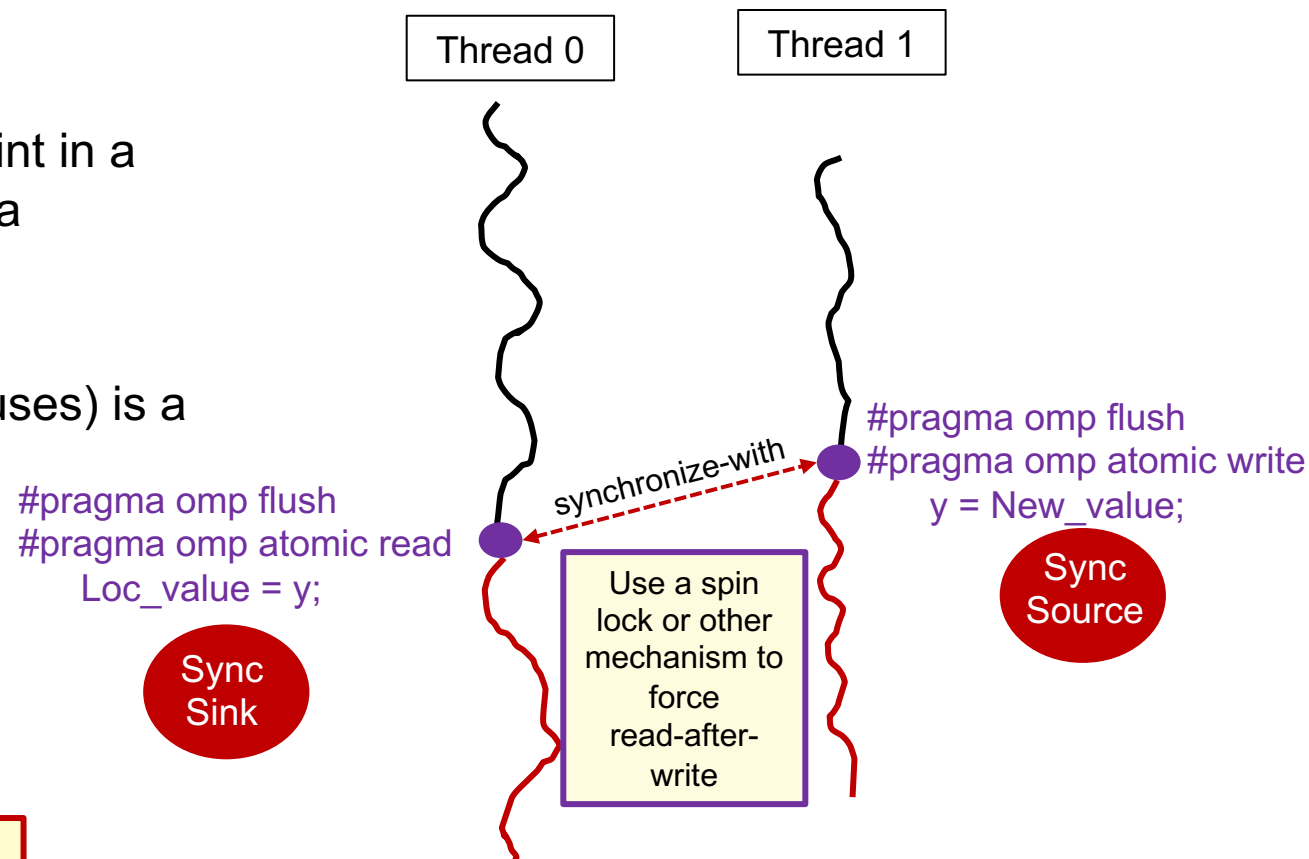


Memory orders

- Memory orders establish which loads and stores can be moved around synchronized-with relations.
- The key construct is **flush**. ... flush defines a point in a program at which a thread is guaranteed to see a consistent view of memory.
- The default case for flush (i.e., no additional clauses) is a **strong flush**:
 - Previous read/writes by this thread have completed and are visible to other threads
 - No subsequent read/writes by this thread have occurred

A strong flush on its own does NOT define a synchronization point. The flush only addresses memory orders.

To synchronize threads, you need a synchronized-with relation which in this case, comes from an atomic write paired with an atomic read



Memory orders defined at the synchronize-with statements define happens-before relationships between Loads/stores in the black/red sections of threads 0 and 1.

Black operations on Thread 1 happen-before Red operations on thread 0.

Memory orders

- Memory orders establish which loads and stores can be moved around synchronized-with relations.
- The strong flush by itself is expensive as it impacts all the shared variables visible to a thread.
- There are more focused forms of memory order
The 2 most fundamental memory orders are:

read-acquire

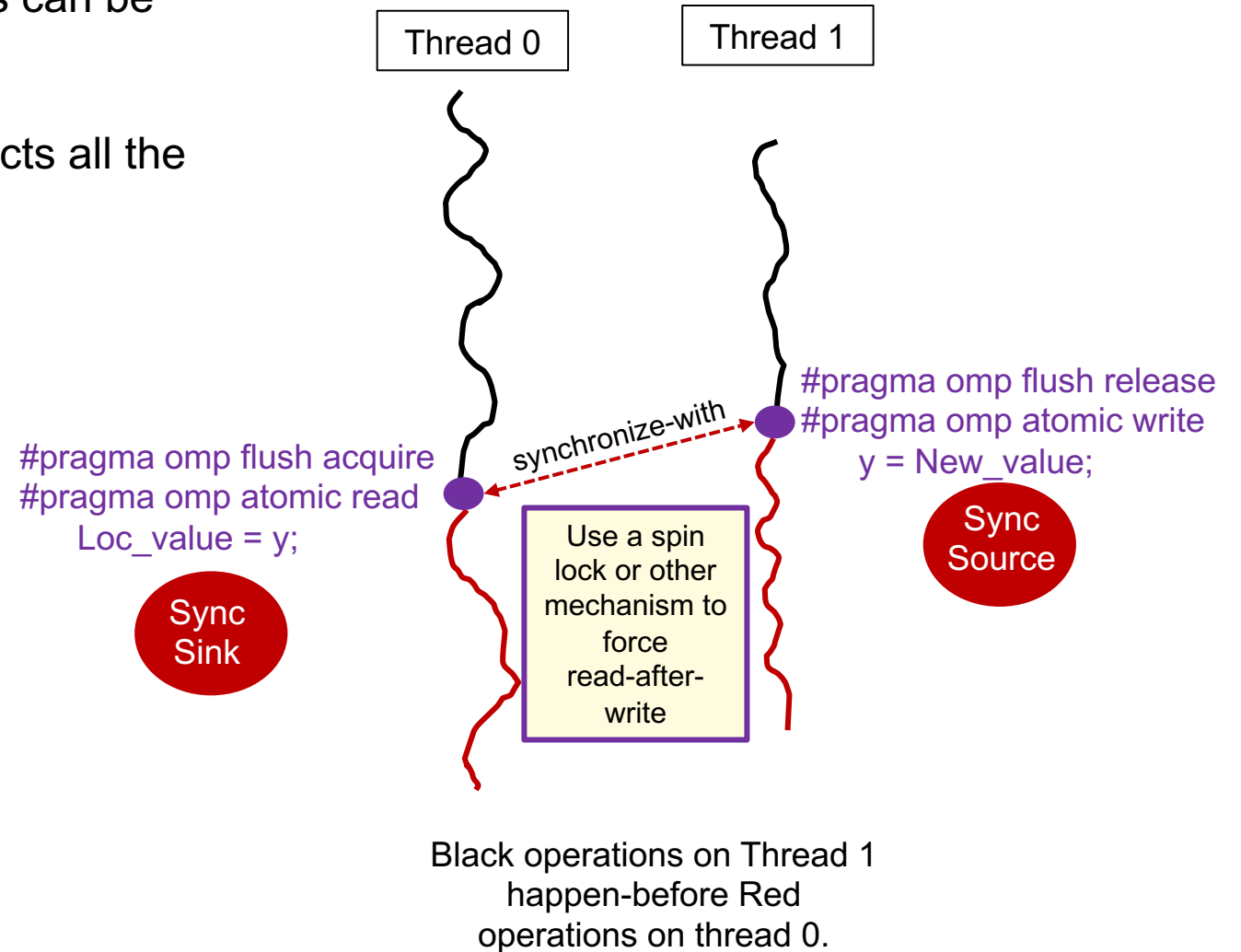
*all memory
operations stay
below the line*

- Acquire: Reads/writes that follow the read-with-acquire cannot happen-before the read-with-acquire operation.

*all memory
operations stay
above the line*

write-release

- Release: Reads/Writes prior to the write-with-release must happen-before the write-with-release.



Memory orders

- Memory orders establish which loads and stores can be moved around synchronized-with relations.
- The strong flush by itself is expensive as it impacts all the shared variables visible to a thread.
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The 2 most fundamental memory orders are:

read-acquire

*all memory
operations stay
below the line*

- Acquire: Reads/writes that follow the read-with-acquire cannot happen-before the read-with-acquire operation.

*all memory
operations stay
above the line*

write-release

- Release: Reads/Writes prior to the write-with-release must happen-before the write-with-release.

`#pragma omp atomic read acquire`
`Loc_value = y;`

`#pragma omp atomic write release`
`y = New_value;`

Sync
Sink

synchronize-with

Use a spin
lock or other
mechanism to
force
read-after-
write

Sync
Source

Black operations on Thread 1
happen-before Red
operations on thread 0.

We can combine the flush and the atomic constructs

producer/consumer program correctly synchronized

```
#include <stdio.h>
#include <omp.h>
#define COUNT 1000000
int main()
{
    int answer = 0, flag= 0,err=0;
    #pragma omp parallel shared(flag,answer) num_threads(2)
    {
        int id = omp_get_thread_num();
        if (id == 0) {
            answer = 42;
            #pragma omp atomic write release
            flag = 1;
        }
        else if (id == 1){
            int fetch = 0;
            while (fetch == 0) {
                #pragma omp atomic read acquire
                fetch = flag;
            }
            if(answer!=42) err++;
        }
    }
    return 0;
}
```

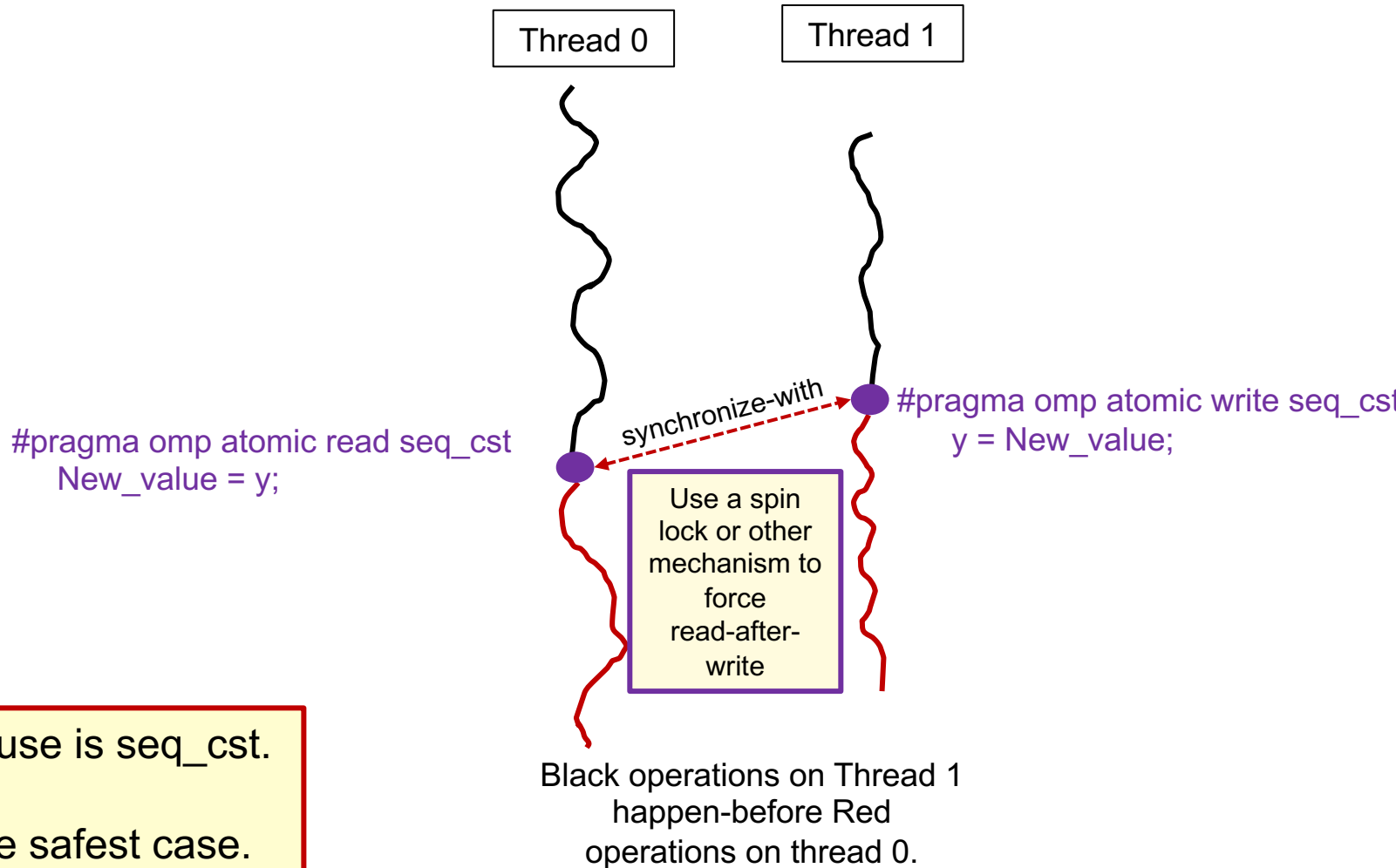
Other Memory orders in OpenMP

- Other OpenMP memory orders

- **acq_rel**: Applies acquire and release memory order constraints at a single point in a program's execution.
- **seq_cst**: sequential consistency. All data accessible to a thread are written to memory, subsequent writes are set to load from memory (akin to the strong flush)

The most important memory order to use is seq_cst.

It can be more expensive, but it is the safest case.



Keep it simple ... let OpenMP take care of Flushes for you

- A flush operation is implied by OpenMP constructs ...

- at entry/exit of parallel regions
- at implicit and explicit barriers
- at entry/exit of critical regions

This has not been a detailed discussion of the full OpenMP memory model. The goal was to explain how memory models work and to understand the subset of features people commonly use.

- OpenMP programs that:

- Do not use non-sequentially consistent atomic constructs;
- Do not rely on the accuracy of a false result from `omp_test_lock` and `omp_test_nest_lock`; and
- Correctly avoid data races

... behave as though operations on shared variables were simply interleaved in an order consistent with the order in which they are performed by each thread. The relaxed consistency model is invisible for such programs, and any explicit flushes in such programs are redundant.

WARNING:

If you find yourself wanting to write code with explicit flushes, stop and get help. It is very difficult to manage flushes on your own. Even experts often get them wrong.

This is why we defined OpenMP constructs to automatically apply flushes most places where you really need them.

Exercise

- In the file hist.c, we provide a program that generates a large array of random numbers and then generates a histogram of values.
- This is a "quick and informal" way to test a random number generator ... if all goes well the bins of the histogram should be the same size.
- Parallelize the filling of the histogram. You must assure that your program is race free and gets the same result as the sequential program.
- Using everything we've covered today, **manage updates to shared data in multiple ways**. Try to minimize the time to generate the histogram.
- Time ONLY the assignment to the histogram. Can you beat the sequential time?

```
#define      num_trials    1000000 // number of x values
#define      num_buckets   50      // number of buckets in hitogram
static long xlow           = 0.0;   // low end of x range
static long xhi            = 100.0; // High end of x range

int main (){
    double x[num_trials]; // array used to assign counters in the histogram
    long   hist[num_buckets]; // the histogram
    double bucket_width;    // the width of each bucket in the histogram
    double time;

    seed(xlow, xhi); // seed the random number generator over range of x
    bucket_width = (xhi-xlow)/(double)num_buckets;

    // fill the array. << code not shown >>

    // initialize the histogram << code not shown >>

    // Assign x values to the right histogram bucket
    time = omp_get_wtime();
    for(int i=0;i<num_trials;i++){

        long ival = (long) (x[i] - xlow)/bucket_width;

        hist[ival]++;

    }

    time = omp_get_wtime() - time;

    // compute statistics and output results << code not shown >>
    return 0;
}
```

Only focus
on this part of
the program

Histogram Program: Critical section

- A critical section means that only one thread at a time can update a histogram bin ... but this effectively serializes the loops and adds huge overhead as the runtime manages all the threads waiting for their turn for the update.

```
#pragma omp parallel for  
for(i=0;i<NVALS;i++){  
    ival = (int) x[i];  
    #pragma omp critical  
    hist[ival]++;  
}
```

Easy to write and
correct, but terrible
performance



Histogram with locks: mutual exclusion but more control

- Example: conflicts are rare, but to play it safe, we must assure mutual exclusion for updates to histogram elements.

```
#pragma omp parallel for
for(i=0;i<NBUCKETS; i++){
    omp_init_lock(&hist_locks[i]);    hist[i] = 0;
}
#pragma omp parallel for
for(i=0;i<NVALS;i++){
    ival = (int) x[i];
    omp_set_lock(&hist_locks[ival]);
    hist[ival]++;
    omp_unset_lock(&hist_locks[ival]);
}
```

One lock per element of hist

Enforce mutual exclusion on update to hist array

A lock implies a memory fence (a “flush”) of all thread visible variables

```
#pragma omp parallel for
for(i=0;i<NBUCKETS; i++)
    omp_destroy_lock(&hist_locks[i]);
```

Free-up storage when done.

Locks with hints were added in OpenMP 4.5 to suggest a lock strategy based on intended use (e.g. contended, uncontended, speculative, unspeculative)

Histogram program: reduction with an array

- We can give each thread a copy of the histogram, they can fill them in parallel, and then combine them when done

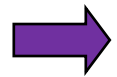
```
#pragma omp parallel for reduction(+:hist[0:Nbins])  
for(i=0;i<NVALS;i++){  
    ival = (int) x[i];  
    hist[ival]++;  
}
```

Easy to write and correct, Uses a lot of memory on the stack, but its fast ... sometimes faster than the serial method.

sequential	0.0019 secs
critical	0.079 secs
Locks per bin	0.029 secs
Reduction, replicated histogram array	0.00097 secs

1000000 random values in X sorted into 50 bins. Four threads on a dual core Apple laptop (Macbook air ... 2.2 Ghz Intel Core i7 with 8 GB memory) and the gcc version 9.1. Times are for the above loop only (we do not time set-up for locks, destruction of locks or anything else)

- Introduction to OpenMP
- Creating Threads
- Synchronization ... simple collective sync
- Parallel Loops
- Data Environment
- Synchronization ... the general case (including Memory Models)



- Irregular Parallelism and Tasks
 - OpenMP for real hardware (NUMA)
 - Recap

Irregular Parallelism

- Let's call a problem “irregular” when one or both of the following hold:
 - Data Structures are sparse or involve indirect memory references
 - Control structures are not basic for-loops
- Example: Traversing Linked lists:

```
p = listhead ;  
while (p) {  
    process(p) ;  
    p=p->next ;  
}
```

- Using what we've learned so far, traversing a linked list in parallel using OpenMP is difficult.

Exercise: Traversing linked lists

- Consider the program linked.c
 - Traverses a linked list computing a sequence of Fibonacci numbers at each node.
- Parallelize this program selecting from the following list of constructs:

```
#pragma omp parallel
#pragma omp for
#pragma omp parallel for
#pragma omp for reduction(op:list)
#pragma omp critical
int omp_get_num_threads();
int omp_get_thread_num();
double omp_get_wtime();
schedule(static[,chunk]) or schedule(dynamic[,chunk])
private(), firstprivate(), default(none)
```

- Hint: Just worry about the while loop that is timed inside main(). You don't need to make any changes to the “list functions”

Linked Lists with OpenMP: My solution

- See the file solutions/linked_notasks.c

```
while (p != NULL) {  
    p = p->next;  
    count++;  
}  
struct node *parr = (struct node*) malloc(count*sizeof(struct node));  
p = head;  
for(i=0; i<count; i++) {  
    parr[i] = p;  
    p = p->next;  
}  
#pragma omp parallel  
{  
    #pragma omp for schedule(static,1)  
    for(i=0; i<count; i++)  
        processwork(parr[i]);  
}
```

Count number of items in the linked list

Copy pointer to each node into an array

Process nodes in parallel with a for loop

Linked Lists with OpenMP (without tasks)

- See the file solutions/linked_notasks.c

```
while (p != NULL) {
    p = p->next;
    count++;
}
struct node *parr = (struct node*) malloc(count*sizeof(struct node));
p = head;
for(i=0; i<count; i++) {
    parr[i] = p;
    p = p->next;
}
#pragma omp parallel
{
    #pragma omp for schedule(static,1)
    for(i=0; i<count; i++)
        processwork(parr[i]);
}
```

Count number of items in the linked list

Copy pointer to each node into an array

Process nodes in parallel with a for loop

Number of threads	Schedule	
	Default	Static,1
1	48 seconds	45 seconds
2	39 seconds	28 seconds

With so much code to add and three passes through the data, this is really ugly.

There has got to be a better way to do this

Solutions from other people

- A much more elegant solution based on the SPMD pattern

```
#pragma omp parallel firstprivate(p)
{
    int id = omp_get_thread_num();
    int nthreads = omp_get_num_threads();
    int count = 0;
    while(p!=NULL && count < id){
        count ++;
        p = p->next;
    }
    while (p != NULL) {
        processwork(p);
        int count = 0;
        while(p!=NULL && count < nthreads){
            count ++;
            p = p->next;
        }
    }
}
```

Each thread has its own copy of the list pointer set to the head of the list

Each thread advances its list pointer by a number equal to the thread ID

Increment the thread's list pointer by the number of threads

Solutions from other people

- A particularly elegant, SPMD solution

```
#pragma omp parallel firstprivate(p)
{
    int i = 0;
    int thread_num = omp_get_thread_num();
    int num_threads = omp_get_num_threads();
    while (p != NULL)
    {
        if(i % num_threads == thread_num)
            processwork(p);
        p = p->next;
        ++i;
    }
}
```

Each thread has its own copy of the list pointer set to the head of the list

Each of the threads traverses the list, but only one thread processes the work for any give node using the modulus of the count with the number of threads

Linked Lists with OpenMP

```
while (p != NULL) {
    p = p->next;
    count++;
}
struct node *parr = (struct node*) malloc(count*sizeof(struct
node));
p = head;
for(i=0; i<count; i++) {
    parr[i] = p;
    p = p->next;
}
#pragma omp parallel
{
    #pragma omp for schedule(static,1)
    for(i=0; i<count; i++)
        processwork(parr[i]);
}
```

```
#pragma omp parallel firstprivate(p) default(none)
{
    int i = 0;
    int thread_num = omp_get_thread_num();
    int num_threads = omp_get_num_threads();
    while (p != NULL)
    {
        if(i % num_threads == thread_num)
            processwork(p);
        p = p->next;
        ++i;
    }
}
```

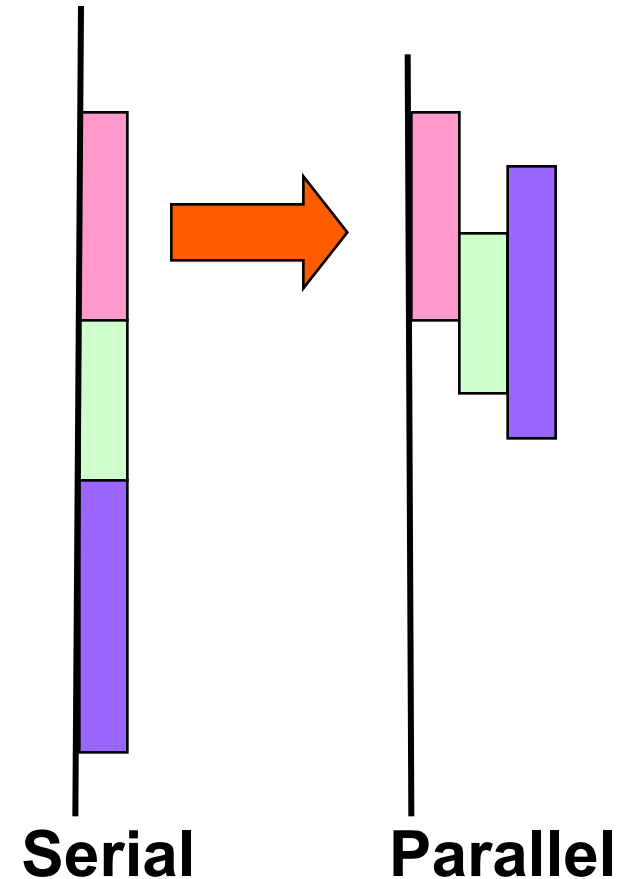
```
#pragma omp parallel firstprivate(p)
{
    int id = omp_get_thread_num();
    int nthreads = omp_get_num_threads();
    int count = 0;
    while(p!=NULL && count < id){
        count ++;
        p = p->next;
    }
    while (p != NULL) {
        processwork(p);
        int count = 0;
        while(p!=NULL && count < nthreads){
            count ++;
            p = p->next;
        }
    }
}
```

With so much code to add and multiple passes through the data, all of these approaches are kind of ugly.

There has got to be a better way to handle irregular problems

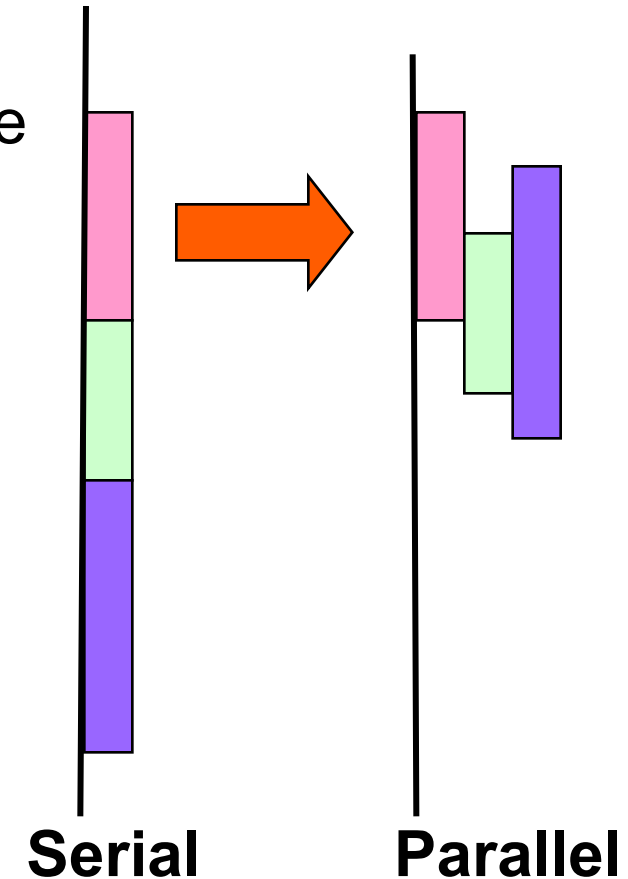
What are Tasks?

- Tasks are independent units of work
- Tasks are composed of:
 - code to execute
 - data to compute with
- Threads are assigned to perform the work of each task.
 - The thread that encounters the task construct may execute the task immediately.
 - The threads may defer execution until later



What are Tasks?

- The task construct includes a structured block of code
- Inside a parallel region, a thread encountering a task construct will package up the code block and its data for execution
- Tasks can be nested: i.e., a task may itself generate tasks.



A common Pattern is to have one thread create the tasks while the other threads wait at a barrier and execute the tasks

Single Worksharing Construct

- The **single** construct denotes a block of code that is executed by only one thread (not necessarily the primary* thread).
- A barrier is implied at the end of the single block (can remove the barrier with a *nowait* clause).

```
#pragma omp parallel
{
    do_many_things();
    #pragma omp single
    {   exchange_boundaries();   }
    do_many_other_things();
}
```

*This used to be called the “master thread”. The term “master” has been deprecated in OpenMP 5.1 and replaced with the term “primary”.

Task Directive

`#pragma omp task [clauses]`
structured-block

```
#pragma omp parallel
```

Create some threads

```
{
```

```
    #pragma omp single
```

One Thread
packages tasks

```
{
```

```
    #pragma omp task  
        fred();
```

```
    #pragma omp task  
        daisy();
```

```
    #pragma omp task  
        billy();
```

Tasks executed by
some thread in some
order

```
}
```

All tasks complete before this barrier is released

```
}
```

Exercise: Simple tasks

- Write a program using tasks that will “randomly” generate one of two strings:
 - “I think “ “race” “car” “s are fun”
 - “I think “ “car” “race” “s are fun”
- Hint: use tasks to print the indeterminate part of the output (i.e. the “race” or “car” parts).
- This is called a “Race Condition”. It occurs when the result of a program depends on how the OS schedules the threads.
- NOTE: A “data race” is when threads “race to update a shared variable”. They produce race conditions. Programs containing data races are undefined (in OpenMP but also ANSI standards C++’11 and beyond).

`#pragma omp parallel`

`#pragma omp task`

`#pragma omp single`

Racey Cars: Solution

```
#include <stdio.h>
#include <omp.h>
int main()
{ printf("I think");
  #pragma omp parallel
  {
    #pragma omp single
    {
      #pragma omp task
      printf(" car");
      #pragma omp task
      printf(" race");
    }
  }
  printf("s");
  printf(" are fun!\n");
}
```

Data Scoping with Tasks

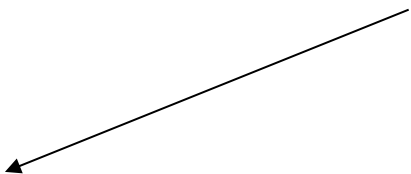
- Variables can be shared, private or firstprivate with respect to task
- These concepts are a little bit different compared with threads:
 - If a variable is **shared** on a task construct, the references to it inside the construct are to the storage with that name at the point where the task was encountered
 - If a variable is **private** on a task construct, the references to it inside the construct are to new uninitialized storage that is created when the task is executed
 - If a variable is **firstprivate** on a construct, the references to it inside the construct are to new storage that is created and initialized with the value of the existing storage of that name when the task is encountered

Data Scoping Defaults

- The behavior you want for tasks is usually firstprivate, because the task may not be executed until later (and variables may have gone out of scope)
 - Variables that are private when the task construct is encountered are firstprivate by default
- Variables that are shared in all constructs starting from the innermost enclosing parallel construct are shared by default

```
#pragma omp parallel shared(A) private(B)
{
    ...
    #pragma omp task
    {
        int C;
        compute(A, B, C);
    }
}
```

A is shared
B is firstprivate
C is private



Exercise: Traversing linked lists

- Consider the program linked.c
 - Traverses a linked list computing a sequence of Fibonacci numbers at each node.
- Parallelize this program selecting from the following list of constructs:

```
#pragma omp parallel
#pragma omp single
#pragma omp task
int omp_get_num_threads();
int omp_get_thread_num();
double omp_get_wtime();
private(), firstprivate()
```

- Hint: Just worry about the contents of main(). You don't need to make any changes to the “list functions”

Parallel Linked List Traversal

```
#pragma omp parallel
{
    #pragma omp single
    {
        p = listhead ;
        while (p) {
            #pragma omp task firstprivate(p)
            {
                process (p) ;
            }
            p=next (p) ;
        }
    }
}
```

Only one thread
packages tasks

makes a copy of **p**
when the task is
packaged

When/Where are Tasks Complete?

- At thread barriers (explicit or implicit)
 - all tasks generated inside a region must complete at the next barrier encountered by the threads in that region. Common examples:
 - **Tasks generated inside a single construct:** all tasks complete before exiting the barrier on the single.
 - **Tasks generated inside a parallel region:** all tasks complete before exiting the barrier at the end of the parallel region.
- At taskwait directive
 - i.e. Wait until all tasks defined in the current task have completed.
`#pragma omp taskwait`
 - Note: applies only to tasks generated in the current task, not to “descendants” .

Example

```
#pragma omp parallel
{
    #pragma omp single
    {
        #pragma omp task
        fred();
        #pragma omp task
        daisy();
        #pragma omp taskwait
        #pragma omp task
        billy();
    }
}
```

fred() and **daisy()** must complete before **billy()** starts, but this does not include tasks created inside **fred()** and **daisy()**

All tasks including those created inside **fred()** and **daisy()** must complete before exiting this barrier

Example

```
#pragma omp parallel
{
    #pragma omp single nowait
    {
        #pragma omp task
        fred();
        #pragma omp task
        daisy();
        #pragma omp taskwait
        #pragma omp task
        billy();
    }
}
```

The barrier at the end of the single is expensive and not needed since you get the barrier at the end of the parallel region. So use **nowait** to turn it off.

All tasks including those created inside **fred()** and **daisy()** must complete before exiting this barrier

Example: Fibonacci numbers

```
int fib (int n)
{
    int x,y;
    if (n < 2) return n;

    x = fib(n-1);
    y = fib (n-2);
    return (x+y);
}
```

```
int main()
{
    int NW = 5000;
    fib(NW);
}
```

- $F_n = F_{n-1} + F_{n-2}$
- Inefficient $O(2^n)$ recursive implementation!

Parallel Fibonacci

```
int fib (int n)
{  int x,y;
   if (n < 2) return n;
```

```
  #pragma omp task shared(x)
```

```
    x = fib(n-1);
```

```
  #pragma omp task shared(y)
```

```
    y = fib (n-2);
```

```
  #pragma omp taskwait
```

```
    return (x+y);
```

```
}
```

```
Int main()
```

```
{  int NW = 5000;
```

```
  #pragma omp parallel
```

```
  {
```

```
    #pragma omp single
```

```
      fib(NW);
```

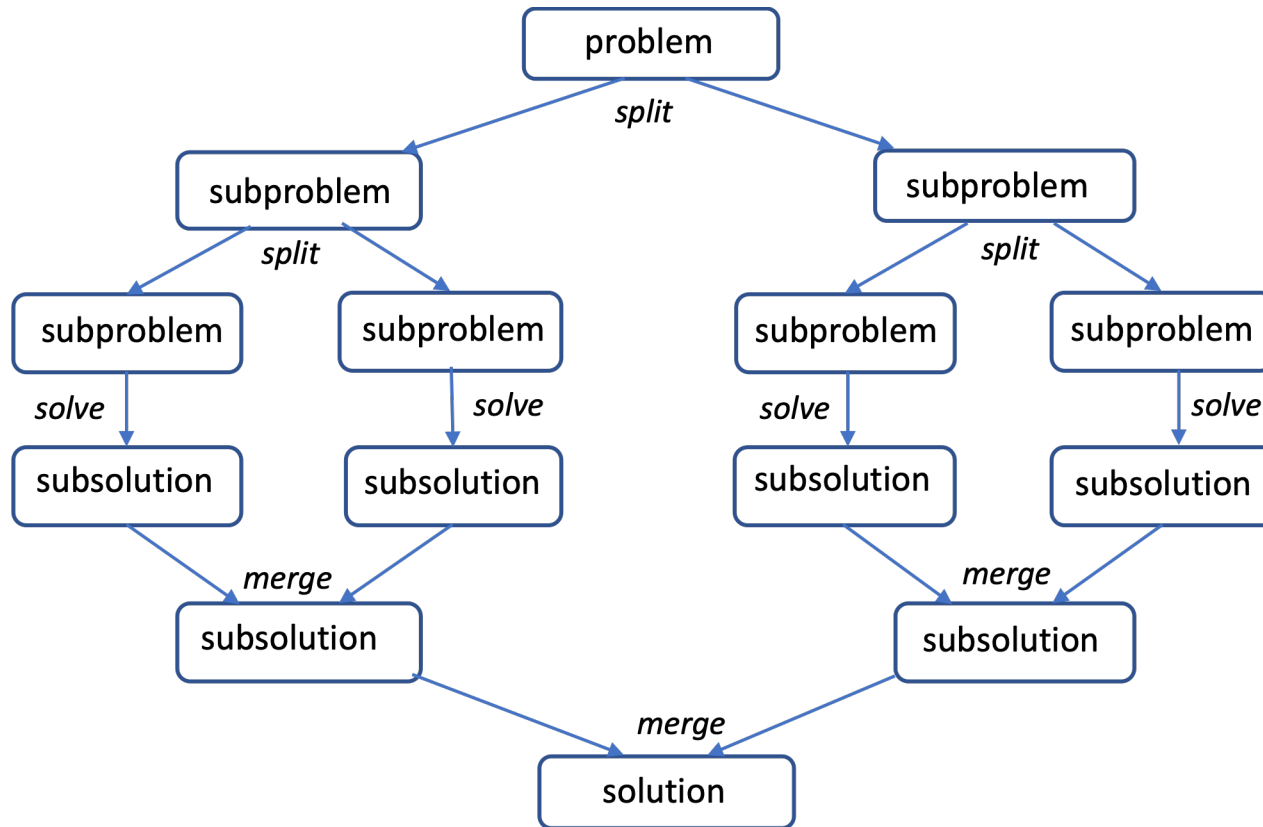
```
  }
```

```
}
```

- Binary tree of tasks
- Traversed using a recursive function
- A task cannot complete until all tasks below it in the tree are complete (enforced with taskwait)
- **x, y** are local, and so by default they are private to current task
 - must be shared on child tasks so they don't create their own firstprivate copies at this level!

Divide and Conquer

- Split the problem into smaller sub-problems; continue until the sub-problems can be solved directly



- 3 Options for parallelism:
 - Do work as you split into sub-problems
 - Do work only at the leaves
 - Do work as you recombine

Exercise: PI with tasks

- Go back to the original pi.c program
 - Parallelize this program using OpenMP tasks

```
#pragma omp parallel
#pragma omp task
#pragma omp taskwait
#pragma omp single
double omp_get_wtime()
int omp_get_thread_num();
int omp_get_num_threads();
```

- Hint: first create a recursive pi program and verify that it works. Think about the computation you want to do at the leaves. If you go all the way down to one iteration per leaf-node, won't you just swamp the system with tasks?

Program: OpenMP tasks

```
include <omp.h>
static long num_steps = 100000000;
#define MIN_BLK 10000000
double pi_comp(int Nstart,int Nfinish,double step)
{  int i,iblk;
   double x, sum = 0.0,sum1, sum2;
   if (Nfinish-Nstart < MIN_BLK){
       for (i=Nstart;i< Nfinish; i++){
           x = (i+0.5)*step;
           sum = sum + 4.0/(1.0+x*x);
       }
   }
   else{
       iblk = Nfinish-Nstart;
       #pragma omp task shared(sum1)
       sum1 = pi_comp(Nstart,      Nfinish-iblk/2,step);
       #pragma omp task shared(sum2)
       sum2 = pi_comp(Nfinish-iblk/2, Nfinish,      step);
       #pragma omp taskwait
       sum = sum1 + sum2;
   }return sum;
}
```

```
int main ()
{
   int i;
   double step, pi, sum;
   step = 1.0/(double) num_steps;
   #pragma omp parallel
   {
       #pragma omp single
       sum =
           pi_comp(0,num_steps,step);
   }
   pi = step * sum;
}
```

Results*: Pi with tasks

threads	1 st SPMD	SPMD critical	PI Loop	Pi tasks
1	1.86	1.87	1.91	1.87
2	1.03	1.00	1.02	1.00
3	1.08	0.68	0.80	0.76
4	0.97	0.53	0.68	0.52

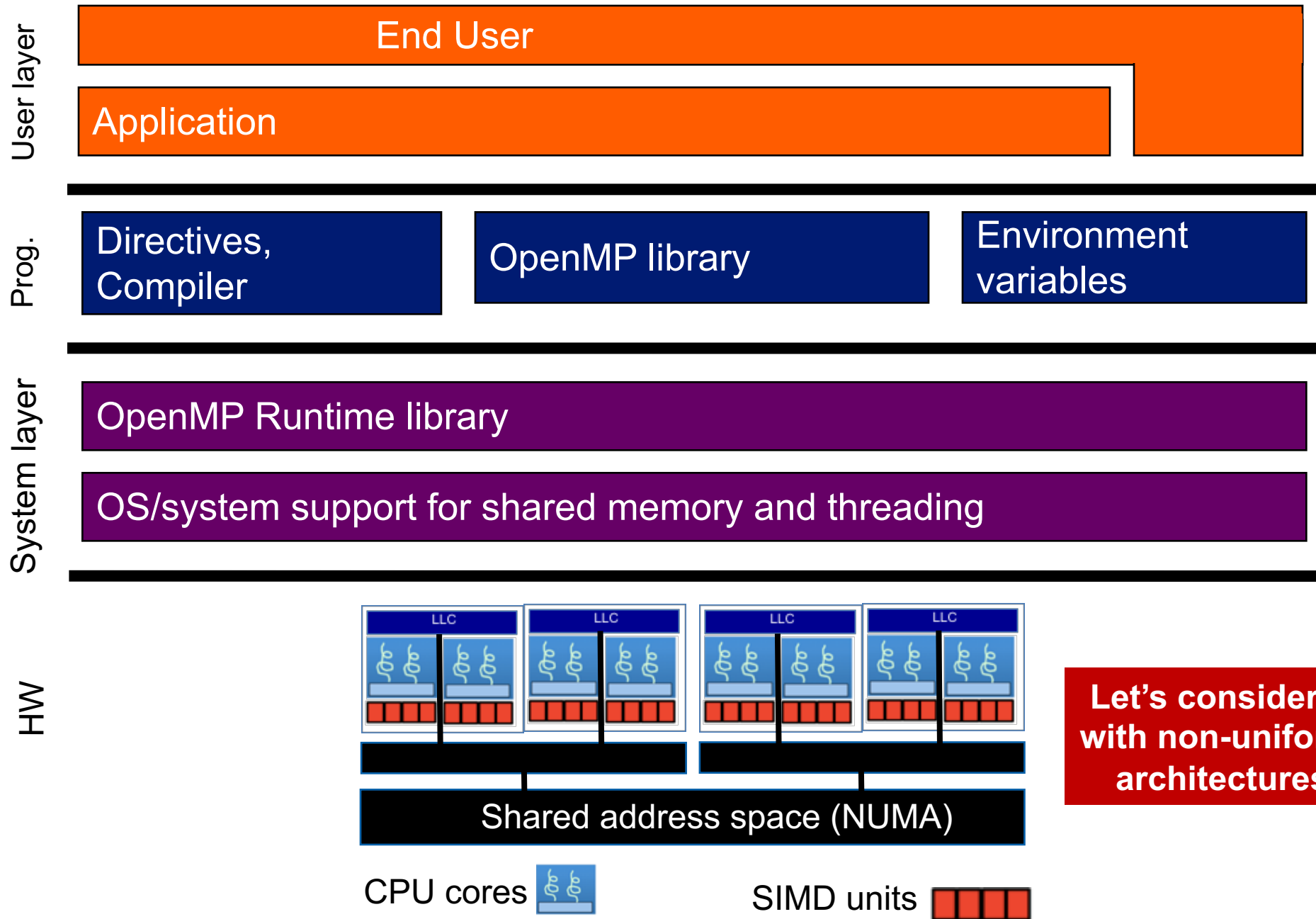
*Intel compiler (icpc) with no optimization on Apple OS X 10.7.3 with a dual core (four HW thread) Intel® Core™ i5 processor at 1.7 Ghz and 4 Gbyte DDR3 memory at 1.333 Ghz.

Using Tasks

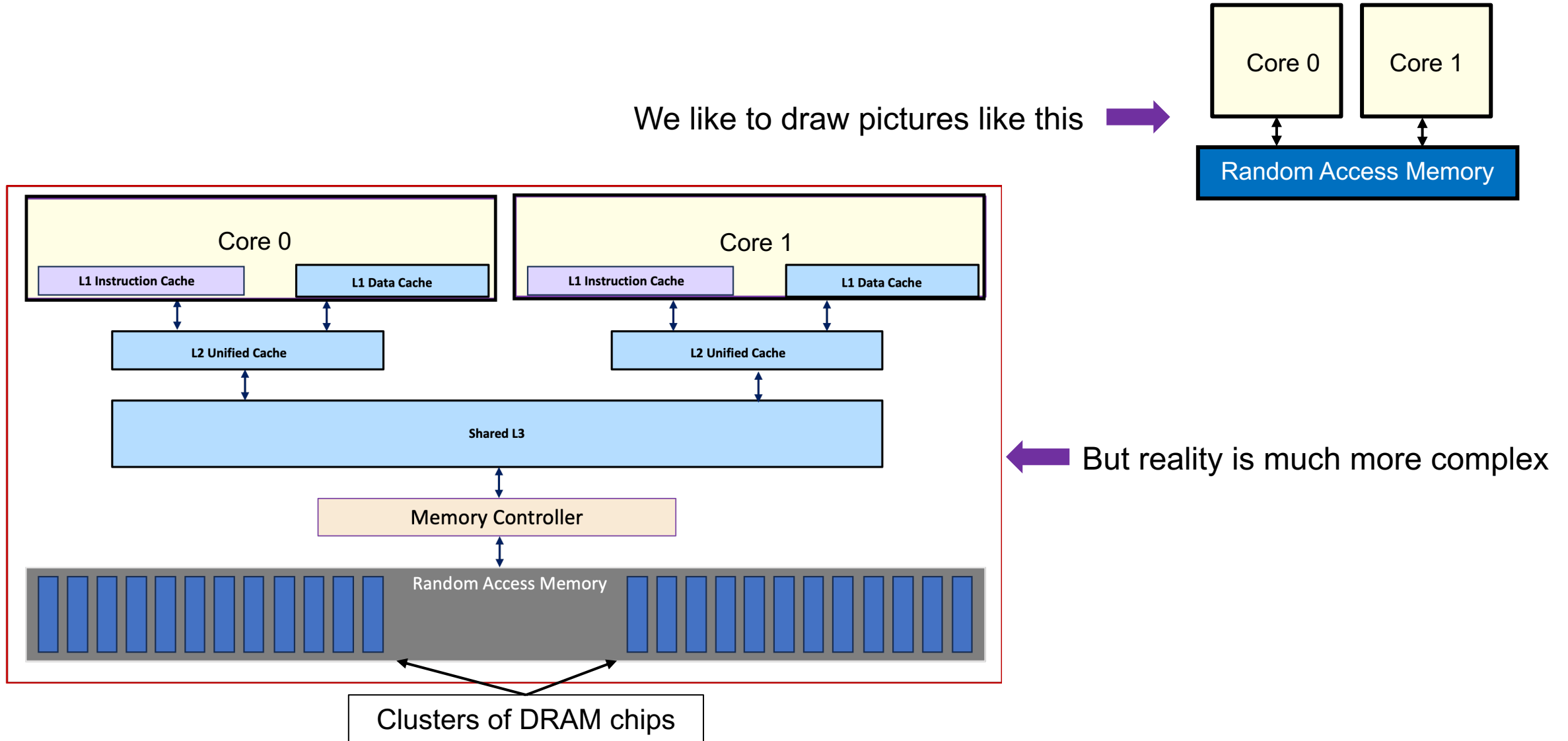
- Don't use tasks for things already well supported by OpenMP
 - e.g. standard do/for loops
 - the overhead of using tasks is greater
- Don't expect miracles from the runtime
 - best results usually obtained where the user controls the number and granularity of tasks

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- Synchronization ... the general case (including Memory Models)
- Irregular Parallelism and Tasks
- ➡ • OpenMP for real hardware (NUMA)
- Recap

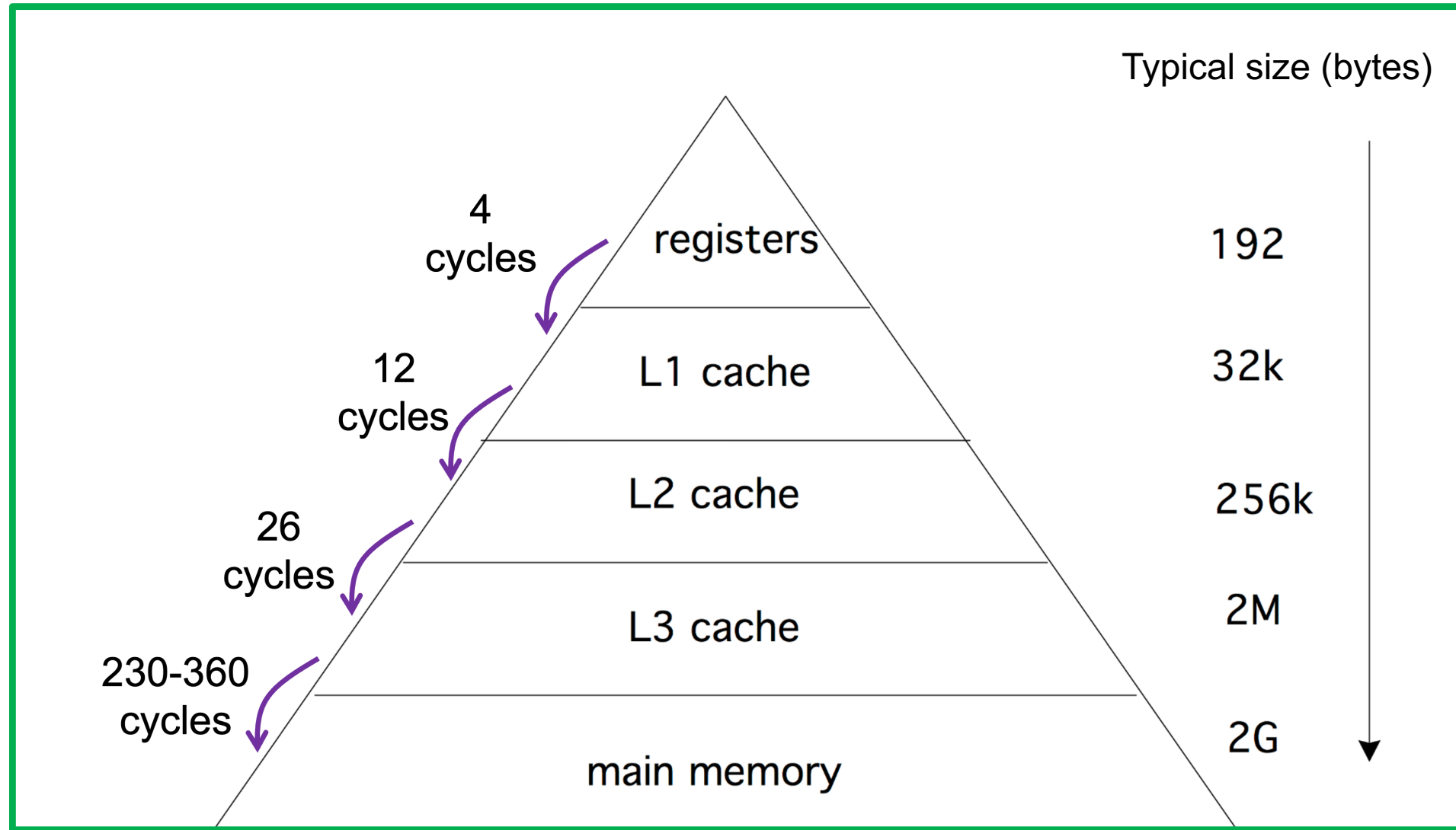
OpenMP Basic Definitions: Basic Solution Stack



SMP is a lie ... The memory is Non-Uniform

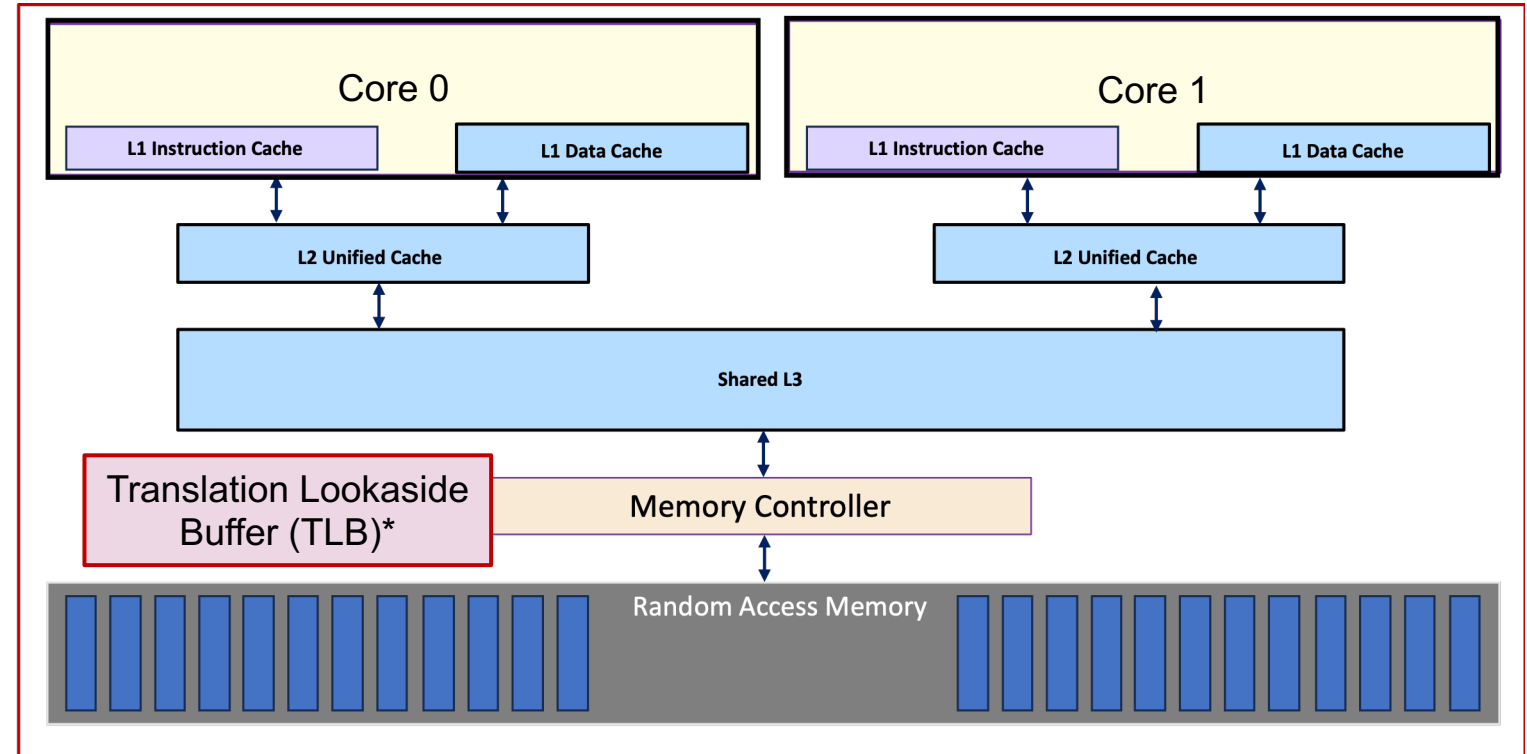


Latencies across the Memory Hierarchy



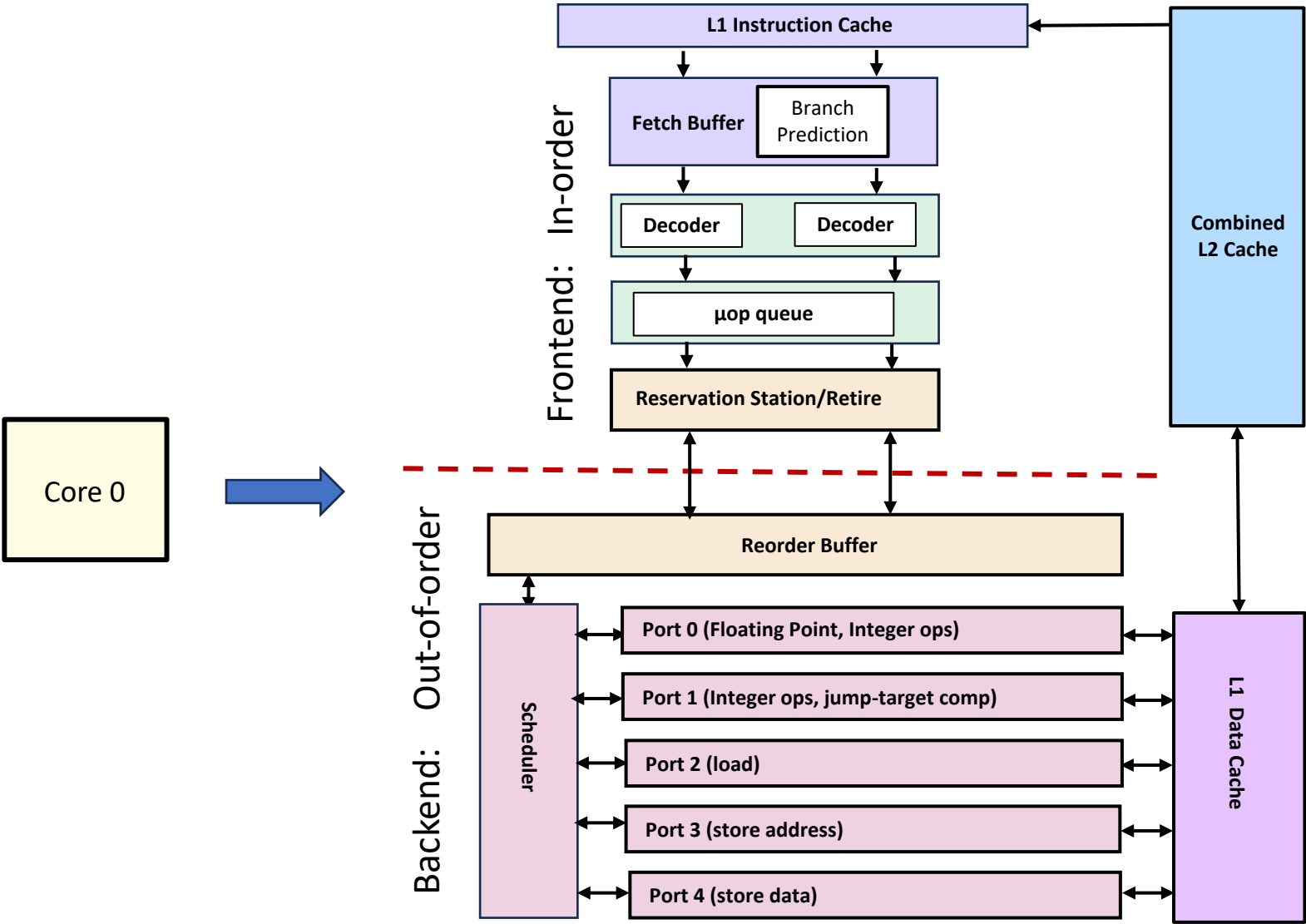
Memory Hierarchies are even more complex: the TLB

- A 64-bit address space can address 16 exabytes of memory. Clearly, that exceeds even the largest RAMs.
- A systems maps the full address space (virtual memory) onto **pages** of physical memory.
- The **TLB (Translation Lookaside Buffer)** implements virtual memory and caches the mapping from virtual addresses to physical addresses
 - If the entry is in the TLB page table, a small overhead of reading the entry and computed the physical address is incurred.
 - If the entry is not in the page table, a page fault exception will be raised. Requires expensive OS operations and stalls the CPU.



*Note: we show just one simple configuration of the TLB where it is associated with the memory controller. The TLB can be hierarchical with separate TLB tied to the cache hierarchy and different TLBs for data and instructions

Once you get past Memory Hierarchy issues, consider the complexity of the cores



Frontend processes instructions for execution.

Maps them into micro-operations (μ op) and manages them (and the register file) so they retire “in order”

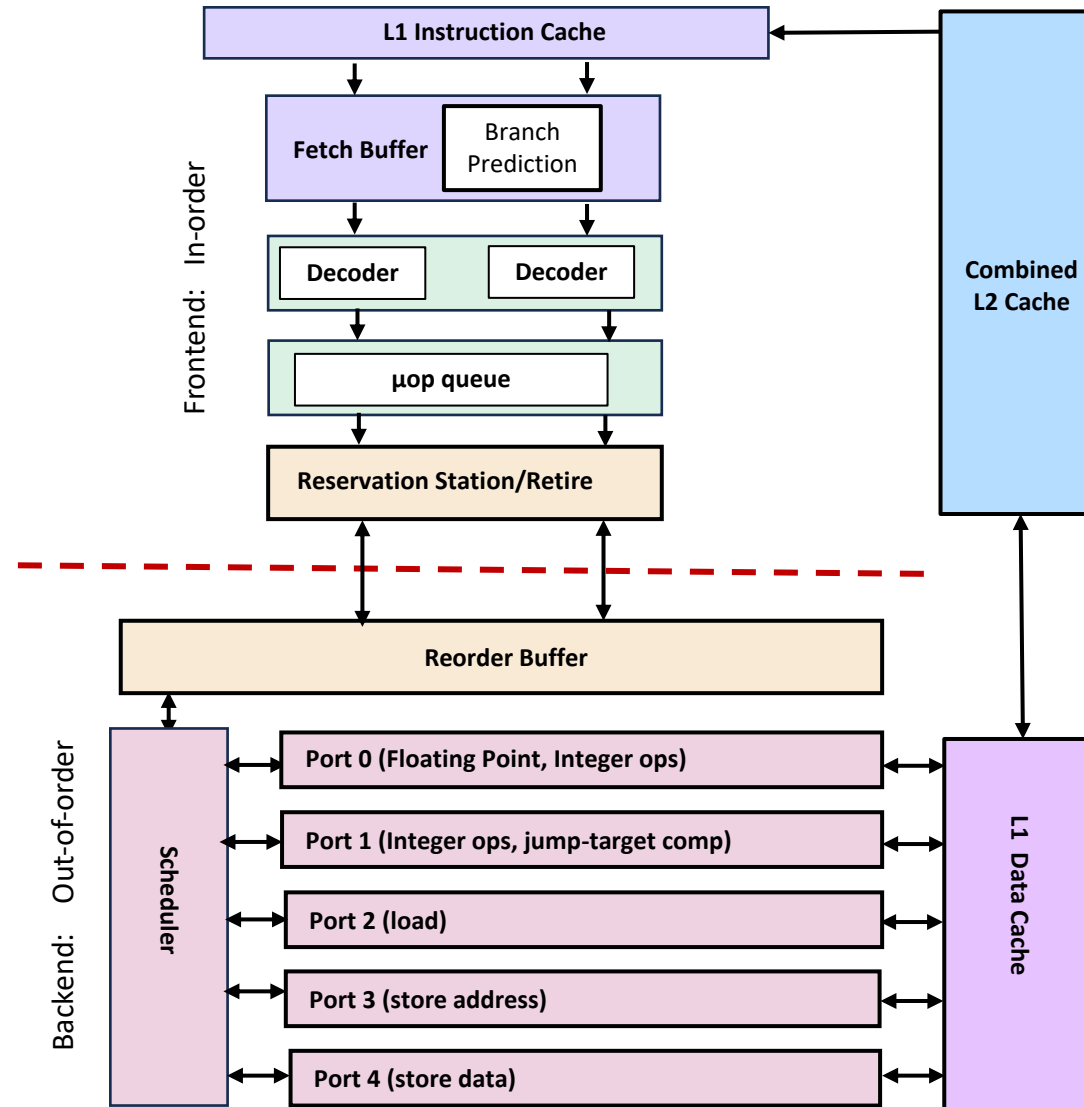
Backend executes decoded instructions.

Instructions are pipelined and include multiple ports and SIMD execution units.

Execution is “out of order” to give the scheduler maximum flexibility to maximize instruction-level parallelism

And its even worse

A single frontend typically can't generate enough decoded instructions to keep all the resource in the backend busy.

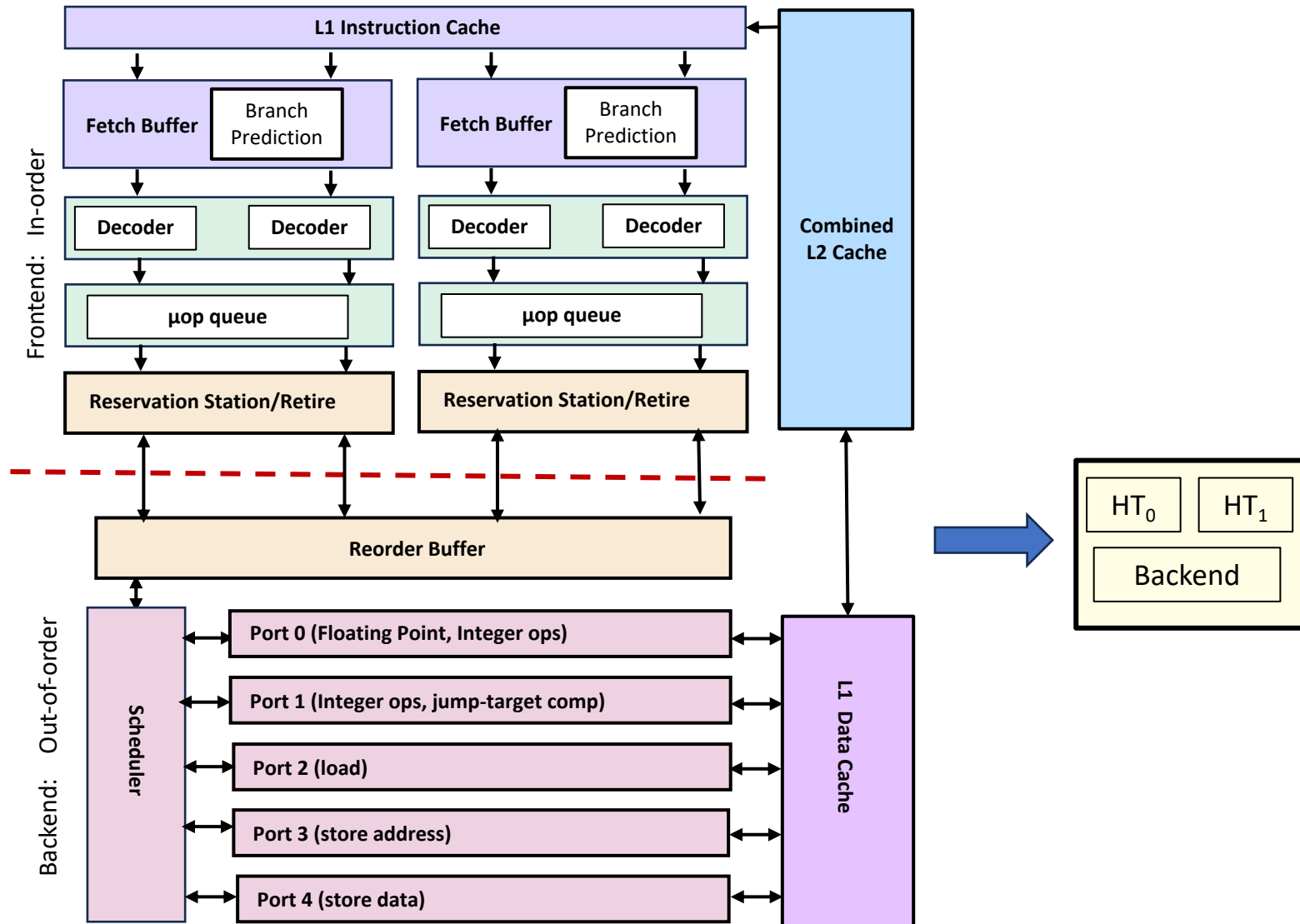


And its even worse

So, we have multiple frontends (in this case, 2) feeding a single back end.

These appear to the OS as an additional core ... or a virtual core.

They maintain state of a thread in hardware so these are often called **Hardware threads**.



HT: hardware thread

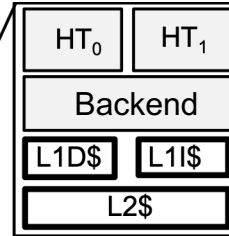
And then we put all this into a real two-socket node

2 Intel® Xeon™ E5-2698 v3 CPUs (Haswell) per node (launched Q3'14)

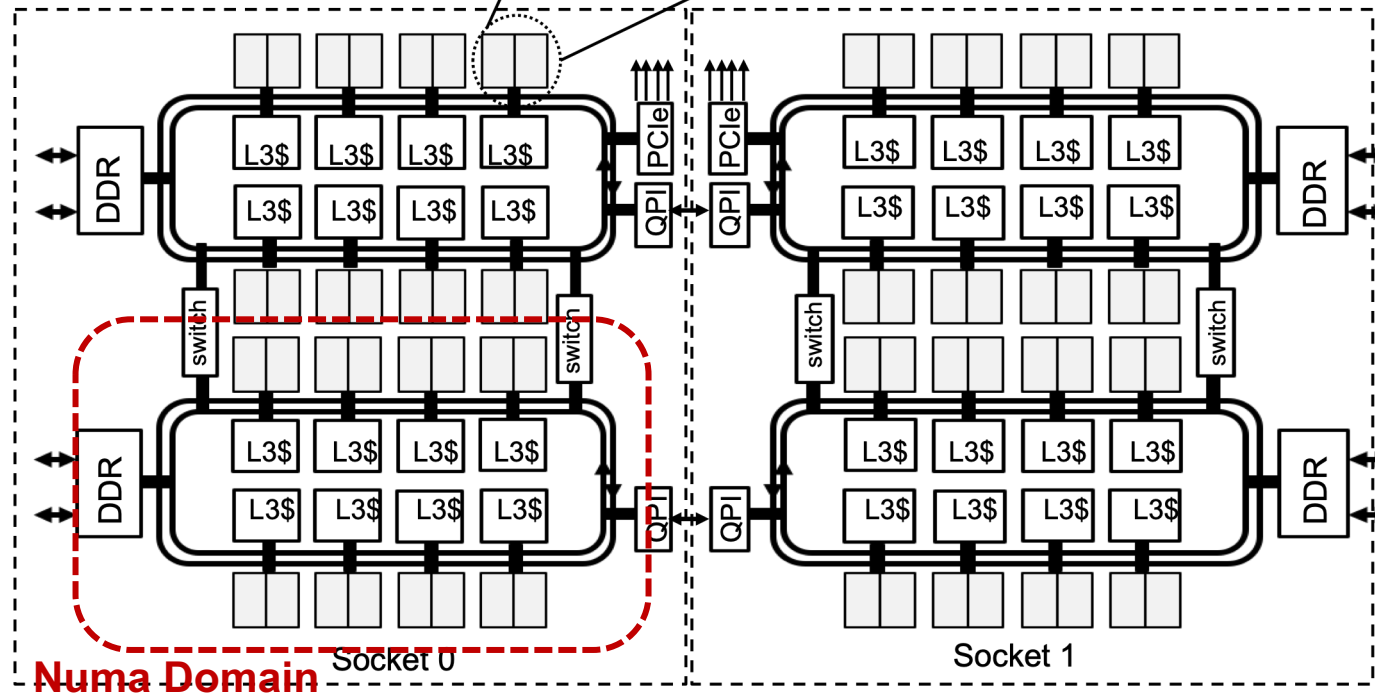
4 blocks of 8 core units connected by an on-chip-network with a DDR memory controller.

Each block is a NUMA domain since memory access from a core to its “own” DDR is less expensive.

2 Hardware threads (HT) per core
Intel® AVX2 (256 bit Vector unit)
L1\$ instruction and data: 32 KB
Unified L2\$ 256 KB



40 MB
shared L3\$



DDR: Double Data Rate
memory controller

PCIe is the connection from
the CPU to other devices in a
node.

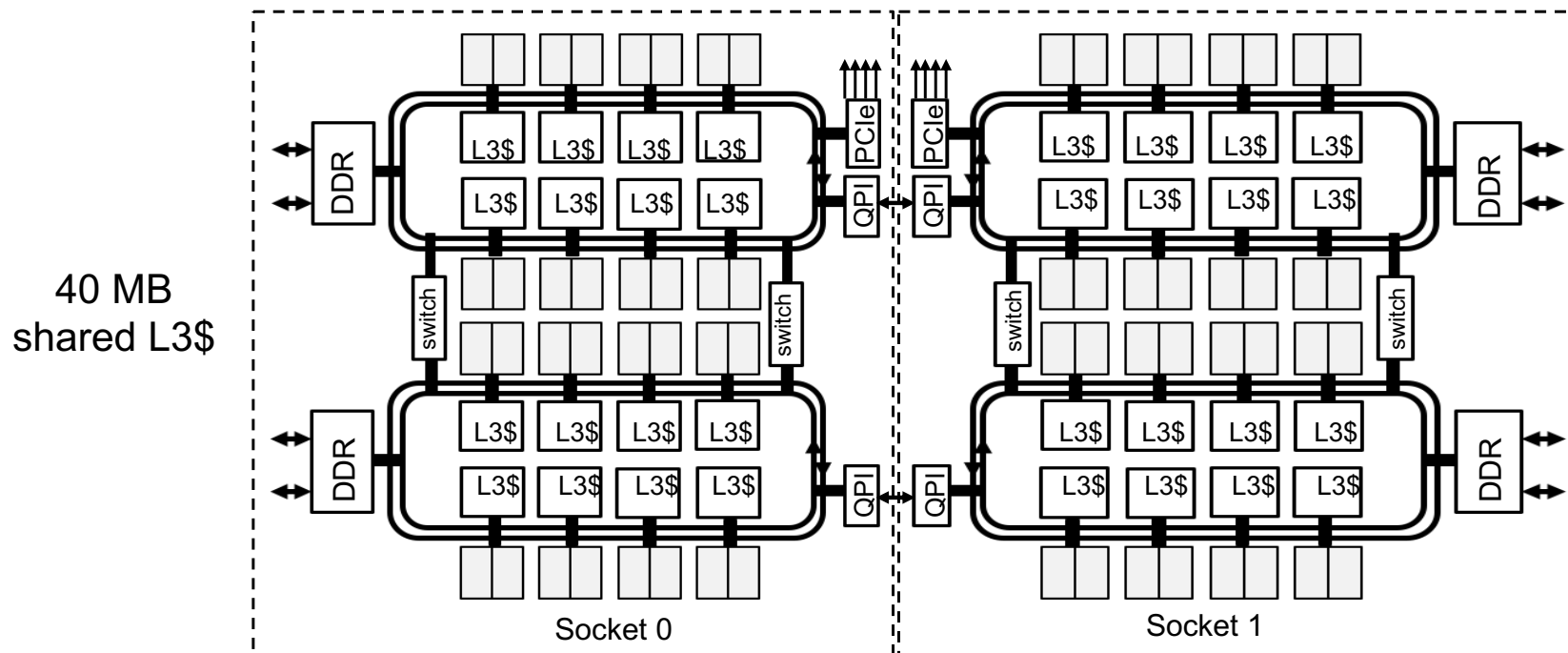
QPI: Quick Path Interconnect.
A coherent interconnect
between CPUs. Makes it
easy to build multi-CPU
nodes.

As configured for Cori at NERSC: CPUs at 2.3 GHz, 2 16 GB DIMMs per DDR memory controller, 16 cores per CPU. 2 CPUs connected by a high-speed interconnect (QPI)

And then we put all this into a real two-socket node

2 Intel® Xeon™ E5-2698 v3 CPUs (Haswell) per node (launched Q3'14)

- A floating-point operation takes $O(\sim 1 \text{ ns})$.
 - L1 Cache $\sim 1.5 \text{ ns}$
 - L2 Cache reference $\sim 5 \text{ ns}$
 - L3 Cache reference $\sim 25 \text{ ns}$
 - Near memory DRAM access $\sim 100 \text{ ns}$
 - Far memory DRAM access $\sim 200 \text{ ns}$
- **The key to performance is to minimize memory movement:** get the memory movement right and the “rest” is easy

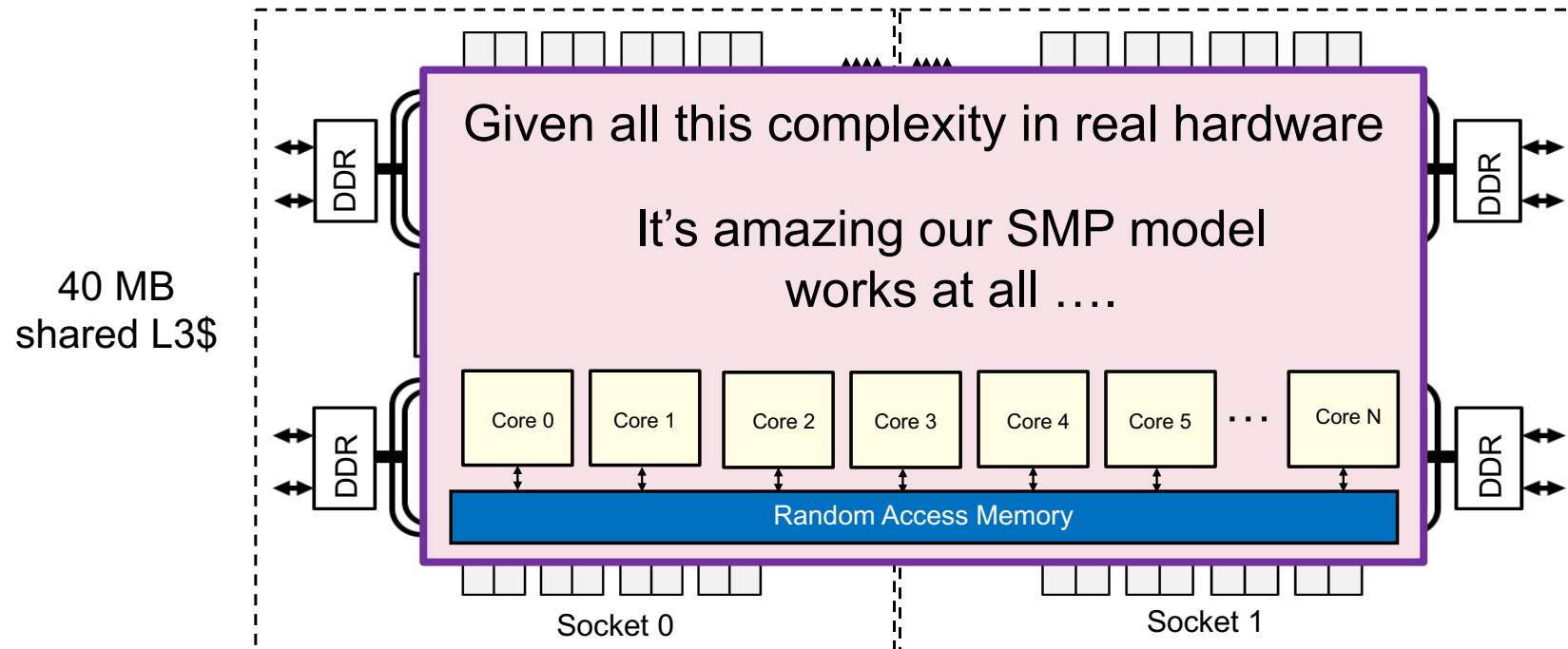


As configured for Cori at NERSC: CPUs at 2.3 GHz, 2 16 GB DIMMs per DDR memory controller, 16 cores per CPU. 2 CPUs connected by a high-speed interconnect (QPI)

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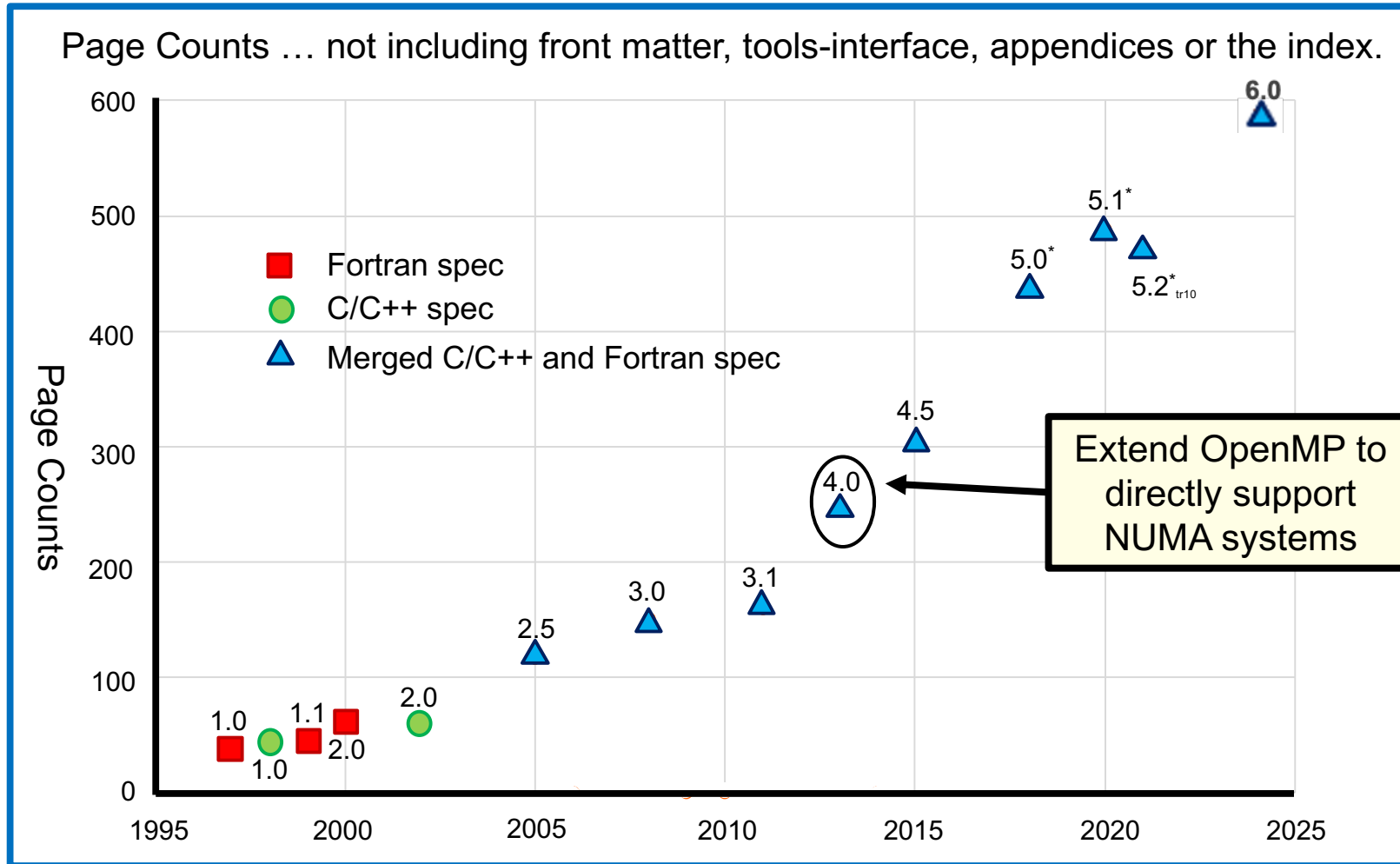


As configured for Cori at NERSC: CPUs at 2.3 GHz, 2 16 GB DIMMs per DDR memory controller, 16 cores per CPU. 2 CPUs connected by a high-speed interconnect (QPI)

Optimizing code for NUMA systems

The Growth of Complexity in OpenMP

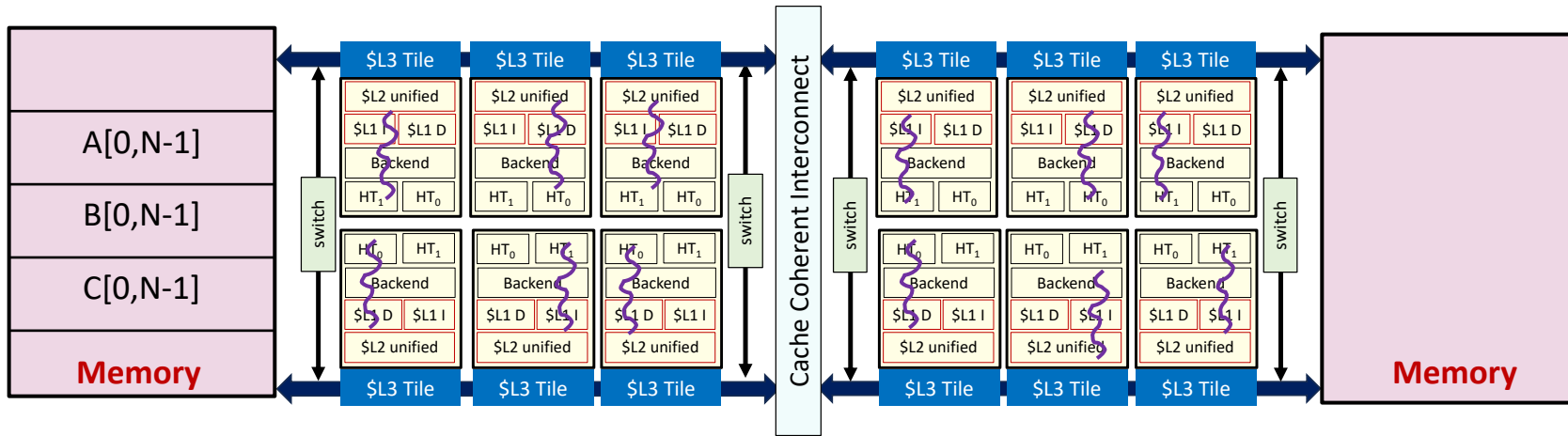
Our goal in 1997 ... A simple interface for application programmers



Consider the Stream Triad Benchmark

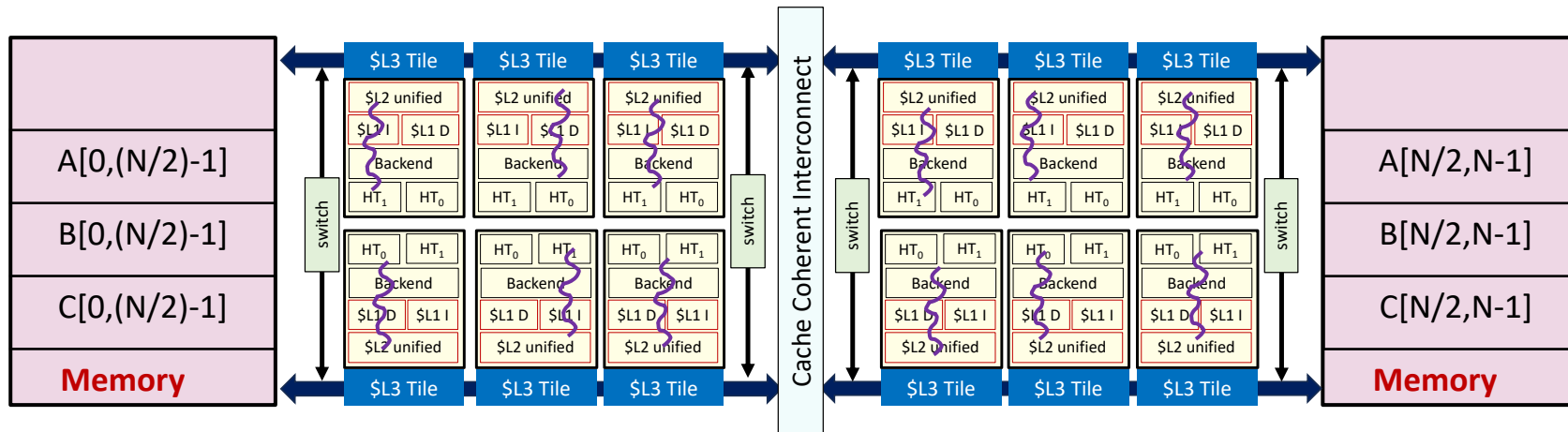
- Stream triad memory-bandwidth benchmark* running on a two-socket Intel® Xeon™ X5675 with 12 threads on 12 cores

```
// Stream Triad benchmark
#pragma omp parallel for
for (int i=0; i<N; i++)
    a[i] = b[i] + scalar * c[i];
```



3 arrays in one NUMA domain

triad
18.2 GB/s



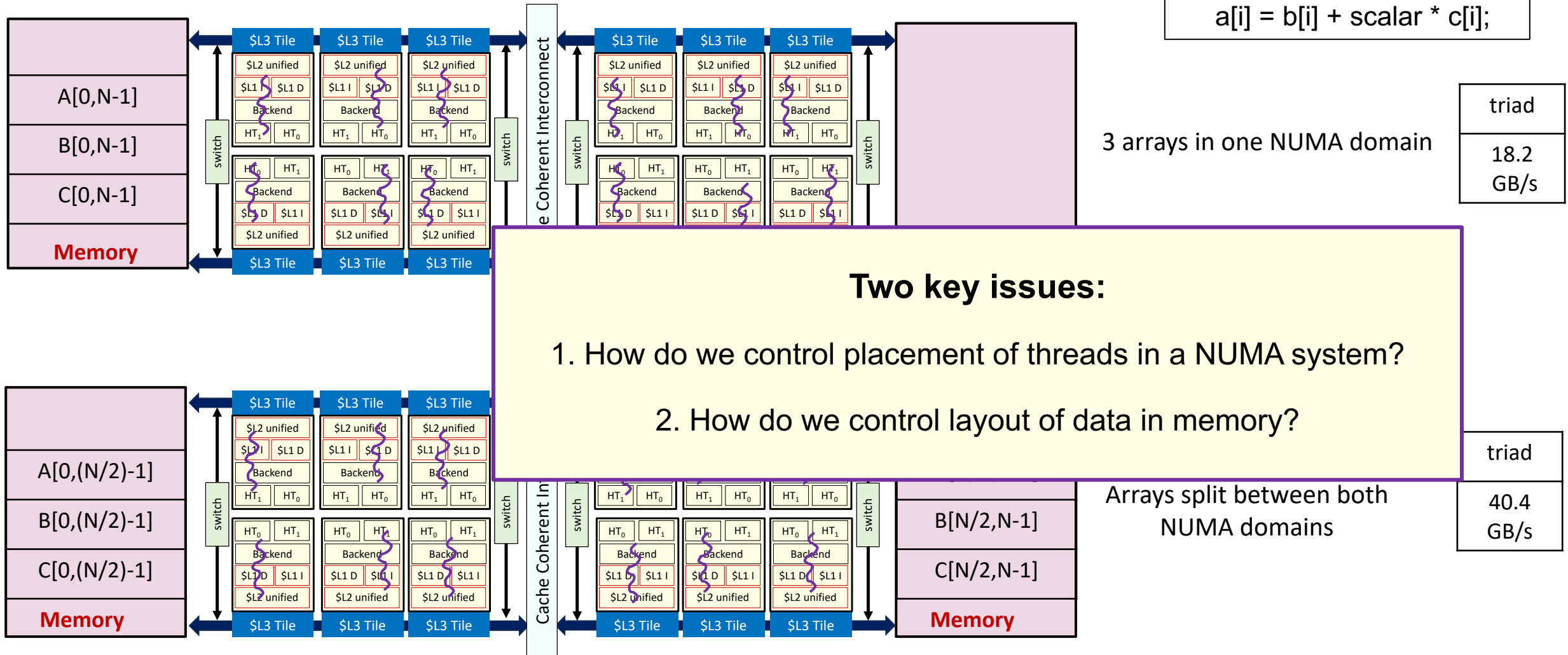
Arrays split between both NUMA domains

triad
40.4 GB/s

Consider the Stream Triad Benchmark

- Stream triad memory-bandwidth benchmark* running on a two-socket Intel® Xeon™ X5675 with 12 threads on 12 cores

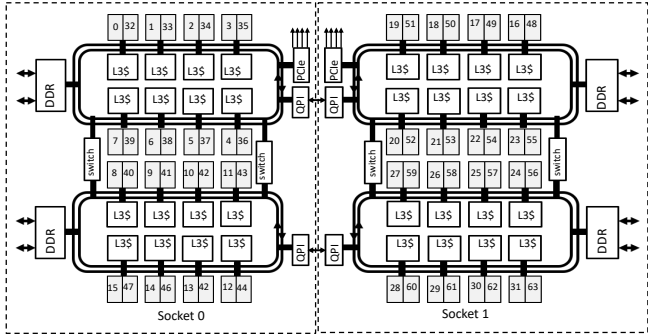
```
// Stream Triad benchmark  
#pragma omp parallel for  
for (int i=0; i<N; i++)  
    a[i] = b[i] + scalar * c[i];
```



Controlling thread layout on a NUMA system:

- 1. How does the OS view components of processors?**
- 2. Where can threads go (places) and how do you line up threads to places (affinity)?**

Discover the OS view of virtual cores



Machine (126GB total)

NUMANode P#0 (63GB)															
Package P#0															
L3 (40MB)															
L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)
L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)
L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)
Core P#0	Core P#1	Core P#2	Core P#3	Core P#4	Core P#5	Core P#6	Core P#7	Core P#8	Core P#9	Core P#10	Core P#11	Core P#12	Core P#13	Core P#14	Core P#15
PU P#0	PU P#1	PU P#2	PU P#3	PU P#4	PU P#5	PU P#6	PU P#7	PU P#8	PU P#9	PU P#10	PU P#11	PU P#12	PU P#13	PU P#14	PU P#15
PU P#32	PU P#33	PU P#34	PU P#35	PU P#36	PU P#37	PU P#38	PU P#39	PU P#40	PU P#41	PU P#42	PU P#43	PU P#44	PU P#45	PU P#46	PU P#47

NUMANode P#1 (63GB)															
Package P#1															
L3 (40MB)															
L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)	L2 (256KB)
L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)	L1d (32KB)
L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)	L1i (32KB)
Core P#0	Core P#1	Core P#2	Core P#3	Core P#4	Core P#5	Core P#6	Core P#7	Core P#8	Core P#9	Core P#10	Core P#11	Core P#12	Core P#13	Core P#14	Core P#15
PU P#16	PU P#17	PU P#18	PU P#19	PU P#20	PU P#21	PU P#22	PU P#23	PU P#24	PU P#25	PU P#26	PU P#27	PU P#28	PU P#29	PU P#30	PU P#31
PU P#48	PU P#49	PU P#50	PU P#51	PU P#52	PU P#53	PU P#54	PU P#55	PU P#56	PU P#57	PU P#58	PU P#59	PU P#60	PU P#61	PU P#62	PU P#63

Graphical output for a dual Socket node with Intel® Xeon™ E5-2698v3 CPUs

PU: processor unit. The smallest physical execution unit that hwloc recognizes.

Based on content from Yun (Helen) He from NERSC)

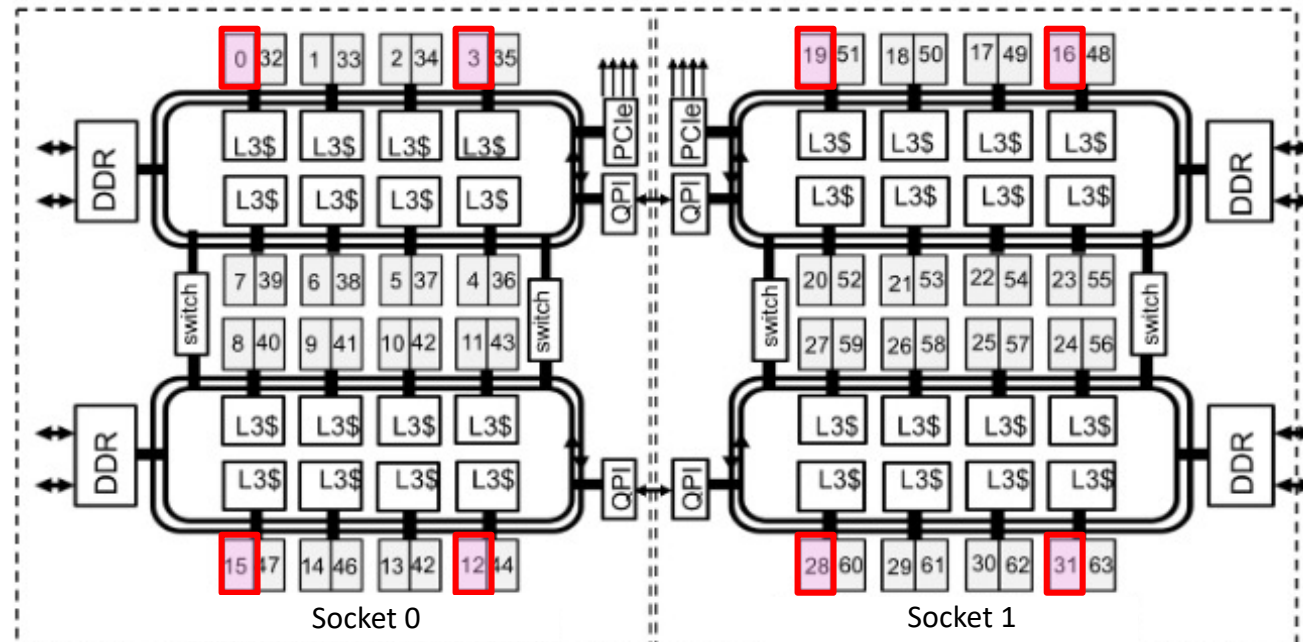
Using OMP_PLACES to select where to map threads

- Use a tool to learn the logical core IDs (processor units or PUs) as understood by the OS.
- OpenMP defines an environment variable (OMP_PLACES) to define where threads can be scheduled to execute.

```
> export OMP_PLACES="{0, 3, 15, 12, 19, 16, 28, 31}"
```

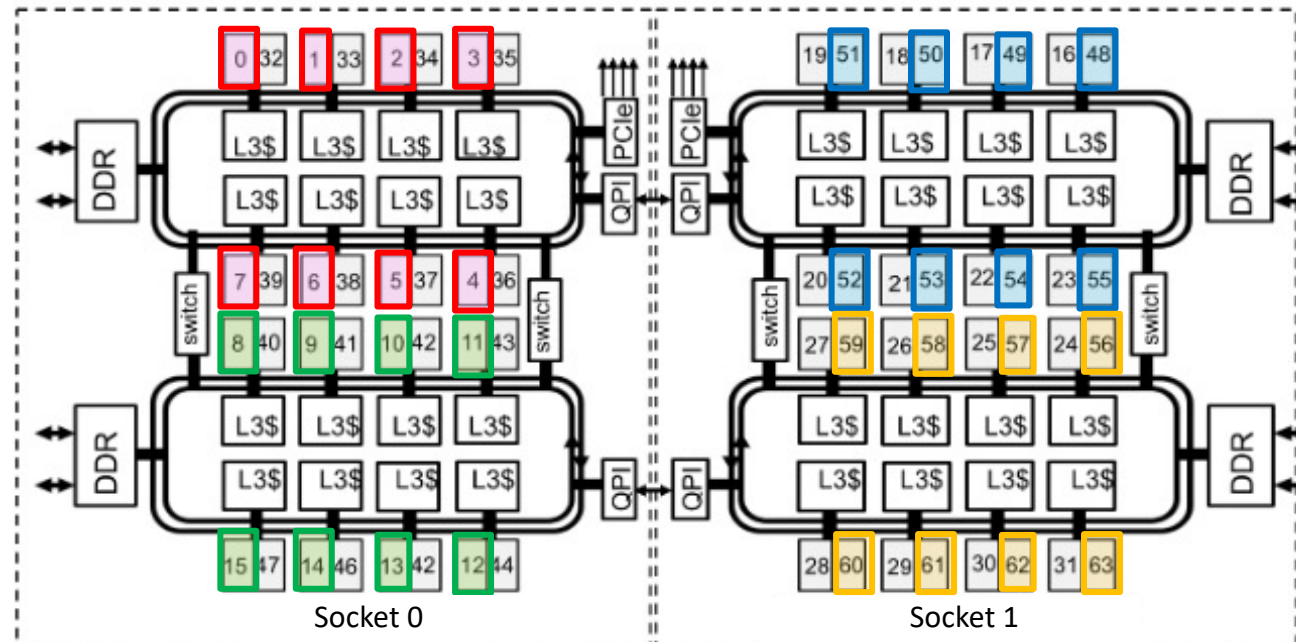
```
> export NUM_THREADS= 6
```

You can't control where the first thread goes, but the others will start at that first thread and wrap around the set of places



Using OMP_PLACES to select where to map threads

- Use a tool to learn the logical core IDs (processor units or PUs) as understood by the OS.
 - OpenMP defines an environment variable (OMP_PLACES) to define where threads can be scheduled to execute.
 - We can organize places into partitions to capture the structure of the NUMA system.
 - In the following, we have one places partition per NUMA domain*
- > export OMP_PLACES="{0:7:1},{8:15:1},{48:55:1},{56:63:1}"

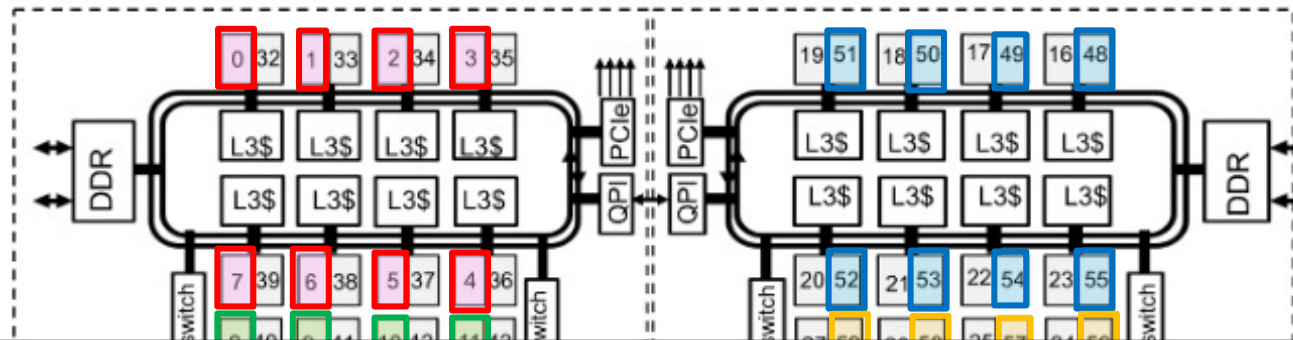


* Note that {0:7:1}. is equivalent to {0, 1, 2, 3, 4, 5, 6, 7}.

The details of what constitutes a NUMA domain is OS dependent ... it depends on how the OS maps memory pages onto DRAM and how caches are managed.

Using OMP_PLACES to select where to map threads

- Use a tool to learn the logical core IDs (processor units or PUs) as understood by the OS.
 - OpenMP defines an environment variable (OMP_PLACES) to define where threads can be scheduled to execute.
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 - In the following, we have one places partition per NUMA domain*
- > export OMP_PLACES="{0:7:1},{8:15:1},{48:55:1},{56:63:1}"



Programmers can use OMP_PLACES for detailed control over the execution-units threads utilize. BUT ...

- The rules for mapping onto physical execution units are complicated.
- PLACES expressed as numbers is non-portable

There has to be an easier and more portable way to describe places

* Note that {0:7:1}. is equivalent to {0, 1, 2, 3, 4, 5, 6, 7}.

The details of what constitutes a NUMA domain is OS dependent ... it depends on how the OS maps memory pages onto DRAM and how caches are managed.

Hardware Abstraction and OMP_PLACES

- OMP_PLACES can use an abstract name instead of a list of numbers
 - **threads**: each place corresponds to a single hardware thread
 - **cores**: each place corresponds to a single core (which may have one or more hardware threads)
 - **sockets**: each place corresponds to a single socket (consisting of one or more cores)

- Examples:
 - `export OMP_PLACES=threads`
 - `export OMP_PLACES=cores`

Mapping threads to places: Thread affinity with OMP_PROC_BIND

- OMP_PROC_BIND controls thread affinity within and between OpenMP places
- Allowed values:
 - **true**: the runtime will not move threads around between processors
 - **false**: the runtime may move threads around between processors
 - **close**: bind threads close to the primary* thread
 - **spread**: bind threads as evenly distributed as possible (i.e., spread them out)
 - **primary**: bind threads to the same place as the primary thread
- The values **primary***, **close**, and **spread** imply the value **true**

Example:

```
export OMP_PROC_BIND=spread
```

*Primary thread: this is the thread with ID=0 that encountered the parallel construct and created the team of threads

Mapping threads to places: Thread affinity with `proc_bind()` clause

- The `proc_bind()` clause controls thread affinity within and between OpenMP places
- Allowed values:
 - `true`: the runtime will not move threads around between processors
 - `false`: the runtime may move threads around between processors
 - `close`: bind threads close to the primary* thread
 - `spread`: bind threads as evenly distributed as possible (i.e., spread them out)
 - `primary`: bind threads to the same place as the primary thread
- The values **primary***, **close**, and **spread** imply the value **true**

Example ... using clauses on a parallel construct:

```
#pragma omp parallel num_threads(4) proc_bind(spread)
```

*Primary thread: this is the thread with ID=0 that encountered the parallel construct and created the team of threads

Note: You **cannot** set places as a clause on a directive since the OS defines places when it launches the process to run a program

Examples: OMP_PROC_BIND

Assume 4 cores, 2 hardware threads per core, and 4 OpenMP threads

Set 4 place partitions: OMP_PLACES={0,1},{2,3},{4,5},{6,7}

- **none**: no affinity setting
- **close**: Bind threads as close to each other as possible

Node	Core 0		Core 1		Core 2		Core 3	
	HT ₀	HT ₁	HT ₀	HT ₁	HT ₀	HT ₁	HT ₀	HT ₁
	PU 0	PU 1	PU 2	PU 3	PU 4	PU 5	PU 6	PU 7
Thread	0	1	2	3				

- **spread**: Bind threads as far apart as possible

Node	Core 0		Core 1		Core 2		Core 3	
	HT ₀	HT ₁	HT ₀	HT ₁	HT ₀	HT ₁	HT ₀	HT ₁
	PU 0	PU 1	PU 2	PU 3	PU 4	PU 5	PU 6	PU 7
Thread	0		1		2		3	

We have no control of where the initial thread is placed, but for these examples, we assume it is placed on HT₀ of Core 0.

With close, threads placed in consecutive locations

With spread, threads placed in first place of each partition

- **primary**: bind threads to the same place as the primary thread

OMP_PROC_BIND Choices for STREAM Benchmark

OMP_NUM_THREADS=32

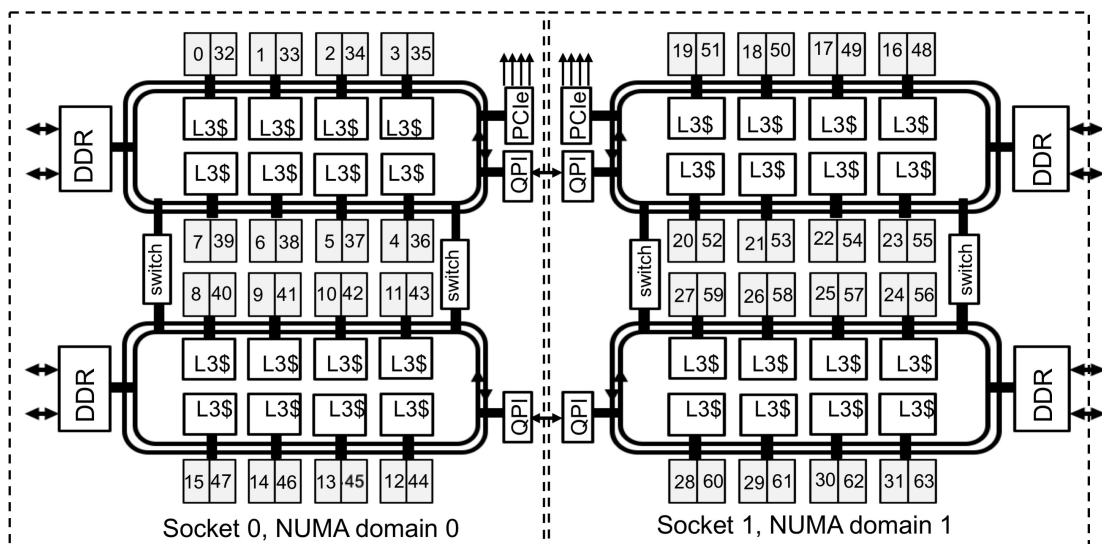
OMP_PLACES=threads

OMP_PROC_BIND=close

Threads 0 to 31 bind to cores
(0,32),(1,33),(2,34),... (15,47). All threads are in the first socket. The second socket is idle. Not optimal.

OMP_PROC_BIND=spread

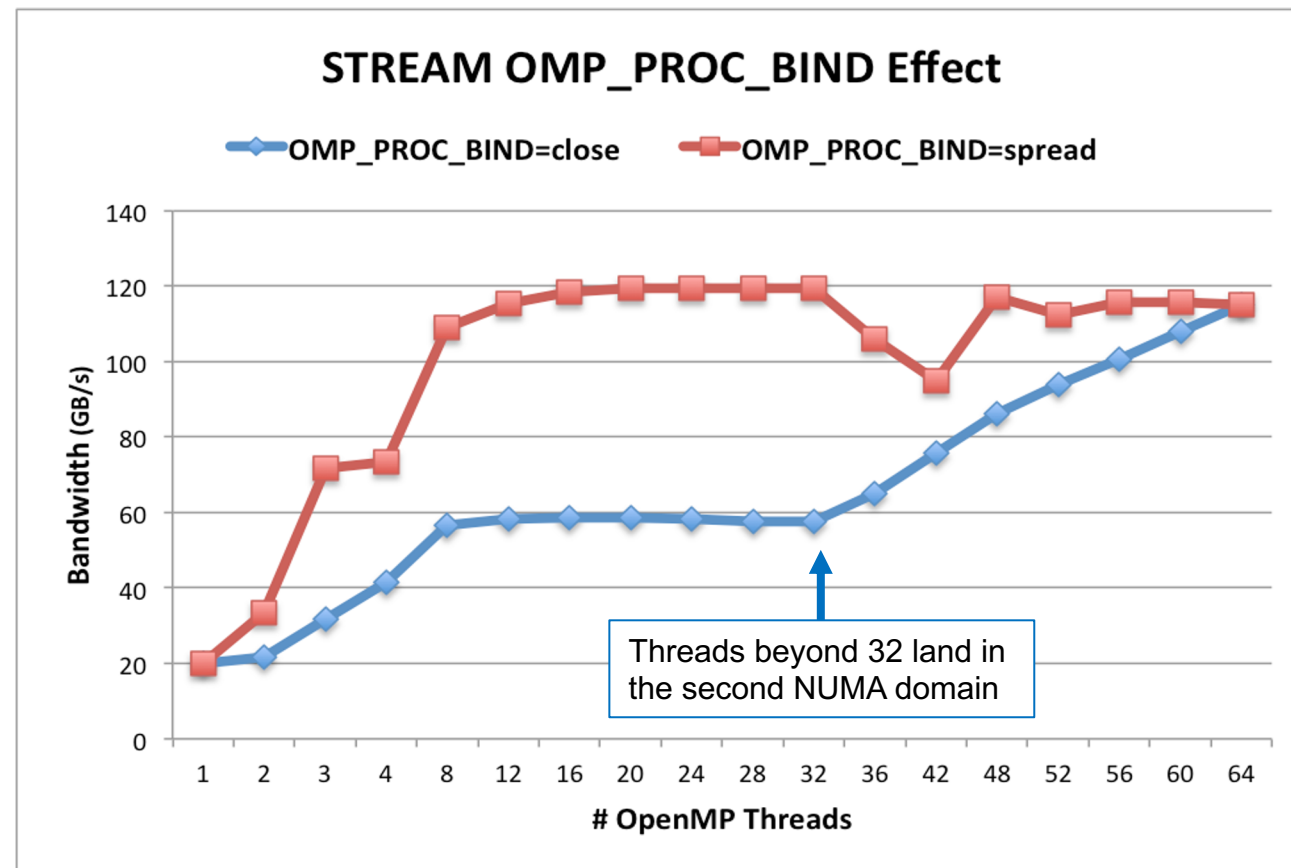
Threads 0 to 31 bind to cores 0,1,2,... to 31. Both sockets and memory are used to maximize memory bandwidth.



Blue: OMP_PROC_BIND=close

Red: OMP_PROC_BIND=spread

In both cases, assume the arrays are spread out across all memory domains (a topic we'll address soon)



Stream is a well-known memory bandwidth benchmark based on simple vector operations on huge vectors

Aligning memory to threads ... First touch

Memory Affinity: Exploiting “First Touch” page mapping policy

Step 1.1 Initialization by primary thread only

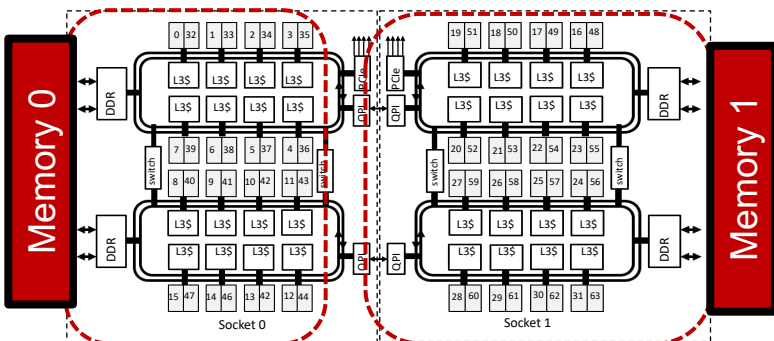
```
for (j=0; j<VectorSize; j++) {  
    a[j] = 1.0; b[j] = 2.0; c[j] = 0.0;}
```

Step 1.2 Initialization by all threads

```
#pragma omp parallel for  
for (j=0; j<VectorSize; j++) {  
    a[j] = 1.0; b[j] = 2.0; c[j] = 0.0;}
```

Step 2 Compute

```
#pragma omp parallel for  
for (j=0; j<VectorSize; j++)  
    a[j]=b[j]+d*c[j];
```



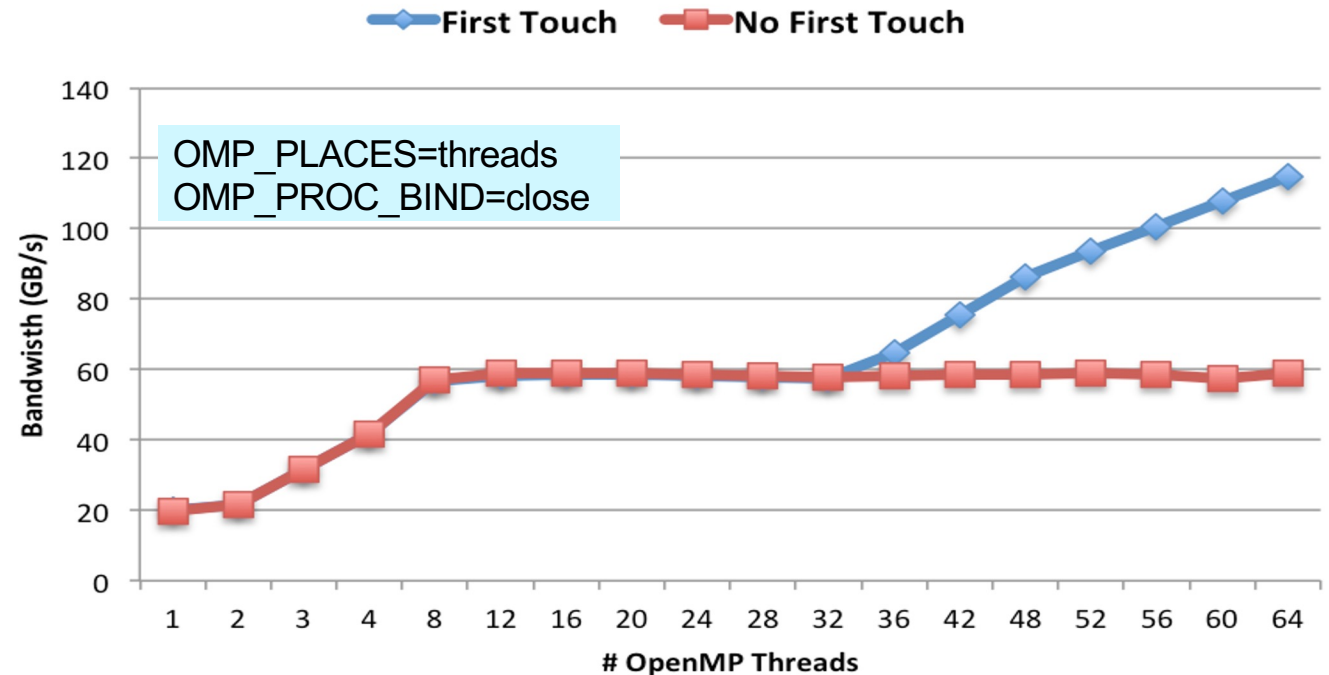
A dual Socket node with Intel® Xeon™ E5-2698v3 CPUs

- The OS maps pages of memory based on a **first touch** policy.
- Hence, Affinity to memory is not defined when memory is allocated ... it is defined when the memory is initialized.
- The result: memory is local to the thread that initializes it.

Red: step 1.1 + step 2. Arrays come from Memory 0 only

Blue: step 1.2 + step 2. Arrays come from both Memory domains

STREAM NUMA Effect



Based on content from Yun (Helen) He from NERSC)

Example: working with the First Touch Policy

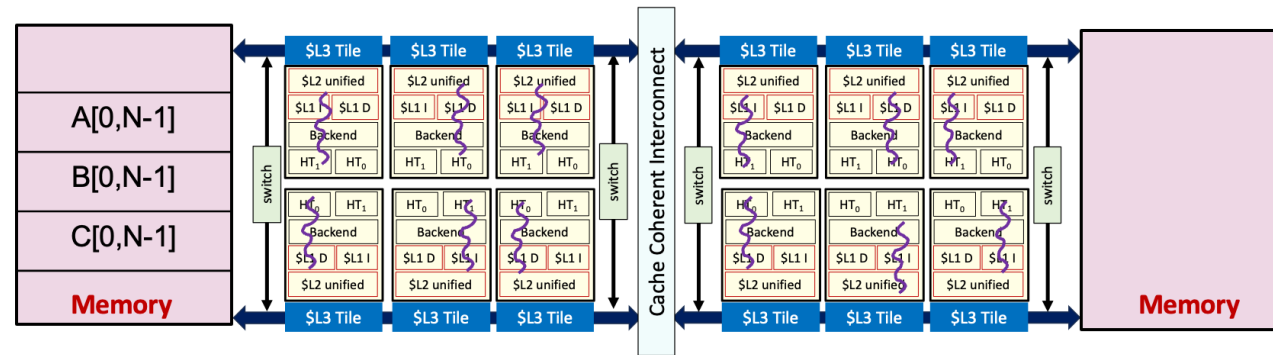
Remember this slide?

Consider the Stream Triad Benchmark

- Stream triad memory-bandwidth benchmark* running on a two-socket Intel® Xeon™ X5675 with 12 threads on 12 cores

```
// Stream Triad benchmark
#pragma omp parallel for
for (int i=0; i<N; i++)
    a[i] = b[i] + scalar * c[i];
```

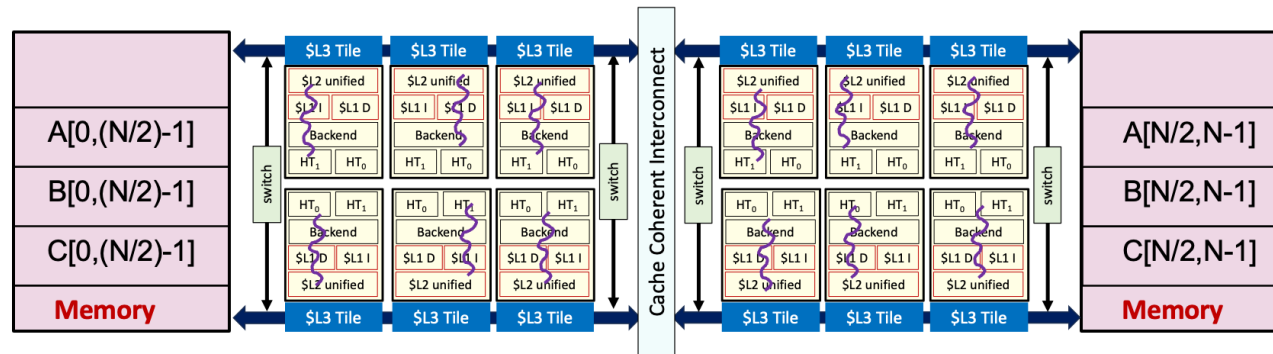
Arrays A, B, and C
initialized on
primary thread



3 arrays in one NUMA domain

triad
18.2 GB/s

Arrays A, B, and C
initialized with the
same “parallel for
schedule” as used
for the computation



Arrays split between both
NUMA domains

triad
40.4 GB/s

Exercise: Matrix Transpose

- Consider the program transpose.c
 - Generates a square matrix, transposes the matrix and tests the result.
- This is all memory movement. Optimize the transpose operation:

```
export OMP_PROC_BIND=spread
#pragma omp parallel proc_bind(spread)
export OMP_PLACES="{0,21, 42, 64}"
export OMP_PLACES="{0:7:1},{8:15:1},{48:55:1},{56:63:1}"
export OMP_PLACES=threads

#pragma omp parallel
#pragma omp for
#pragma omp parallel for
#pragma omp for reduction(op:list)
#pragma omp critical
int omp_get_num_threads();    int omp_get_thread_num();  int omp_set_num_threads();
double omp_get_wtime();
```

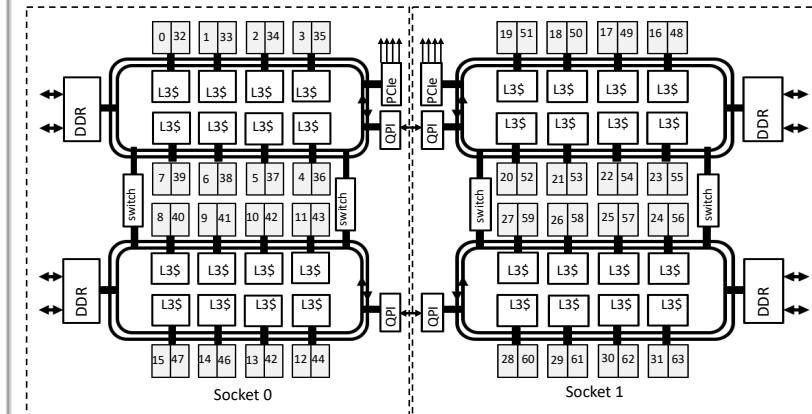
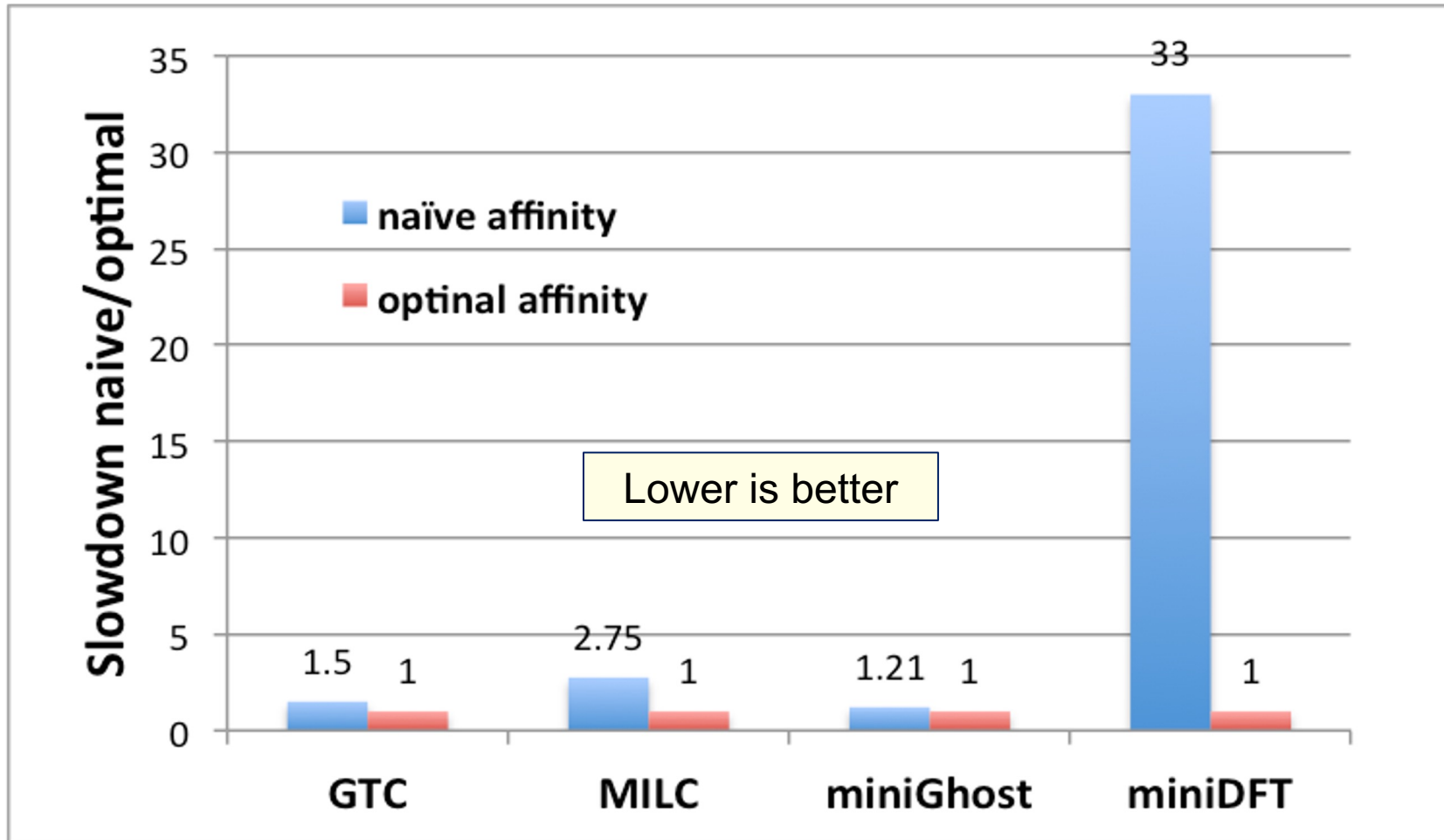
Annotations:

- Box 1: true, false, close, spread, primary (points to `spread` in `OMP_PROC_BIND=spread` and `proc_bind(spread)`)
- Box 2: threads, cores, sockets (points to `threads` in `OMP_PLACES=threads`)

**Wrapping up our discussion of taking NUMA
features of a system into account in your
multithreaded programs ...**

Getting the affinity right can have serious impacts on performance

Application Benchmark Performance for a number of benchmarks at NERSC

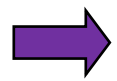


Results running on the Cori system at NERSC which has dual Socket nodes with Intel® Xeon™ E5-2698v3 CPUs

Finding the best strategy for thread affinity

- Experiment to find the best combinations of OMP_PLACES and OMP_PROC_BIND.
 - Using the environment variables makes it easy to try many options
- The best approach depends on the system but also on the features of an application
 - Putting threads far apart ... on different sockets
 - May improve aggregate memory bandwidth available to an application
 - May improve combined cache size for the application
 - May increase synchronization overhead
 - Putting threads close together ... on adjacent cores that may share some caches
 - May reduce synchronization overhead
 - May decrease memory bandwidth and total cache size
- Vendors have their own constructs for controlling NUMA features of a system.
 - Avoid vendor-specific constructs if you can ... use portable OMP_PLACES and OMP_PROC_BIND

- Introduction to OpenMP
- Creating Threads
- Synchronization ... simple collective sync
- Parallel Loops
- Data Environment
- Synchronization ... the general case (including Memory Models)
- Irregular Parallelism and Tasks
- OpenMP for real hardware (NUMA)

 • Recap

The OpenMP Common Core: Most OpenMP programs only use these 21 items

OpenMP pragma, function, or clause	Concepts
#pragma omp parallel	Parallel region, teams of threads, structured block, interleaved execution across threads.
void omp_set_thread_num() int omp_get_thread_num() int omp_get_num_threads()	Default number of threads and internal control variables. SPMD pattern: Create threads with a parallel region and split up the work using the number of threads and the thread ID.
double omp_get_wtime()	Speedup and Amdahl's law. False sharing and other performance issues.
setenv OMP_NUM_THREADS N	Setting the internal control variable for the default number of threads with an environment variable
#pragma omp barrier #pragma omp critical	Synchronization and race conditions. Revisit interleaved execution.
#pragma omp for #pragma omp parallel for	Worksharing, parallel loops, loop carried dependencies.
reduction(op:list)	Reductions of values across a team of threads.
schedule (static [,chunk]) schedule(dynamic [,chunk])	Loop schedules, loop overheads, and load balance.
shared(list), private(list), firstprivate(list)	Data environment.
default(none)	Force explicit definition of each variable's storage attribute
nowait	Disabling implied barriers on workshare constructs, the high cost of barriers, and the flush concept (but not the flush directive).
#pragma omp single	Workshare with a single thread.
#pragma omp task #pragma omp taskwait	Tasks including the data environment for tasks.

There is Much More to OpenMP than the Common Core

- Synchronization mechanisms
 - locks, synchronizing flushes and several forms of atomic
- Data environment
 - lastprivate, threadprivate, default(private|shared)
- Fine grained task control
 - dependencies, tied vs. untied tasks, task groups, task loops ...
- Vectorization constructs
 - simd, uniform, simdlen, inbranch vs. nobranch,
- Map work onto an attached device (such as a GPU)
 - target, teams distribute parallel for, target data ...
- ... and much more. The OpenMP 5.0 specification is over 618 pages!!!

Don't become overwhelmed. Master the common core and move on to other constructs when you encounter problems that require them.

Resources

- The OpenMP Architecture review Board (ARB) has a wealth of helpful resources on its web site: www.openmp.org

OpenMP

The OpenMP API specification for parallel programming

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Specifications Home > Specifications

OpenMP 5.2 Specification

- OpenMP API 5.2 Specification – Nov 2021
 - Softcover Book on Amazon
- OpenMP API Additional Definitions 2.0 – Nov 2020
- OpenMP API 5.2 Reference Guide (English) (Japanese)
- OpenMP API 5.2 Supplementary Source Code
- OpenMP API 5.2 Examples – April 2022
 - Softcover Book on Amazon
- OpenMP API 5.2 Stack Overflow

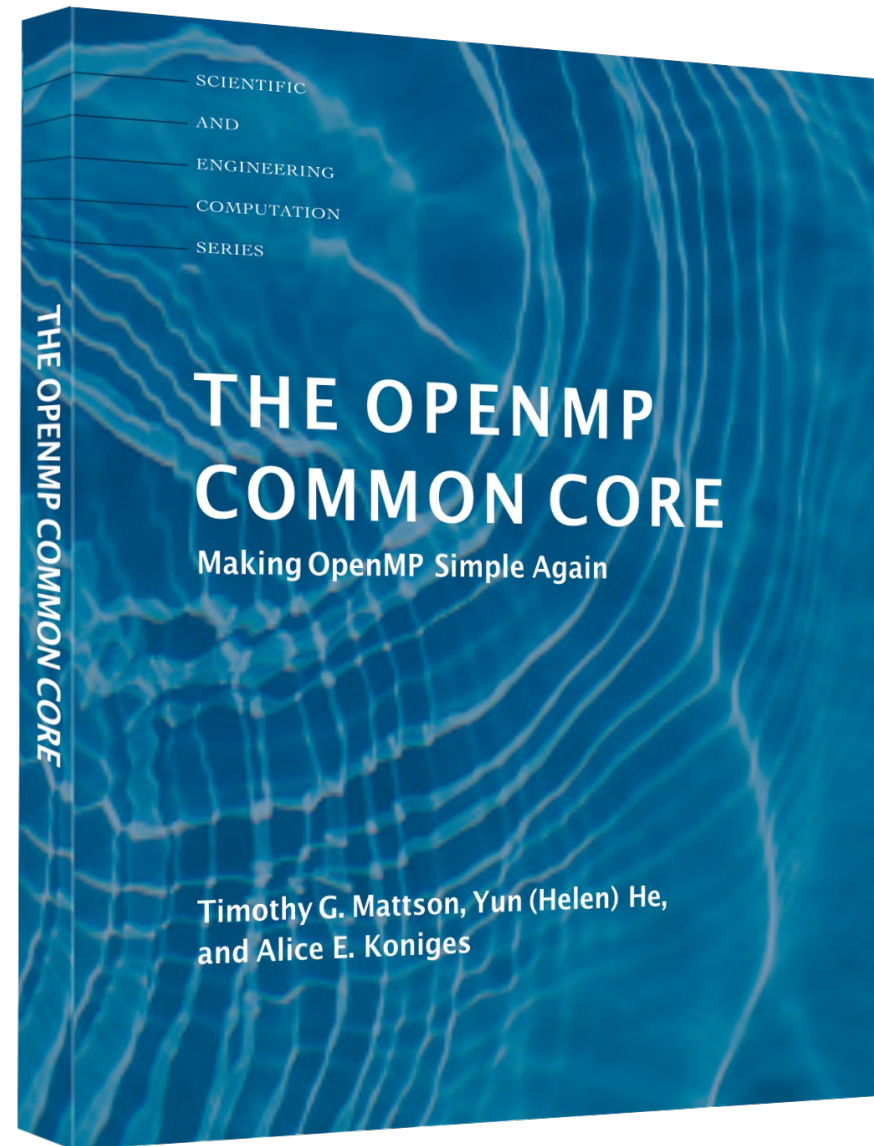
OpenMP 5.1 Specification

- OpenMP API 5.1 Specification – Nov 2020
 - HTML Version Softcover Book on Amazon
- OpenMP API Additional Definitions 2.0 – Nov 2020
- OpenMP API 5.1 Reference Guide
- OpenMP API 5.1 Supplementary Source Code
- OpenMP API 5.1 Examples – August 2021
- OpenMP API 5.1 Stack Overflow

Including a comprehensive collection of examples of code using the OpenMP constructs

To learn OpenMP:

- An exciting new book that Covers the Common Core of OpenMP plus a few key features beyond the common core that people frequently use
- It's geared towards people learning OpenMP, but as one commentator put it ... **everyone at any skill level should read the memory model chapters.**
- Available from MIT Press



Books about OpenMP

A great book that covers
OpenMP features beyond
OpenMP 2.5

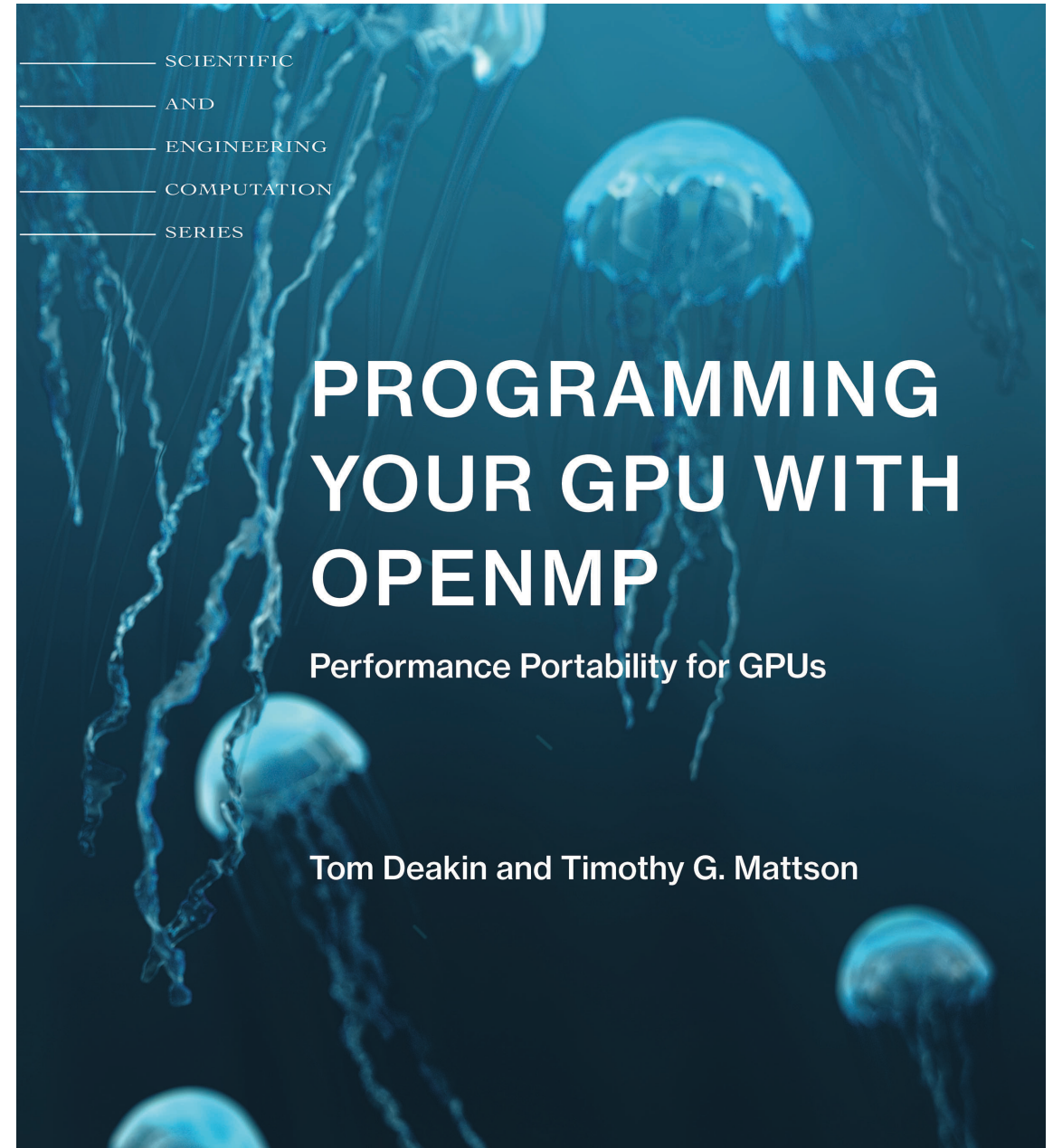


Books about OpenMP

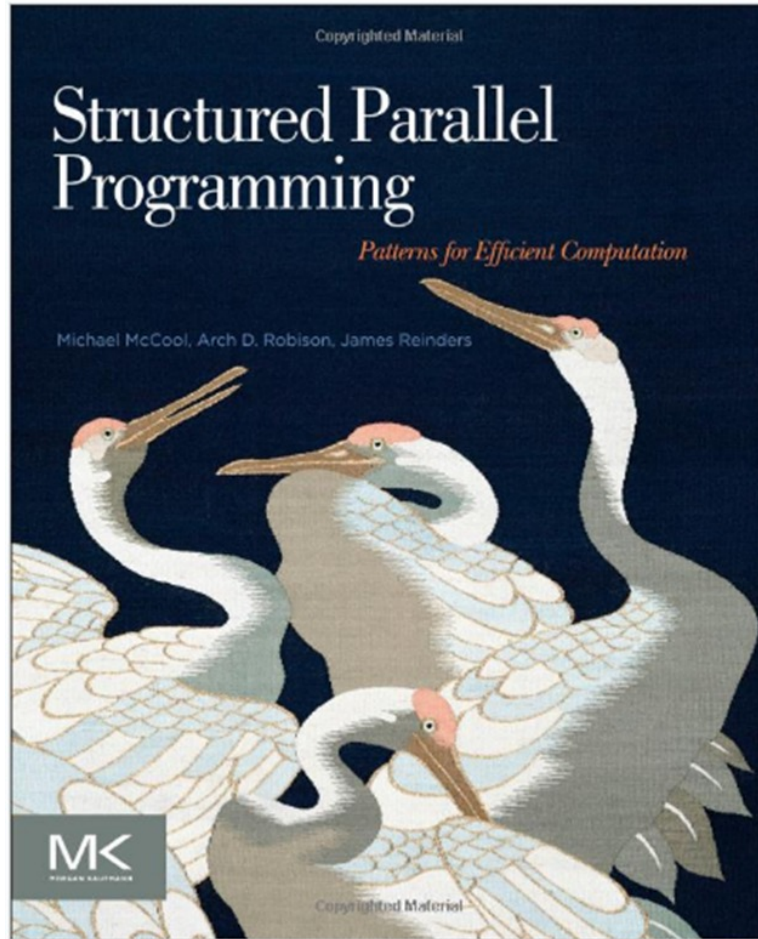
The latest book on OpenMP ...

Released in November 2023.

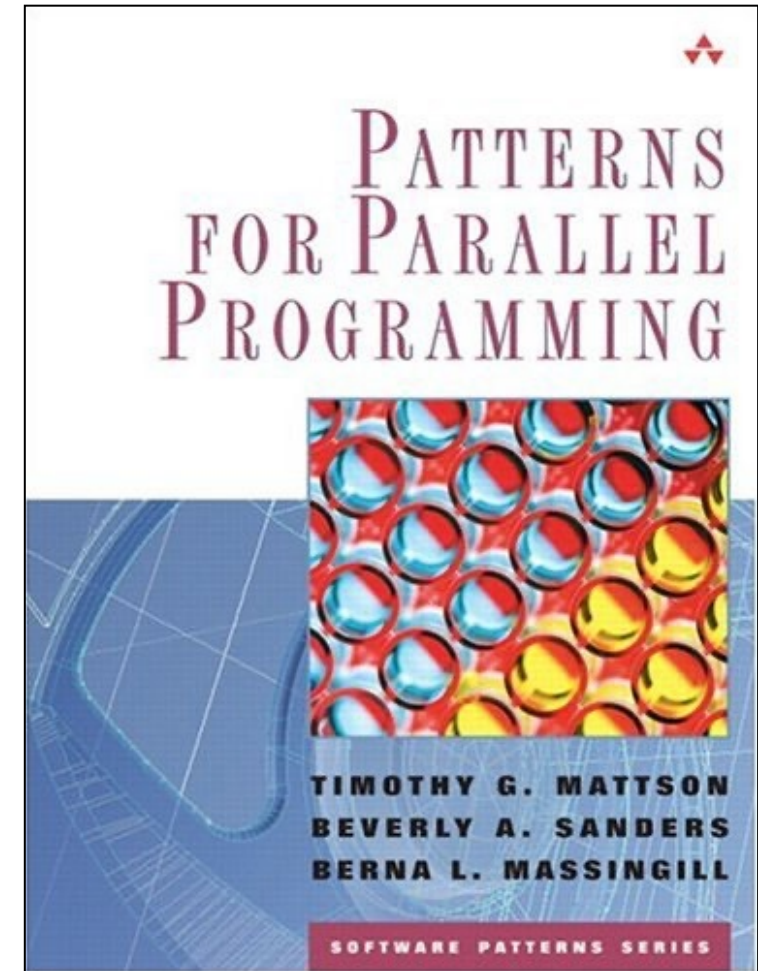
A book about how to use OpenMP to
program a GPU.



Background references



A great book that explores key patterns with Cilk, TBB, OpenCL, and OpenMP (by McCool, Robison, and Reinders)



- A book about how to “think parallel” with examples in OpenMP, MPI and java